



T-L IRRIGATION CO.

Hastings, Nebraska



Precision Point Control III System

INSTALLATION, OPERATION AND TROUBLESHOOTING MANUAL

CD90433

Feb. 2021
Added v7.04 Firmware
v6.58 & 7.04

CONTENTS

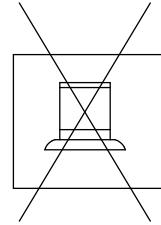
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Notes On Run Lights: Run Lights used with the Precision Point Control III should be supplied by T-L with a Manufacturer's Model No. IS3006. This model has special electronic noise suppression built into it. Electronic noise can interfere with the performance of the Precision Point Control III.

Run Lights at the Pivot Point CANNOT be used with a Remote Mounted Control Panel UNLESS a separate wire is used.



Note On Spark Ignition Engines: To reduce electronic noise, spark ignition engines should have:

- resistor plugs
- suppresser plug wires

General Information:

The Precision Point Control III system uses electro-hydraulic components to provide reliable speed and direction control from the pivot point. Proper installation procedures and precautions will ensure years of continued operation. Observe the following during installation:

1. Read this entire manual before beginning with the installation.
2. Be sure all electrical power is off when wiring the system.
3. Use care in handling and stripping all electrical cables and wires. Watch for nicked wires that may cause a short.
4. Use drip loops with all electrical cable to prevent moisture infiltration.

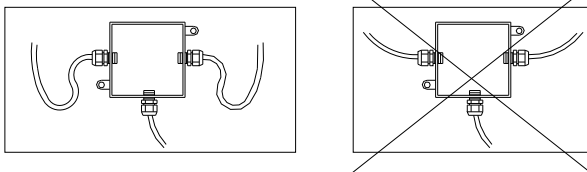


Figure 1. Use Drip Loop.

5. Keep hydraulic lines and ports free from contamination.

Precision Point Control III System Components and Location (Figure 2)

Refer to Figure 1 to help you become familiar with *The Precision Point Control III* system and its major components. You should be familiar with these items and where they are located on the system.

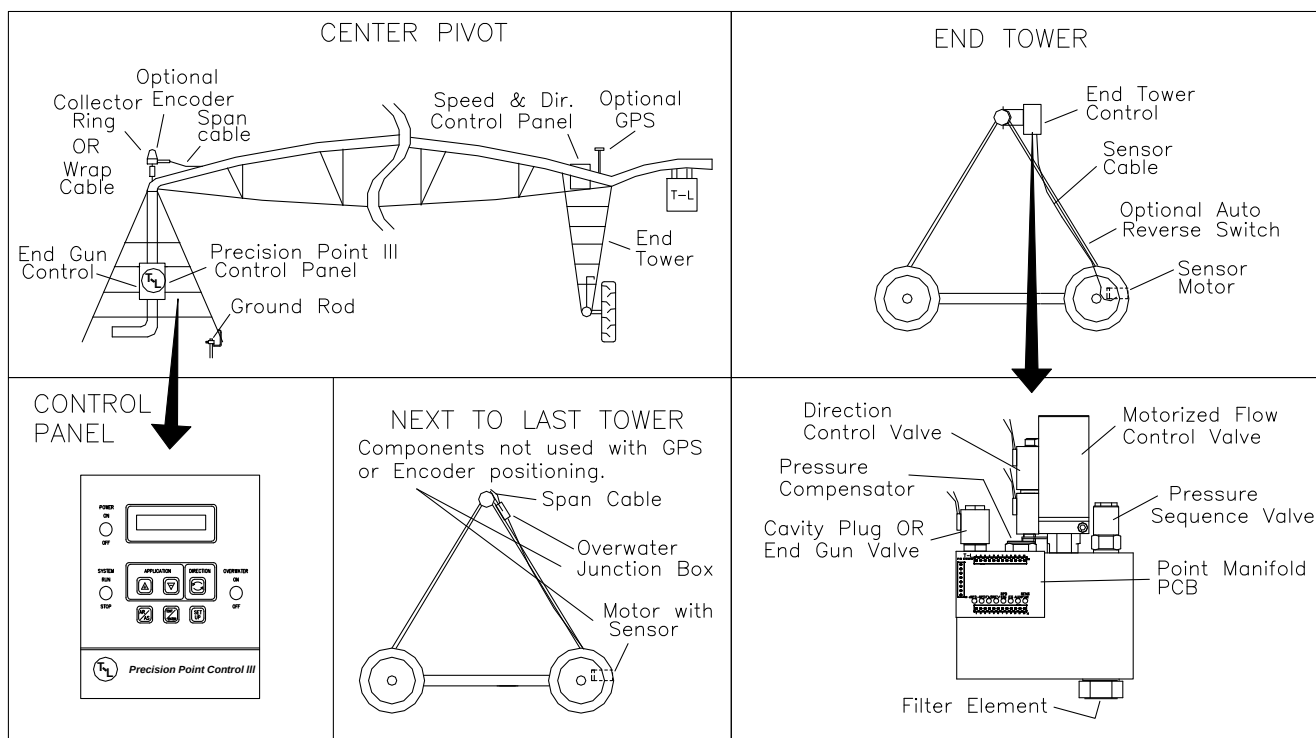


Figure 2. Component Identification.

The Precision Point Control III Panel Mounting (Figures 3, 4, 5, 6)

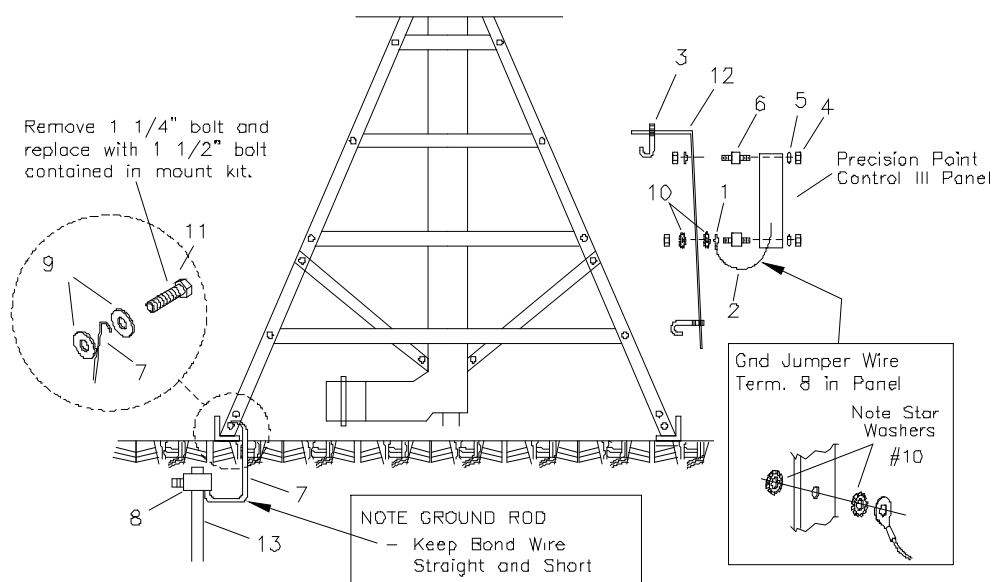
There are 4 different mounting locations for *The Precision Point Control III* Panel:

- Swivel Pivot Point
- Hose Wrap Pivot Point
- Quick Tow Pivot Point
- Post Mount



A. Swivel Pivot Point MB52016 (Figure 3)

1. Decide which side of the pivot point the Precision Point Control III Panel is to be mounted on.
2. Attach 2 Panel Mount Brackets (AB2937) to Pivot Point Steps #3 and #1 using 4 J-Bolts, 5/16" x 3" with nuts. Leave J-Bolts loose until Precision Point Control III Panel has been attached.
3. Attach Precision Point Control III Panel to Panel Mount Bracket using Rubber Mounts, 5/16" Lock Washers, and 5/16" Nuts. On bottom mount, attach Ground Jumper Wire from frame with Star Washers to inside panel ground terminal 8 of 12 terminal block.
4. Center Precision Point Control III Panel and Brackets. Tighten nuts on J-Bolts.



Ref.	Part No.	Description	# Req'd.
1	EH5291	Term Ring 5/16 X 16-14	1
2	EW5383	16 Ga. Wire - Black	1 ft.
3	FB54385	J-Bolt 5/16 X 3" w/Nut	4
4	FN5400	Hex Nut 5/16	8
5	FW5650	Lock Washer 5/16	8
6	UD57701	Rubber Mount 5/16 Studs	4
7	EW5397	#6 Solid Copper Wire	10 ft.
8	FC5739	Ground Rod Clamp 1/2"	1
9	FW5663	Flat Washer 5/8"	2
10	FW56513	Star Washer 5/16"	2
11	FB5455	HHCS 5/8" x 1 1/2" Grd. 5	1
Shipped separate on Load			
12	AB2937	Mount Bracket	2
13	RR5394	Ground Rod 1/2" X 8' Copper	1

Figure 3. The Precision Point Control III Panel Installation – Swivel Pivot Point

HARDWARE INSTALLATION



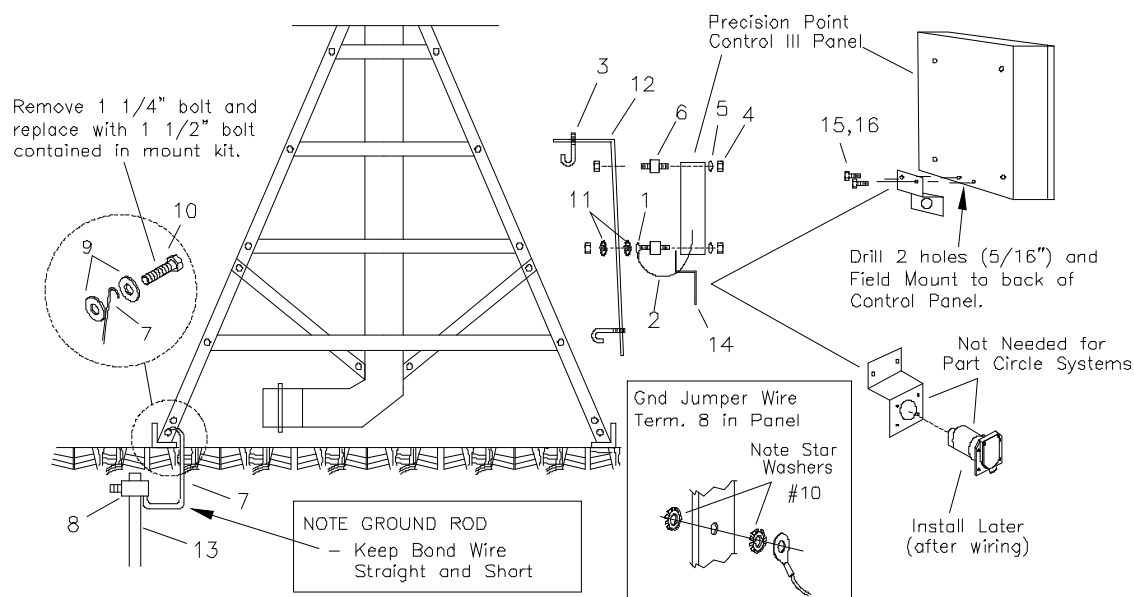
B. Hose Wrap Pivot Point MB52016 & MB45978 (Figure 4)

1. Decide on Control Panel mounting location. Consider Auto Reverse.

NOTE: If the system is a Hose Wrap with Auto Reverse and will never make a full circle, the Span Cable can be wired directly into the Control Panel. (Plug Mount Bracket not needed)

2. Attach Control Panel as shown in Figure 4. Field drill holes and attach Plug Mount Bracket if needed to lower back of Precision Point Control Panel.

CAUTION: On Full Circle Hose Wrap system, make sure wrap hose or wrap cable do not get caught on panel. Rewrap often.



Ref.	Part No.	Description	# Req'd.
1	EH5291	Term Ring 5/16 X 16-14	1
2	EW5383	16 Ga. Wire - Black	1 ft.
3	FB54385	J-Bolt 5/16 X 3" w/Nut	4
4	FN5400	Hex Nut 5/16	8
5	FW5650	Lock Washer 5/16	8
6	UD57701	Rubber Mount 5/16 Studs	4
7	EW5397	#6 Solid Copper Wire	10 ft.
8	FC5739	Ground Rod Clamp 1/2"	1
9	FW5663	Flat Washer 5/8"	2
10	FW56513	Star Washer 5/16"	2
11	FB5455	HHCS 5/8" x 1 1/2" Grd. 5	1
Shipped separate on Load			
12	AB2937	Mount Bracket	2
13	RR5394	Ground Rod 1/2" X 8' Copper	1
MB45978			
14	AB15724	Point Cable Wrap Plug Mount	1
15	FB5435	HHCS 1/4 x 1	2
16	FN5399	Nut Hex 1/4-20	2

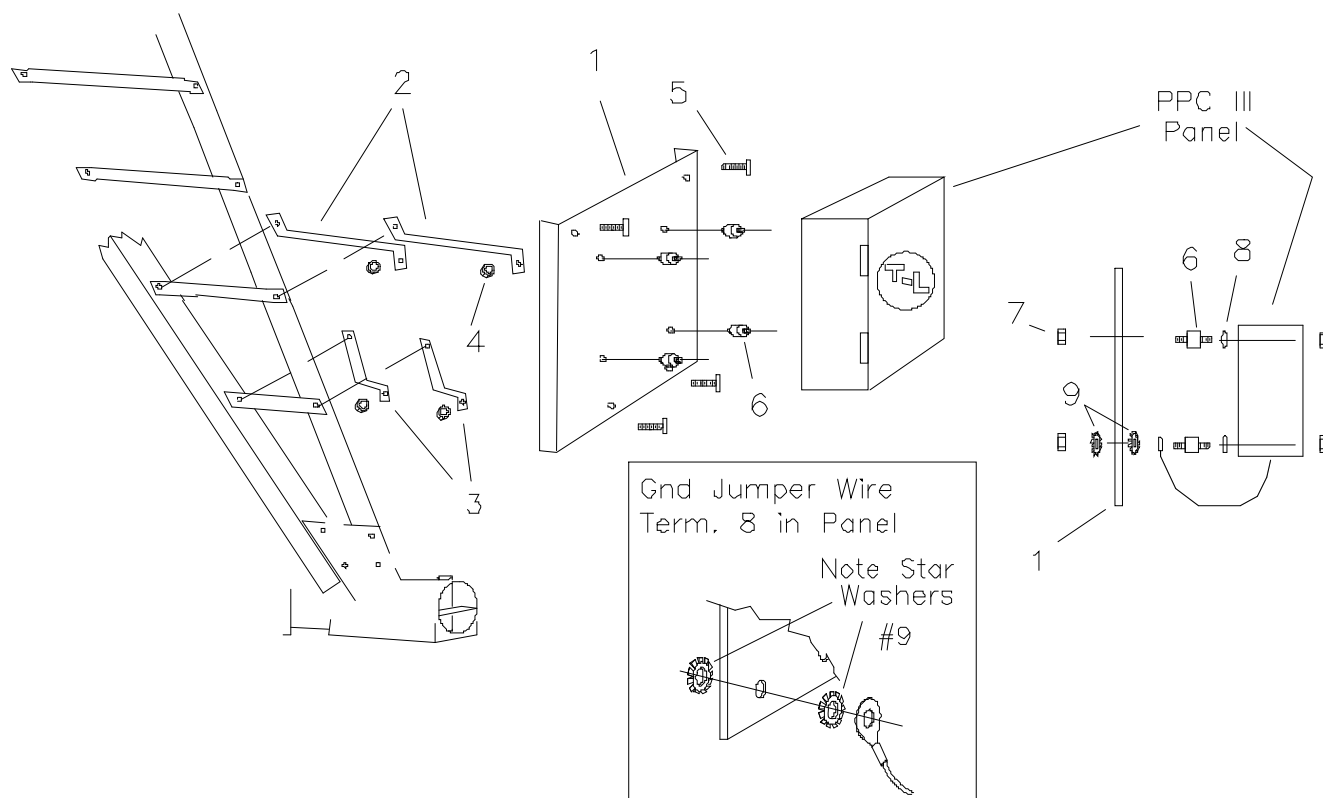
Figure 4. The Precision Point Control Panel Installation – Hose Wrap Pivot Point



C. Quick Tow Pivot Point (Figure 5)

On a Quick Tow Pivot Point, the Control Panel mounts in the location of the No. 1 and 2 Tower Steps.

1. Bolt four panel mount straps (2 & 3) to Tower legs at the #1 and #2 Tower Step locations.
2. Bolt Panel Mount Plate (1) to the Panel Mount Straps using 3/8 x 1 bolts and nuts (4 & 5).
3. Attach the Control Panel to the mounting plate using 4 Rubber Mount 5/16 Studs (6 & 7).



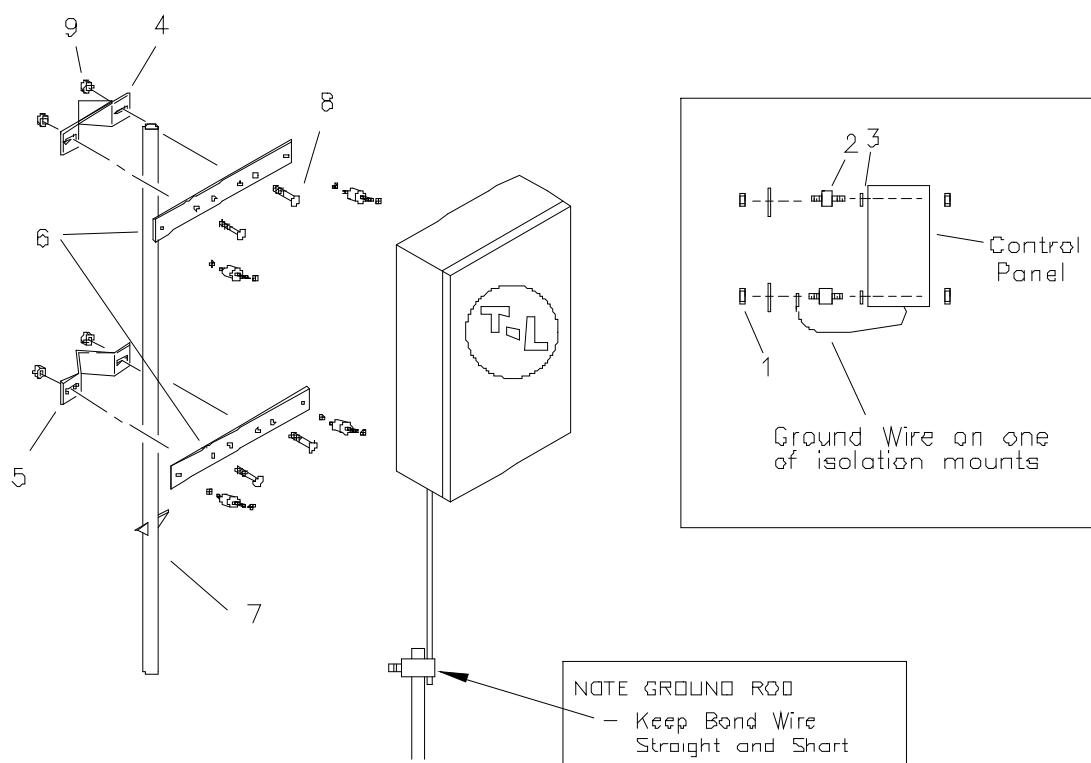
Ref.	Part No.	Description	# Req'd.
1	AP27023	Mount Plate for Precision Point Control III Panel	1
2	AB1663	Panel Upper Mount Strap	2
3	AB1690	Panel Lower Mount Strap	2
4	FN5401	Nut Hex 3/8 - 16	4
5	FB5439	HHCS 3/8 x 1	4
In MB52016			
6	UD57701	Rubber Mount 5/16 Studs	4
7	FN5400	Hex Nut 5/16	8
8	FW5650	Lock Washer 5/16	8
9	FW56513	Star Washer 5/16"	2

Figure 5. The Precision Point Control III Panel Installation – Quick Tow Pivot Point

D. Post Mount MD4105 (Figure 6)

Utilizing Post Mount kit MD4105 you can remote mount the control panel.

1. Set Panel Post into the ground.
2. Attach the Control Panel to the Panel Mount Straps using four Rubber Radiator Mounts
3. Attach the Mount Straps to the Pole using the Top and Flat bracket and ½” bolts and nuts.



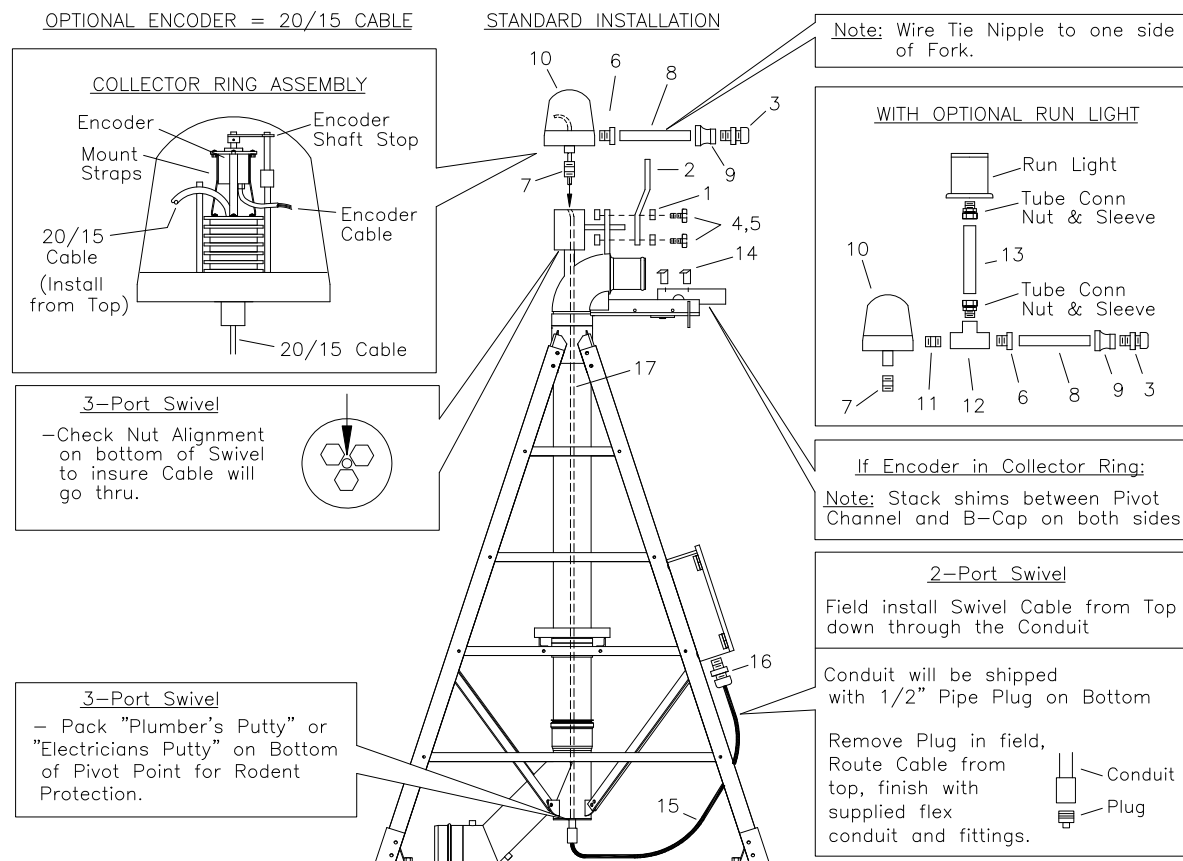
<u>Ref.</u>	<u>Part No.</u>	<u>Description</u>	<u># Req'd.</u>
In MB52016			
1	FN5400	Hex Nut 5/16	8
2	UD57701	Rubber Mount 5/16 Studs	4
3	FW5650	Lock Washer 5/16	8
MD4105	TLC PPC II Panel Post Mount		
4	AW25083	Top Bracket	1
5	AP25082	Clamp Bracket	1
6	AP20724	Panel Mount Strap	2
7	AW25080	Post	1
8	FB5449	HHCS, ½” x 2”	4
9	FN5404	Nut, ½”-13	4

Figure 6. The Precision Point Control III Panel Installation – Post Mount

Collector Ring Mounting with Swivel Pivot Point MB45214 (Figure 7)

Route Swivel Cable through the Swivel Pivot Point and Hydraulic Swivel. (Use fish tape if needed.)

- Swivel Cable: either combo/11 or 20/15, see below.
- see notes in figure below for 2-Port or 3-Port Hydraulic Swivel.



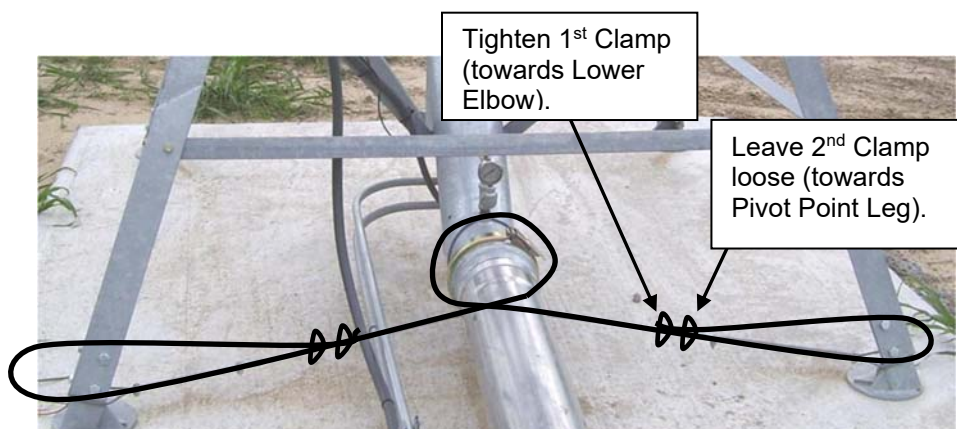
Ref.	Part No.	Description	# Req'd.
1	AP3227	Bolting Bar Collector Ring Mounting	4
2	AP32581	Collector Ring Stop Bar	1
3	EH5242	Strain Relief Connector	1
4	FB5449	HHCS 1/2 x 2	2
5	FN5404	Hex Nut 1/2	2
6	PB8003	Bushing 1 x 3/4 Galv	1
7	PN8213	Nipple 1 x 2 BEH	1
8	PN8612	Nipple 3/4 x 8 Galv	1
9	PR79650	Bell Reducer 3/4 x 1/2 Galv	1
10	EC53232	Collector Ring 11-Ring (6/11 -)	1
	EA50809	Collector Ring Assy/With Optional Encoder	1
11	PN8246	Nipple 1 x CI Galv (in Run Light "LT" Box)	1
12	PT8166	Tee 1" Galv (in Run Light "LT" Box)	1
13	TY10181	Tube 1" x 12" (in Run Light "LT" Box)	1
14	AB3232	Pivot Point Shim (shipped loose)	8
15	EH53915	Conduit 1/2" Non-Metalic Flex	6 ft
16	EH53916	1/2" Strain Liquid Tight Fitting	2
17	EW53758	combo/11 = without Position Sensor or with GPS for Pos. Sensor	25 ft
	EW53756	20/15 = with Encoder for Position Sensor	25 ft

Figure 7. Collector Ring Installation

Swivel Pivot Point with Encoder Position Sensor – Add Cable to Lower Elbow

Install 1/4" Cable on Lower Elbow to stabilize Swivel Pivot Point and ensure that Encoder has maximum movement.

- Step 1. Wrap Cable around Lower Elbow as shown.
- Step 2. Slide 2 Cable Clamps on one end of Cable.
- Step 3. Wrap Cable around Pivot Point Leg and insert end through both Clamps. Tighten Clamp closest to Lower Elbow.
Leave other clamp towards Pivot Point Leg loose.
- Step 4. Repeat with other end of cable.
- Step 5. Pull Cable tight by hand and tighten Clamp closest to Lower Elbow.



- Step 6. Slide the loose clamps towards Tower Legs and tighten, should make cable tight.



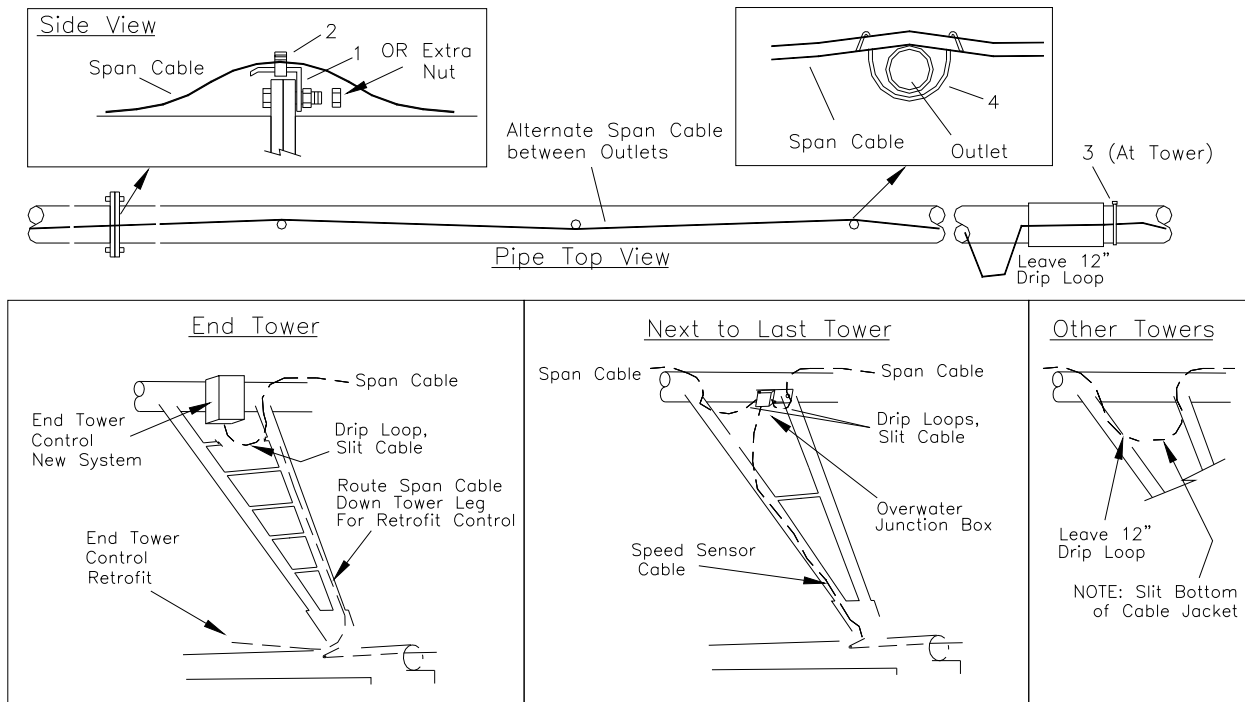
<u>Ref.</u>	<u>Part No.</u>	<u>Description</u>	<u># Req'd.</u>
	FD2082	1/4" Cable 14 ft Long	1
	FC4911	1/4" Cable Clamp	4

Figure 7.1. Add Cable to Stabilize Lower Elbow – Encoder Pivot Point.

Span Cable Installation (Figure 8)

The Pivot Span Cable is attached to the top of the pivot spans, from the Pivot Point to the End Tower Control Panel, with a junction at the Next to Last Tower (see Figure 8). The span cable is easiest to install after the main pivot system is spanned but not yet stood.

1. Start at the pivot point, allow enough span cable at pivot point as follows:
 - Swivel Pivot Point with Collector Ring allow approximately 18"
 - Hose Wrap Pivot Point allow approximately 20 ft. (clip to length later)
1. Route Span Cable alternately between Sprinkler Outlets. At each Sprinkler Outlet (or every 9' 10" for multi-outlet pipe), secure span cable to Outlet with Span Cable Clip (4).
2. At each Span Pipe Flange, attach a Cable Holder (1). Use existing nut (or additional nut, not provided). Wire tie the Span Cable to the Cable Holder with the wire ties provided.
3. At each Tower, attach Span Cable with two 28" Wire Tie (3).
 - allow a 12" Drip Loop at each Tower
5. At the Next to Last Tower, allow enough cable to reach Overwater Junction Box. (Overwater Junction Box should be positioned so Speed Sensor Cable will reach it.)
6. At the end tower, route the cable as shown, depending on new installation or retrofit.



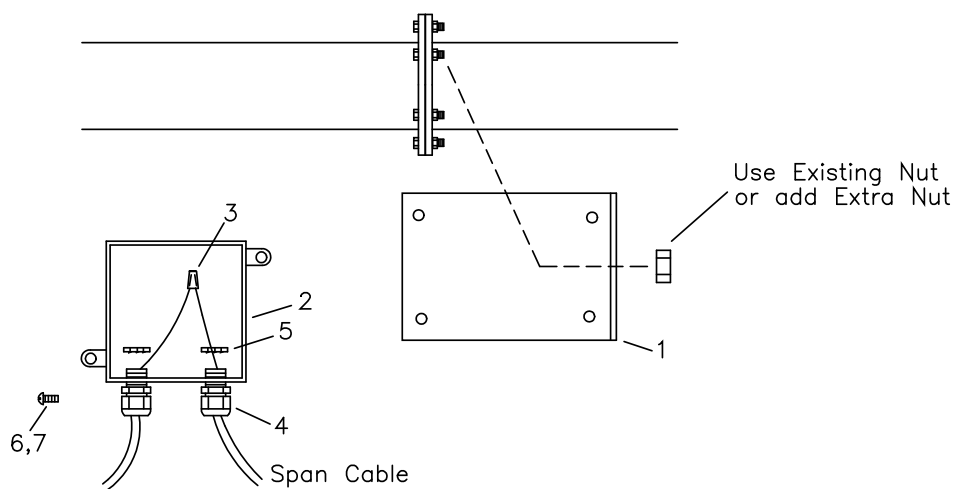
Ref.	Part No.	Description	# Req'd.
1	AB1572	Span Cable Holder	1 per flange
2	FT5702	Wire Tie	1 per flange
3	FT5705	Wire Tie 32"	1 per Tower
4	FT57058	Span Cable Outlet Clip	1 per 9'10" outlet

Figure 8. Span Cable Installation

Span Cable Splice Kit MB45956 (Figure 9)

On longer systems, it may be necessary to splice two lengths of span cable together. A span cable splice kit is used for this as shown in Figure 9.

1. Attach first span cable from the pivot point out as described in the previous section.
2. At the end of the first span cable, locate nearest flange back from the cable end at which to splice the cable. Cut first cable allowing at least 6 to 8 inches extra for splicing.
3. Splice second span cable into junction box and continue attaching second span cable from that point to the end tower.



Ref.	Part No.	Description	# Req'd.
1	AB15728	Point Cable Splice Mount Bracket	1
2	EH50808	Point Cable Splice Junction Box	1
3	EH52350	Wire Connector Orange	8
4	EH5242	1/2" Strain Relief Connector	2
5	EH5248	1/2" Sealing Lock Nut	2
6	FN5424	Nut 10-24	2
7	FS5481	SRHMS 10-24 x 1/2	2

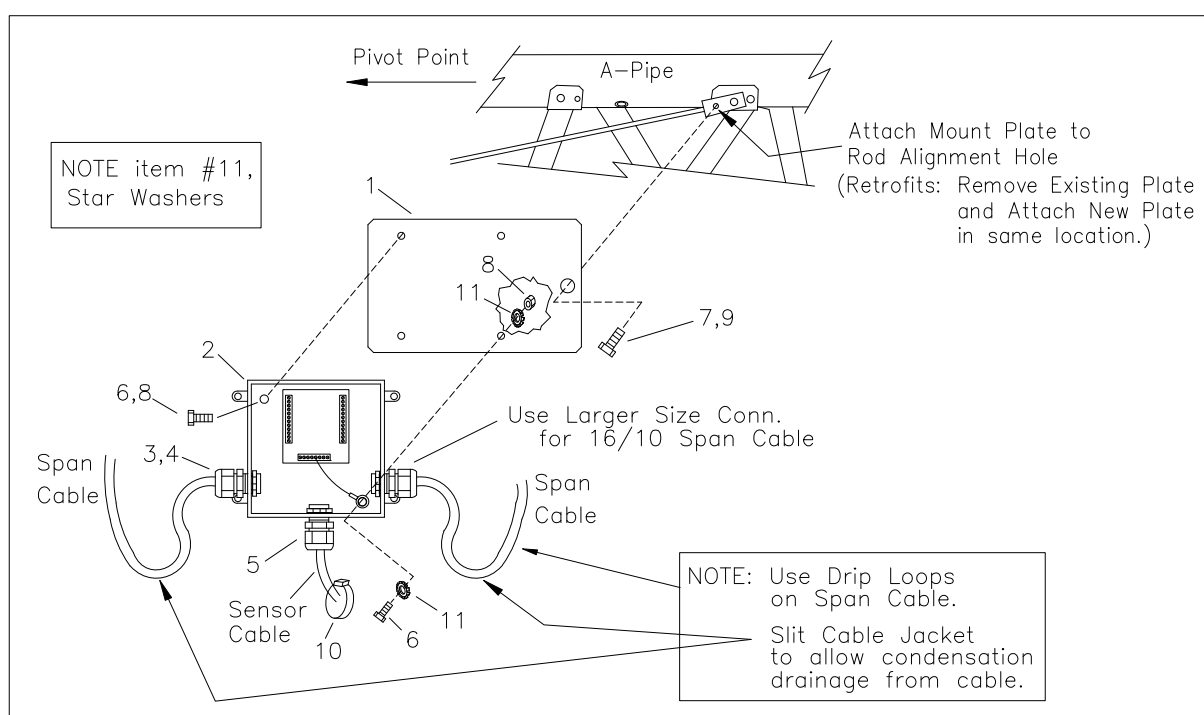
Figure 9. Span Cable Splice

Overwater Junction Box MB74662 (Figure 10) (If No Position Sensor)

With the *Precision Point Control III*, over watering is detected by an additional Speed Sensor Motor installed at the Next to Last Tower unless GPS or Encoder positioning is installed. Install the Overwater Junction Box as shown in Figure 10. As of January 22, 2015, the Overwater Junction Box is 8 X 8 X 4 and utilizes the same PCB (Printed Circuit Board) as the End Tower Manifold.

June 1, 2004 – January 22, 2015, the Overwater Junction Box was 6 X 6 X 4.

Prior to June 1, 2004, MB74661 was used with a 4 X 4 X 2 Junction Box and Terminal Strip.



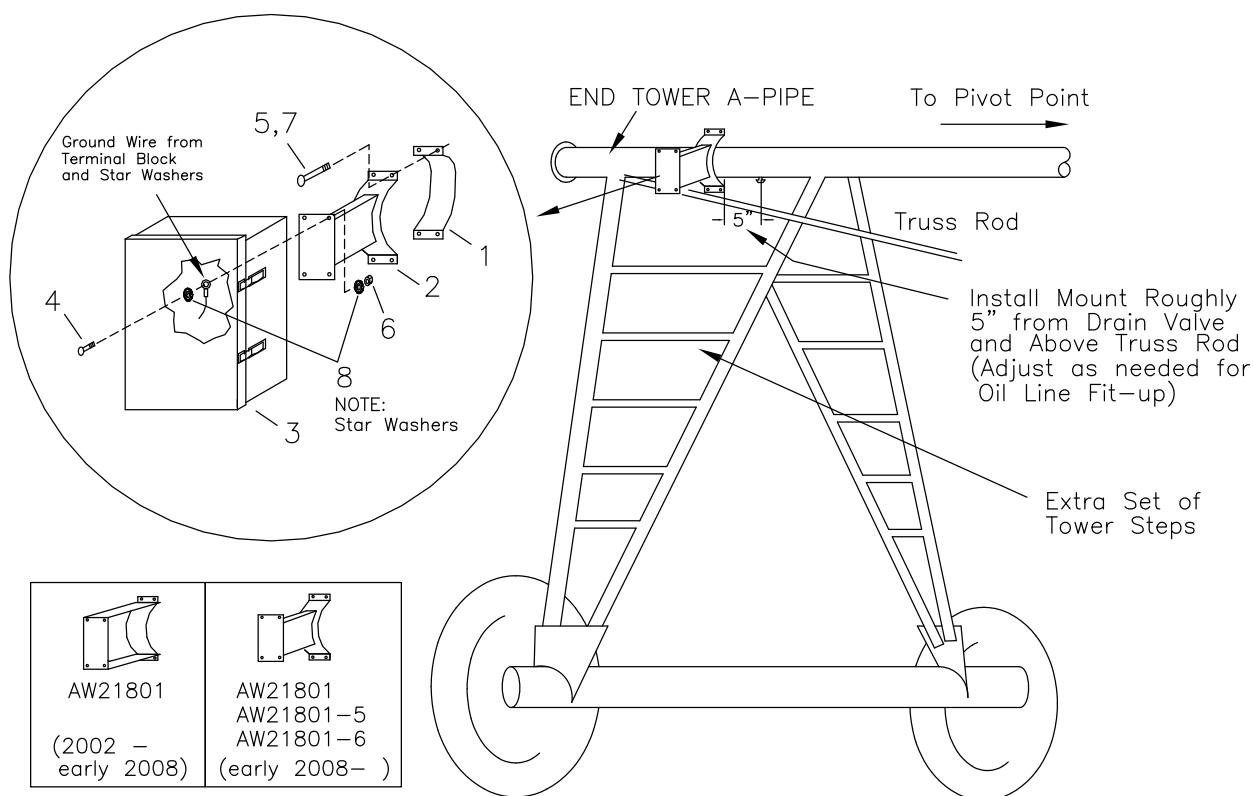
Ref.	Part No.	Description	# Req'd.
1	AB15730	Overwater 6" J-Box Mount Plate	1
2	EA50813	Overwater Sensor PCB J-Box Assy	1
3	EH52394	1/2" Conduit Locknut	2
4	EH5242	1/2" Cord Conn. .170-.470	2
5	EH5244	1/2" Cord Conn. .100-.300	1
6	FB4808	HHCS 1/4 X 5/8"	2
7	FB5446	HHCS 1/2 X 1-1/4"	1
8	FN5399	Hex Nut 1/4-20	2
9	FN5404	Hex Nut 1/2-13	1
10	FT5702	Wire Tie	8
11	FW56512	Star Washer 1/4	2

Figure 10. Overwater Junction Box

Attaching Manifold Panel To A-Pipe MB45246 (Figure 11)

The End Tower Controls are contained in an enclosure that mounts to the End Tower A-Pipe as shown.

1. Attach Manifold Panel Assy (3) to Manifold Mount Weldment (2) using four 1/4" x 5/8" Bolts, Nuts (4,6). Be sure to attach frame Ground Wire to one of the 4 bolts.
2. Attach Manifold Mount Weldment (2) to left side of End A-Pipe with Half-Band (1) using four 3/8" x 2" Bolts, Nuts (5,7).
3. Leave Half-Band loose for final positioning during end tower plumbing installation. Note rough dimension shown.



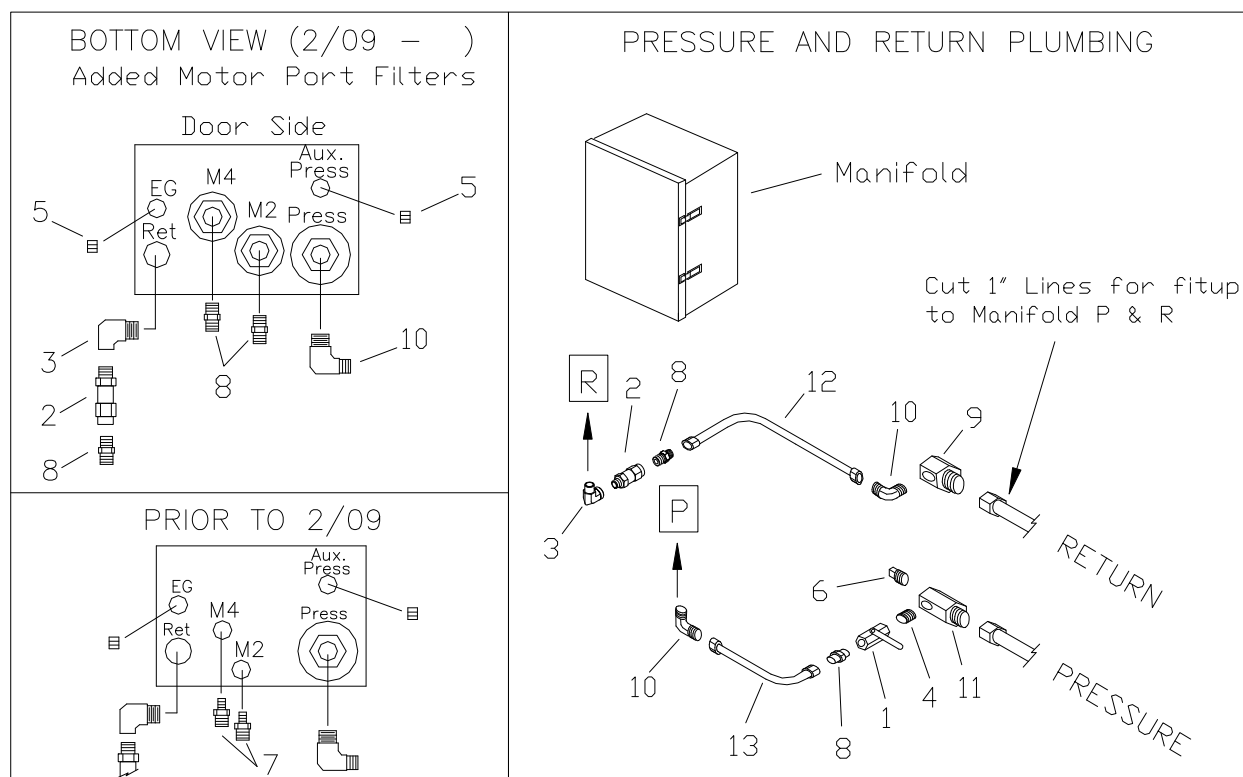
Ref.	Part No.	Description	# Req'd.
1	AP641	Half Band 6-5/8"	1
2	AW21801	Point Manifold Mount Weldment (Design change early 2008)	1
	AW21801-5	Manifold Mnt Weldment for 5" Galv Pipe (2008 -)	
	AW21801-6	Manifold Mnt Weldment for 6" Alum Pipe (2008 -)	
3	EA53247	Point Manifold Panel Assy	1
4	FB4808	HHCS 1/4" x 5/8"	4
5	FB5444	HHCS 3/8" x 2"	4
6	FN5399	Hex Nut 1/4"	4
7	FN5401	Hex Nut 3/8"	4
8	FW56512	Star Washer 1/4"	2

Figure 11. Attachment of Manifold Panel Assembly to A-Pipe

Pressure and Return Oil Line Plumbing (Figure 12)

NOTE: If Lift Jacks are used, see Plumbing diagram Figure 16 also.

1. Remove plastic plugs from bottom of Manifold. Install fittings into bottom of Manifold as shown. Do not install EG Plug if End Gun Control to be used (non-booster).
2. Dry fit parts together to determine distance to cut 1" System Pressure/Return Lines. Cut back 1" lines and install Pressure and Return fittings to Manifold Pressure and Return Ports.

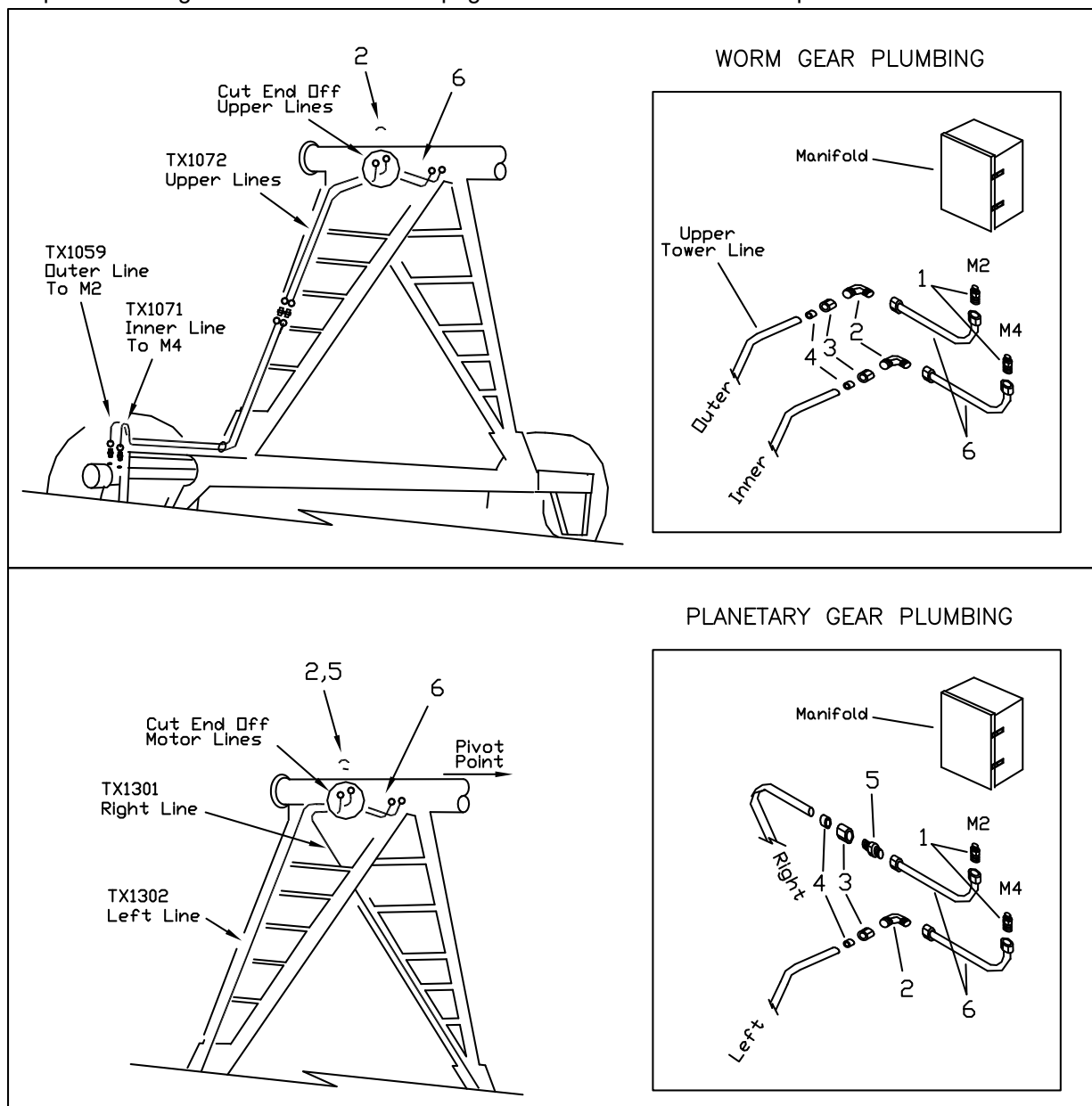


Ref.	Part No.	Description	# Req'd.
1	HV9707	Ball Valve 1/2" Steel	1
2	HV97381	Check Valve 5 psi, 1/2"MPT X 1/2"FPT (6/04 -)	1
3	PE81255	Street Elbow 1/2" 90 Deg Plated (6/04 -)	1
	TE9629	(factory installed 10/07 -)	
	TE9629	Elbow 1/2"MPT x 1/2"Erm (1/02 – 5/04)	1
4	PN8232	Nipple 1/2" x 1 1/2" BEH	1
5	PP8185	Pipe Plug 1/4" Galv.	2
6	PP8187	Pipe Plug 3/4" Galv.	AR
7	TC9613	Conn 1/4"MPT x 1/2"Erm (not used after Feb. '09)	2
8	TC9614	Conn 1/2"MPT x 1/2"Erm (was qty 2 before Feb. '09)	4
9	TE9456	Elbow 1"Erm x 1/2"FPT	1
10	TE9629	Elbow 1/2"MPT x 1/2"Erm	2
11	TT94567	Tee 1"Erm x 3/4" FPT x 1/2"FPT	1
12	TX22419	Manifold Return Line 10" x 14" (6/04 -)	1
	TX22412	Manifold Return Line 14" X 14" (1/02 – 5/04)	1
13	TX525	Manifold Pressure 1/2" Line 4" x 8-5/8"	1

Figure 12. Pressure and Return Oil Line Plumbing (eff 2/09).

Motor Oil Line Plumbing (Figures 13, 14, 15)

See pictures in Figures 14 and 15 on next page for motor line installation help.



Ref.	Part No.	Description	# Req'd.
1	TC9613	Conn 1/4" MPT x 1/2" Erm (prior to Feb. '09)	2
	TC9614	Conn 1/2" MPT X 1/2" Erm (after Feb. '09)	2
2	TE9631	Union Elbow 1/2" Erm	2
3	TN9604	Tube Nut 1/2" Erm	2
4	TS9608	Tube Sleeve 1/2" Erm	2
5	TU9622	Union Conn. 1/2" Erm	1
6	TX22413	Manifold Motor Lines 5" x 15-1/2"	2

Figure 13. Motor Line Connections

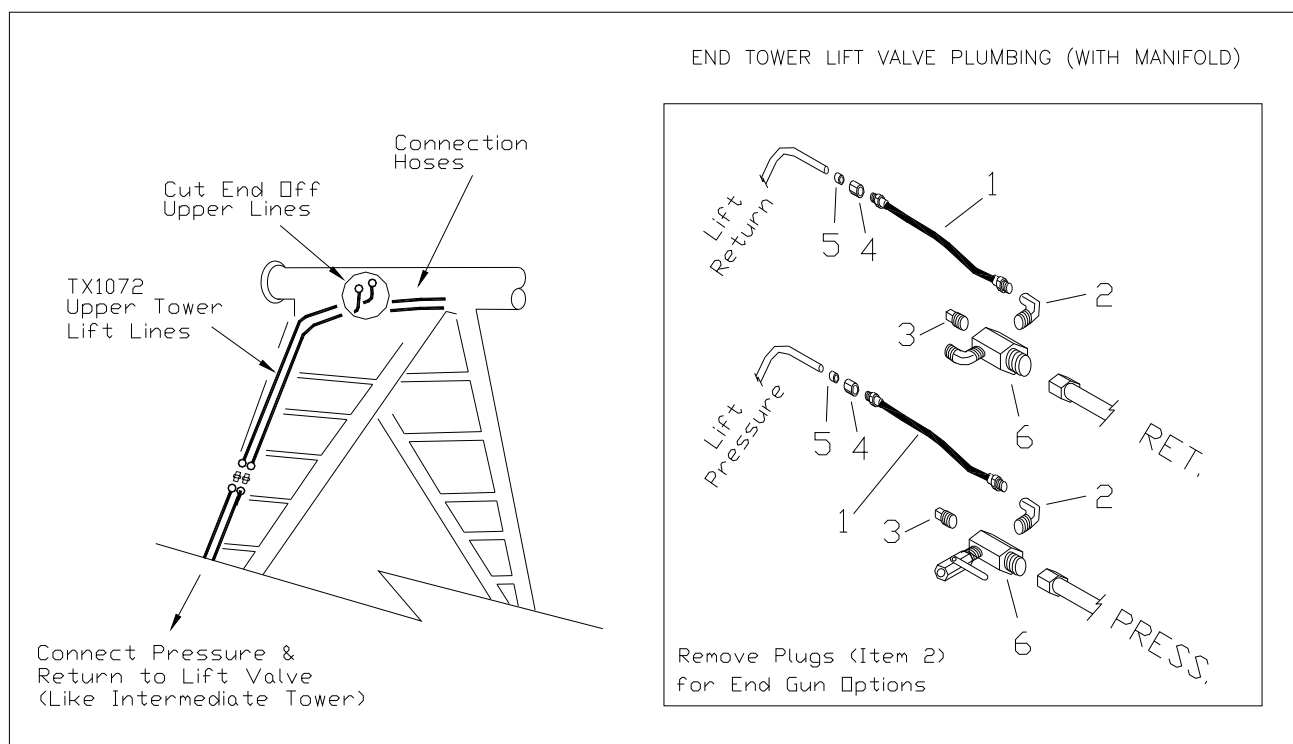


Figure 14. Worm Gear with End Gun Control



Figure 15. Planetary Gear with End Gun Control

Optional End Tower Lift Valve Plumbing (Figure 16)

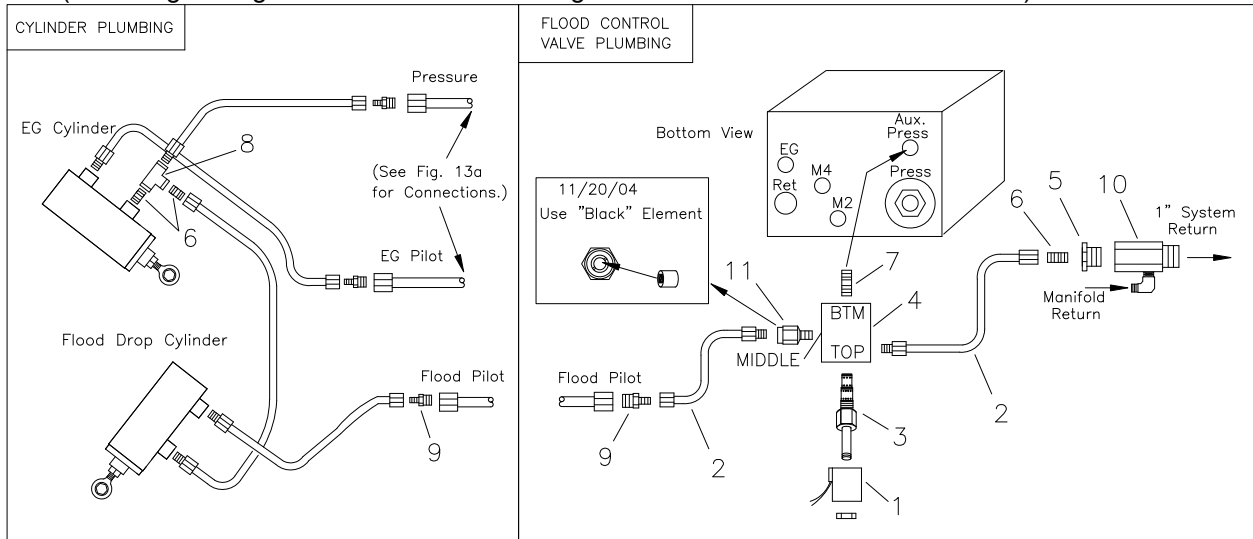


Ref.	Part No.	Description	# Req'd.
1	HH61731	Towable Hose 3/8"-1/2"MP-1/2"Erm-22"	2
2	PE81255	Street Elbow 1/2"	2
3	PP8187	Pipe Plug 3/4"	2
4	TN9604	Tube Nut 1/2" Erm	2
5	TS9608	Tube Sleeve 1/2" Erm	2
6	TT94568	Cross 1"Erm-3/4"FP-1/2"FP-1/2"FP	2

Figure 16. Plumbing for Lift Valve

Manifold Plumbing with Two End Gun Controls (MB45223) (Figure 17)

(Plumbing Change 6/04. Previous Plumbing used a Tee on Manifold Return Port.)



Ref.	Part No.	Description	# Req'd.
1	EC50539	Coil Gold 24 VDC 14 watt (in MB45247 Box)	1
2	HH61205	Hose 1/4" FP xMPX – 30" (in MB45247 Box)	2
3	HV98043	Solenoid Vlv 3-way Cartridge (in MB45247 Box)	1
4	HV98388	Aluminum Body for 3-Way Valve	1
5	PB80412	Bushing 3/4 x 1/4 Galv	1
6	PN8201	Nipple 1/4 x close Heavy	3
7	PN8233	Nipple 1/4 x 1-1/2 Heavy	1
8	PT8164	Tee 1/4" Heavy	1
9	TC9613	Conn 1/4 MPT x 1/2 Erm (in MB45247 Box)	2
10	TT94567	Plated Tee 1"Ern-3/4"FP-1/2"FP	1
11	HF57602	Brass Filter, 1/4" Male X Female (in MB45247 11/04)	1

Figure 17. Plumbing for 2 End Gun Controls, Flood Control Valve

Manifold Plumbing with Corner System (Figure 18)

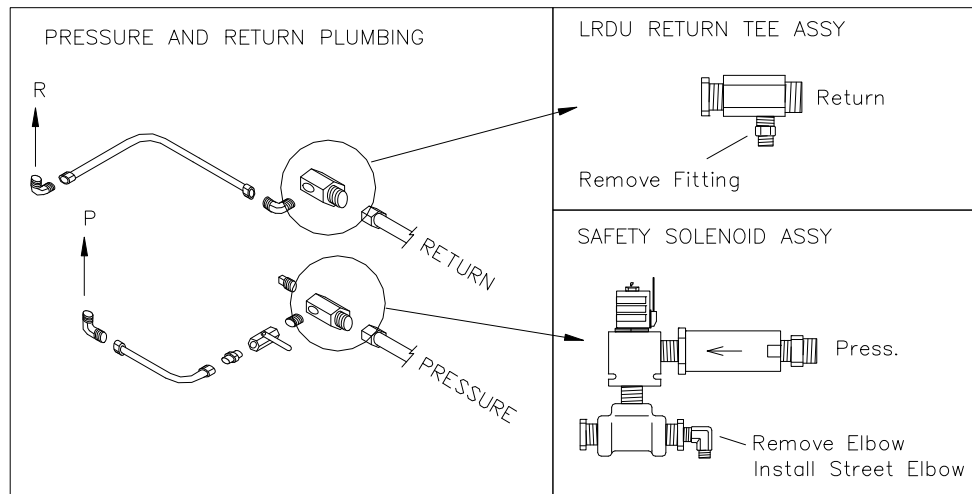


Figure 18. Plumbing for Manifold with Corner System

Retrofit Installation – End Tower Control Manifold (Figure 19, 20, 21)

When installing the End Tower Control on existing systems, the End Tower Control Manifold can be installed on the Tower Base for ease of plumbing connections.

The End Tower Control Manifold can:

- Replace the existing Manual Control (Figure 19) using MB46701 Retrofit ET Kit
- Be added with the existing Manual Control using a Parallel Plumbing Kit. (Figure 20)
- MB45225 Parallel Plumbing Kit with MB46701 Retrofit ET Kit

To Install End Tower Control Manifold:

1. Drill Holes in End Tower Control Mount. (Lay Manifold Panel on Plate to mark holes.)
2. Attach End Tower Control Mount to Tower Base using U-Bolts 3/8" with nuts.
3. Attach Manifold to Control Mount using 1/4" x 5/8" Bolts with nuts. Be sure to attach ground wire from terminal block to one of the bolts.
4. Adjust into position before tightening 3/8" U-bolts.

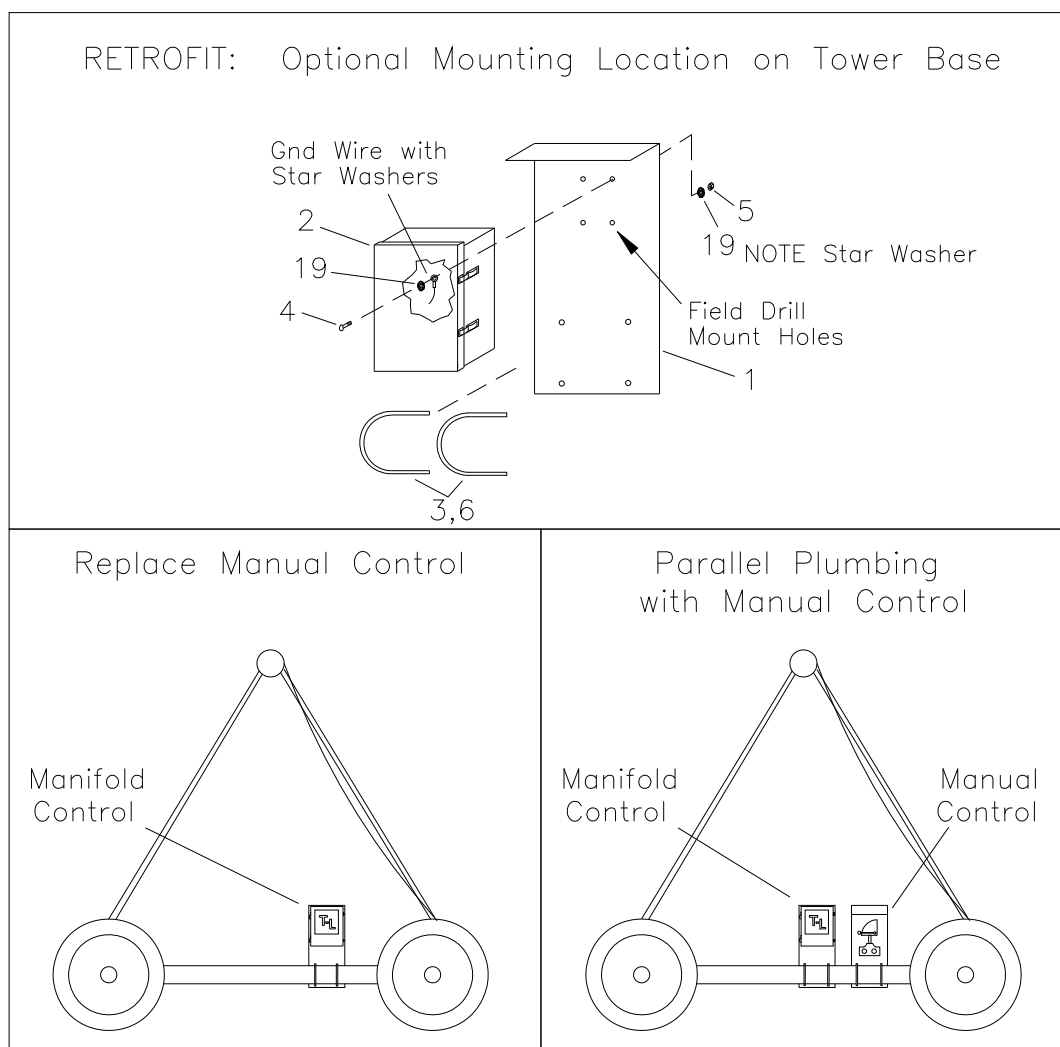
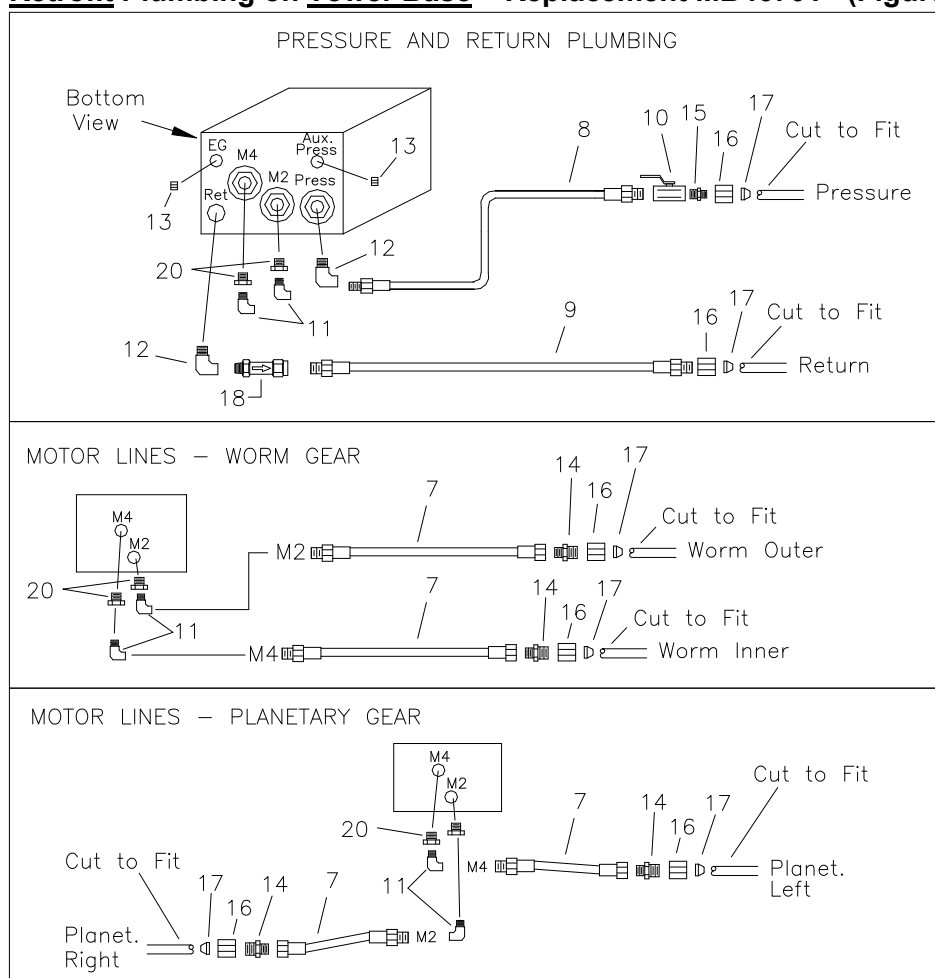


Figure 19. End Tower Control Retrofit Installation on Existing Systems

End Tower Retrofit Plumbing on Tower Base – Replacement MB46701 (Figure 19 & 20)



Ref.	Part No.	Description	# Req'd.
1	AP3035	End Tower Mount Plate	1
2	EA53247	Point Manifold Panel Assy	1
3	FB383	End Tower U-Bolt 3/8"	2
4	FB4808	HHCS 1/4" x 5/8"	4
5	FN5399	Hex Nut 1/4"	4
6	FN5401	Hex Nut 3/8"	4
7	HH61203	Hose 1/4 FP x 1/4 MPX – 24"	2
8	HH61264	Hose 1/2 MP x 1/2 MPX – 24"	1
9	HH61731	Towable Hose 3/8-1/2MP-1/2Erm-22"	1
10	HV9707	Ball Valve 1/2"	1
11	PE8111	Street Elbow 1/4" x 90 Heavy Plated	2
12	PE81255	Street Elbow, 1/2" x 90 Heavy Plated	2
13	PP8185	Pipe Plug 1/4" MP	2
14	TC9613	Connector 1/4"MPT x 1/2"Erm	2
15	TC9614	Connector 1/2"MPT x 1/2"Erm	1
16	TN9604	Tube Nut 1/2"Erm	4
17	TS9608	Tube Sleeve 1/2"Erm	4
18	HV97381	Check Valve 5 psi, 1/2"MPT X 1/2"FPT (6/04 -)	1
19	FW56512	Star Washer 1/4"	2
20	PB7997	Bushing 1/2" X 1/4" (2/09 -)	2

Figure 20. Manifold Retrofit Installation Replacing Existing Manual Control.



End Tower Retrofit Parallel Plumbing on Tower Base MB45225 (Figure 19 & 21)

A retrofit plumbing kit can be installed in cases where the Point Control is being added to an existing system. The End Tower Manifold is located on the Tower Base next to the existing manual control.

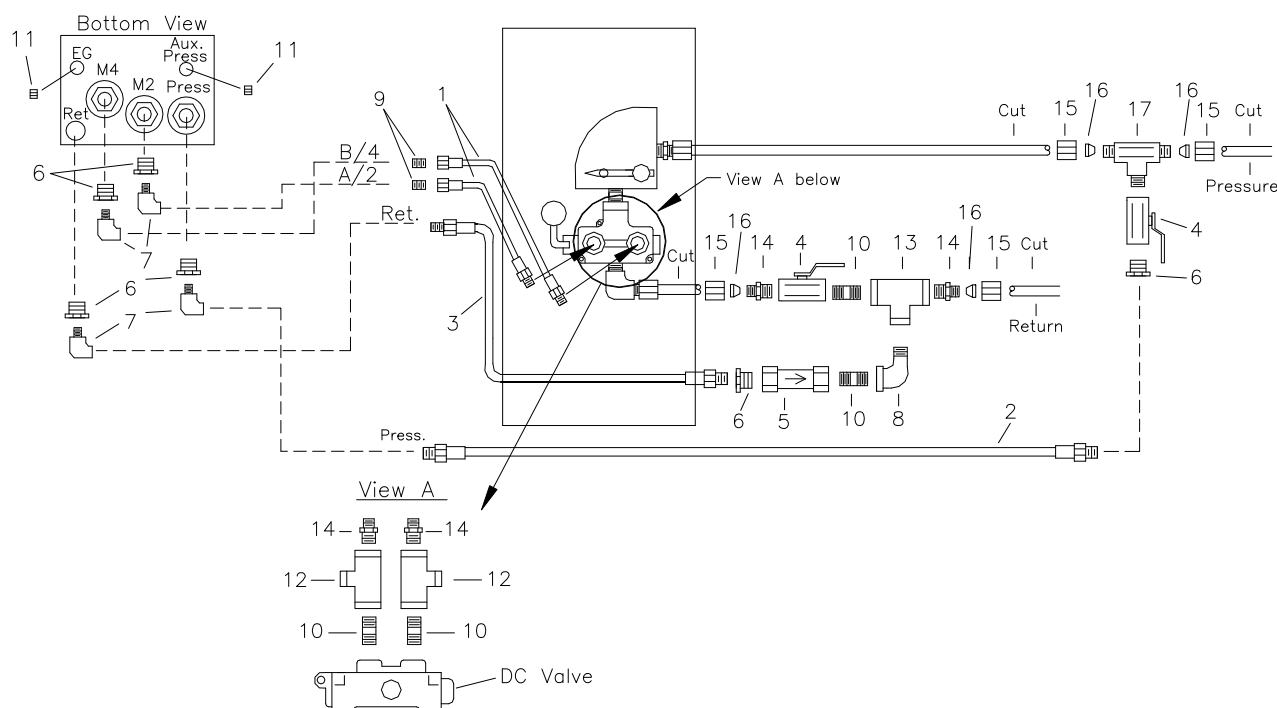
Pressure Ball Valve
Return Ball Valve
End Tower FC
End Tower DC

Pivot Point Control

Open
Closed
Set to 0
Neutral

End Tower Control

Closed
Open
Set from 0 to 10 per speed desired
Fwd or Rev per direction desired
Be sure Point Control Switch is in STOP



Ref.	Part No.	Description	# Req'd.
1	HH61203	Hose ¼ FP x ¼ MPX – 24"	2 (in MB46701)
2	HH6151	Hose ¼MP x ¼MPX – 36"	1
3	HH6154	Hose ¼MP x ¼MPX – 28"	1
4	HV9707	Ball Valve ½"	2 (1 in MB46701)
5	HV9738	Check Valve C30S	1
6	PB7997	Bushing ½ x ¼	6 (2 added 2/09)
7	PE8111	Street Elbow ¼" x 90 BEH	4 (2 in MB46701)
8	PE81255	Street Elbow, ½" x 90 BEH	1 (in MB46701)
9	PN8201	Nipple ¼" x Close BEH	2
10	PN8232	Nipple ½ x 1½ BEH	4
11	PP8185	Pipe Plug ¼" MP	2 (in MB46701)
12	PT8150	Tee ½ x ½ x ¼ BEH	2
13	PT8163	Tee ½" BEH	1
14	TC9614	Connector ½"MPT x ½"Erm	4 (1 in MB46701)
15	TN9604	Tube Nut ½"Erm	4 (in MB46701)
16	TS9608	Tube Sleeve ½"Erm	4 (in MB46701)
17	TT9637	Male Tee ½"Erm x ½"Erm x ½"MPT	1

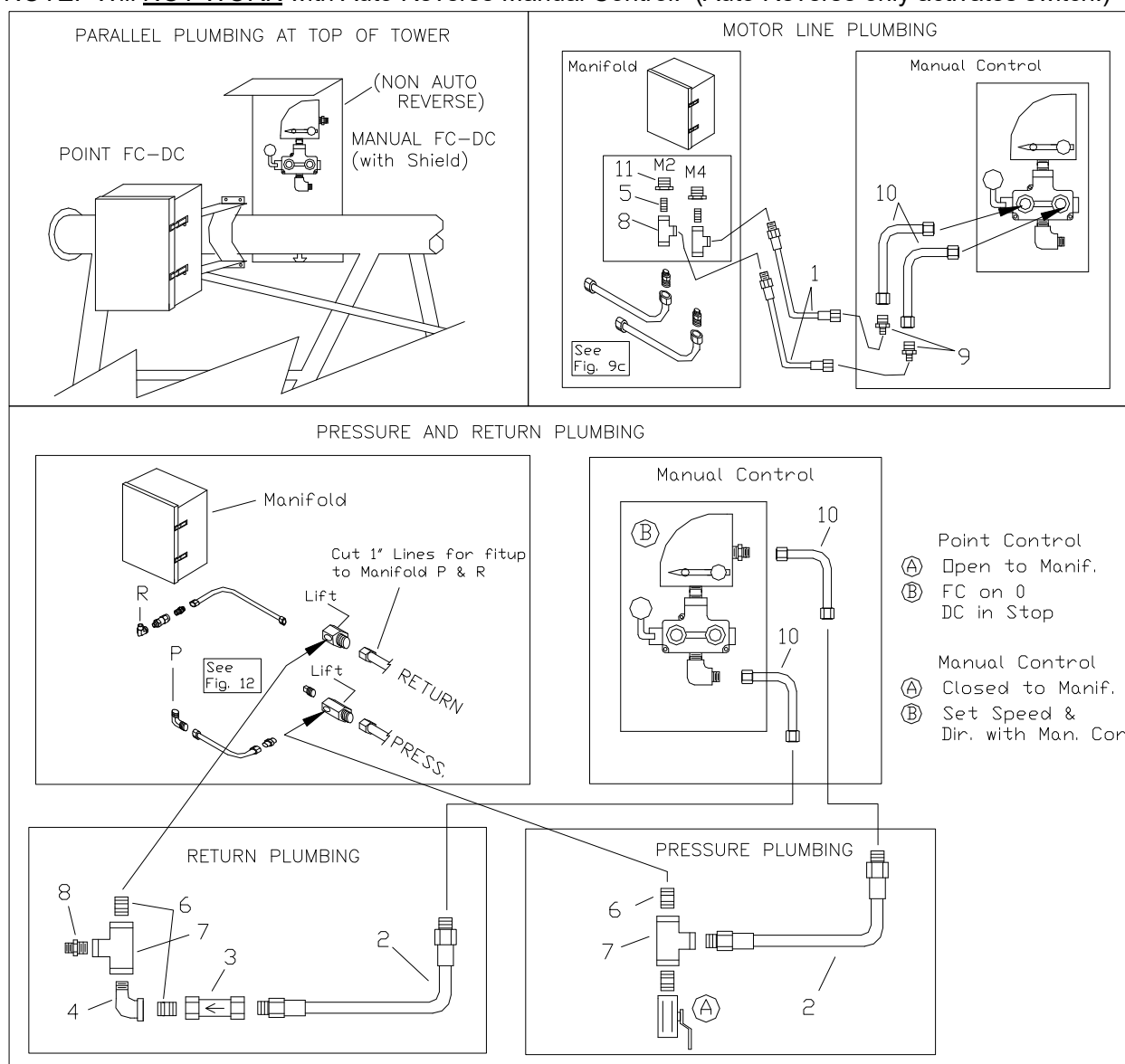
Figure 21. Retrofit End Tower Panel Plumbing on Tower Base

HARDWARE INSTALLATION



End Tower Parallel Plumbing on A-Pipe for New Systems MB45227 (Figure 22)

NOTE: Will NOT WORK with Auto Reverse Manual Control. (Auto Reverse only activates switch.)



Ref.	Part No.	Description	# Req'd.
1	HH61205	Hose 1/4 FP x 1/4 MPX - 30"	2
2	HH61731	3/8" Hose 1/2 ERM X 1/2 MPT - 22"	2
3	HV9738	Check Valve C30S	1
4	PE81255	Street Elbow, 1/2" x 90 BEH	1
5	PN8201	Nipple 1/4" x Close BEH	2
6	PN8204	Nipple 1/2" x Close BEH	3
7	PT8163	Tee 1/2" Plated	2
8	PT8164	Tee 1/4" Plated	2
9	TC9613	Connector 1/4"MPT x 1/2"Erm	2
10	TX1001	Base Line End Planetary	4
11	PB7997	Bushing 1/2" X 1/4" (2/09 -)	2

Figure 22. End Tower Parallel Plumbing on A-Pipe

Motors With Integral Speed Sensor (Figure 23)

Precision Point Control = 2 Speed Sensor Motors IF NO Position Sensor for Overwater Timer.

- 1 Motor at End Tower for travel speed
- 1 Motor at Next to Last Tower for Overwater Timer
- Planetary drives, see Figure below for Sensor Motor location (non-towable vs. towable)
- Install Sensor Shield on motor, route cable to box at top of tower. Attach cable with wire ties.
- Towable systems, route sensor cable along one of the hoses. Allow slack for wheel pivoting.

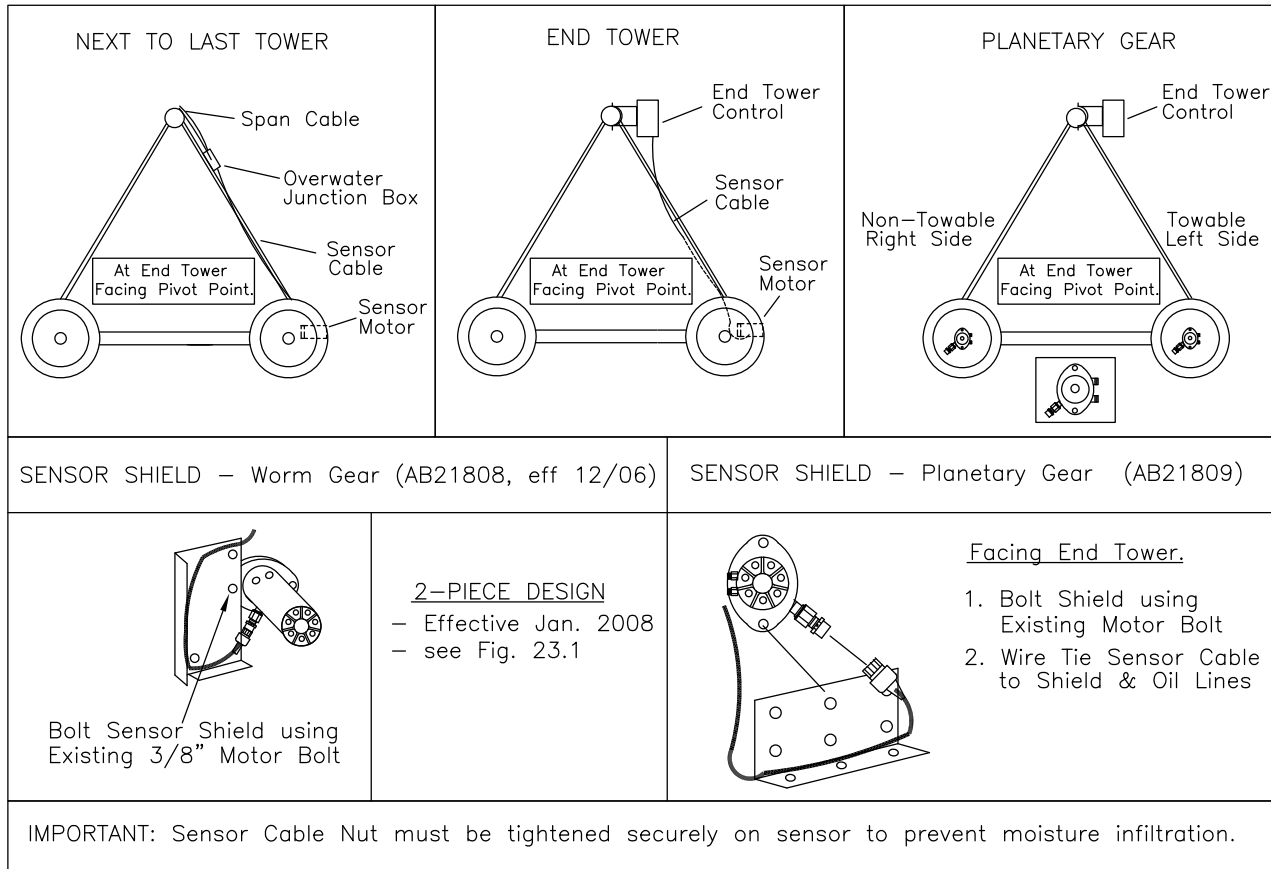


Figure 23. Motor with Speed Sensor Installation

Worm Gear Sensor Shield 2-Piece, Jan. 2008 - (Figure 23.1)

Ref.	Part No.	Description	Req'd.
1	AB21806	Sensor Shield Worm Cover Plate	1
2	AP21807	Sensor Shield Worm Motor Plate	1
3	FB5444	HHCS, 3/8" x 2" Grd. 5	1
4	FN5401	Hex, Nut 3/8" - 16	1
5	FT5706	Wire Tie 6" Long	2
6	FW5652	Lock Washer 3/8"	1
	CD91156	I.S. Worm 2-Piece Shield	1

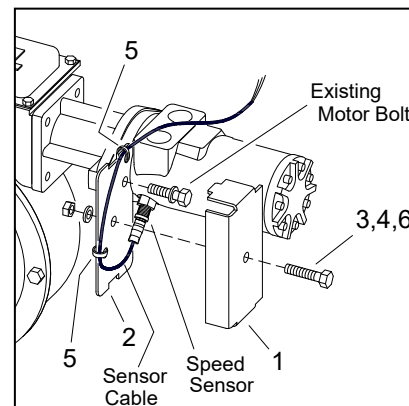


Figure 23.1 Worm Gear 2-Piece Sensor Shield

End Gun or End Gun Booster Control Installation - Pivot Point (Figures 24, 25, 26)

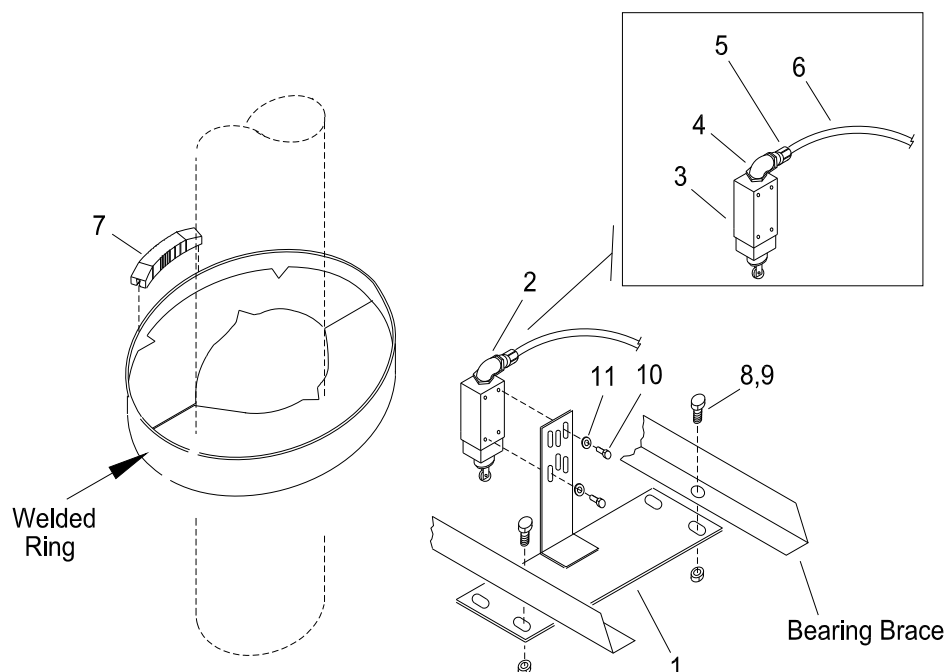
The Optional End Gun Control uses a 3-way electro-hydraulic valve to control pilot pressure to the standard Cylinder shutoff. An electric switch mounted at the pivot point controls the 3-way solenoid which is at the end tower. There are two different controls at the pivot point:

- A. Swivel or Hose Wrap Pivot Point Control
- B. Quick Tow Pivot Point Control

A. Swivel or Hose Wrap Pivot Point Control (MB52018)

1. Bolt Actuator Stand (1) to the Pivot Bearing Braces using $\frac{1}{2}$ " x $1\frac{1}{4}$ " Bolts W/Nuts (7,12).
2. Bolt Roller Switch Assy (2) to Actuator Stand (2) using 10-32 x $\frac{1}{2}$ " Screws, FW (14,15).
3. Position two Cam Ring Halves (1) around the Pivot Pipe. Bolt the rings together using four $\frac{3}{8}$ " x $1\frac{1}{2}$ " Bolts W/Nuts (6,11). Adjust the ring up to the End Gun Switch and level.
4. Refer to Cam Wafer Valve Installation sheet for installation of black plastic wafers. Position End Gun Cams & sections (8,9) on ring where the End Gun should be **OFF**.

Note: IF control is to be Off more than $\frac{1}{2}$ way, reverse the logic to minimize the number of black wafers installed. Rewire the switch to NO, 3 & 4. Place wafers where control is to be **ON**.



	Ref.	Part No.	Description	# Req'd.
	1	AW6041	EG-PS Actuator Mount	1
	2	EA53471	Top Roller Switch Assy with Elbow	1
In Assy	3	ES52545	Top Roller Switch LSD1A (SWITCH ONLY)	1
In Assy	4	EH42406	$\frac{1}{2}$ " Conduit Street Elbow	1
In Assy	5	EH5244	Strain Relief Conn (12/22/03)	1
In Assy	6	EW5356	Cable 14/2 TC (12/22/03)	10 ft
	7	FD60430	6" Molded End Gun Cam	5
	8	FB5446	HHCS $\frac{1}{2}$ x $1\frac{1}{4}$	2
	9	FN5404	Hex Nut $\frac{1}{2}$	2
	10	FS5537	SRHMS 10-32 x $\frac{1}{2}$	2
	11	FW5646	Washer 3/16 Flat	2

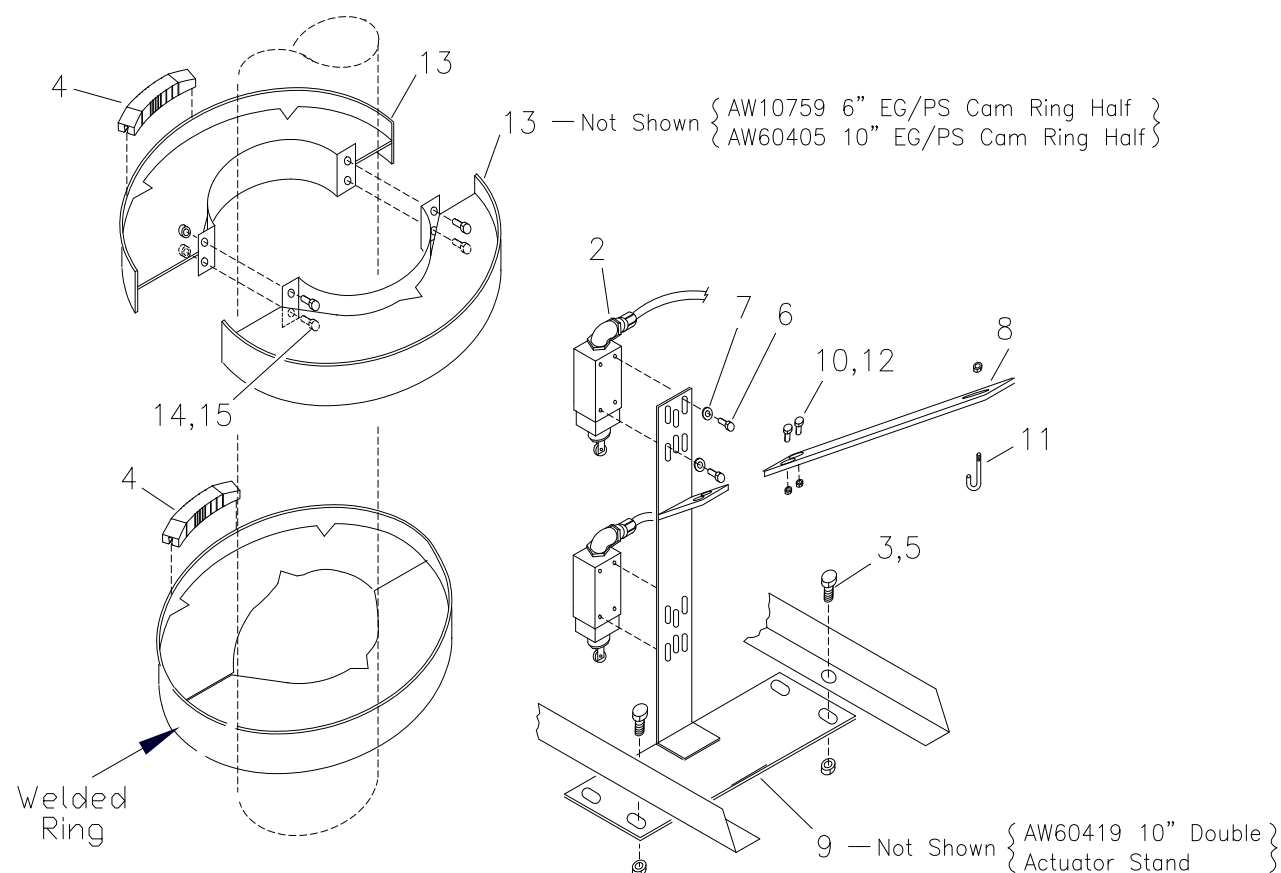
Figure 24. End Gun Control Switch – Swivel or Hose Wrap Pivot Point



Double Actuator Mount MB45224 – 8” (Figure 25)

Double Actuator Mount MB45236– 6” (Not Shown)

Double Actuator Mount MB45239– 10” (Not Shown)



Parts included in 2 of MB52018 EG Control Pivot Point Box

Ref.	Part No.	Description	# Req'd.
1	AW6041	EG-PS Actuator Mount (Not Shown)	0 (2 extra)
2	EA53471	Top Roller Switch Assy with Elbow	2
3	FB5446	HHCS ½ x 1¼	2 (2 extra)
4	FD60430	6” Molded End Gun Cam	10
5	FN5404	Hex Nut ½	2 (2 extra)
6	FS5537	SRHMS 10-32 x ½	4
7	FW5646	Washer 3/16 Flat	4

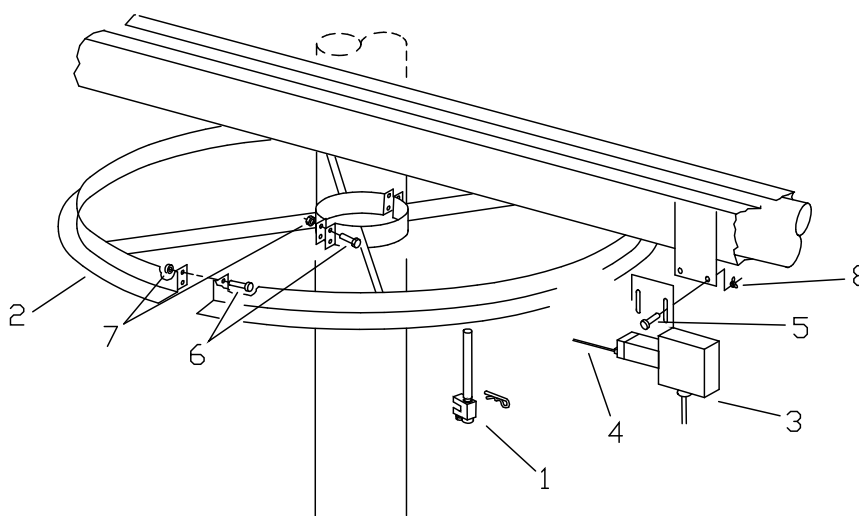
Parts included in MB45224 Double Actuator Mount Kit

8	AP60414	Double Actuator Strap	1
9	AW60410	8” Double Actuator Stand	1
10	FB5435	HHCS ¼ x 1”	2
11	FB54385	J-Bolt 5/16 x 3 with nut	1
12	FN5399	Hex Nut ¼”	2
MB48080 8” Pivot Point Control Ring Box			
13	AW6040	EG-PS Half Ring	2
14	FB54650	HHCS 5/16 x 1½	4
15	FN5400	Hex Nut 5/16 - 18	4

Figure 25. Double Actuator Installation

B. Quick Tow Pivot Point Control (MB55807)

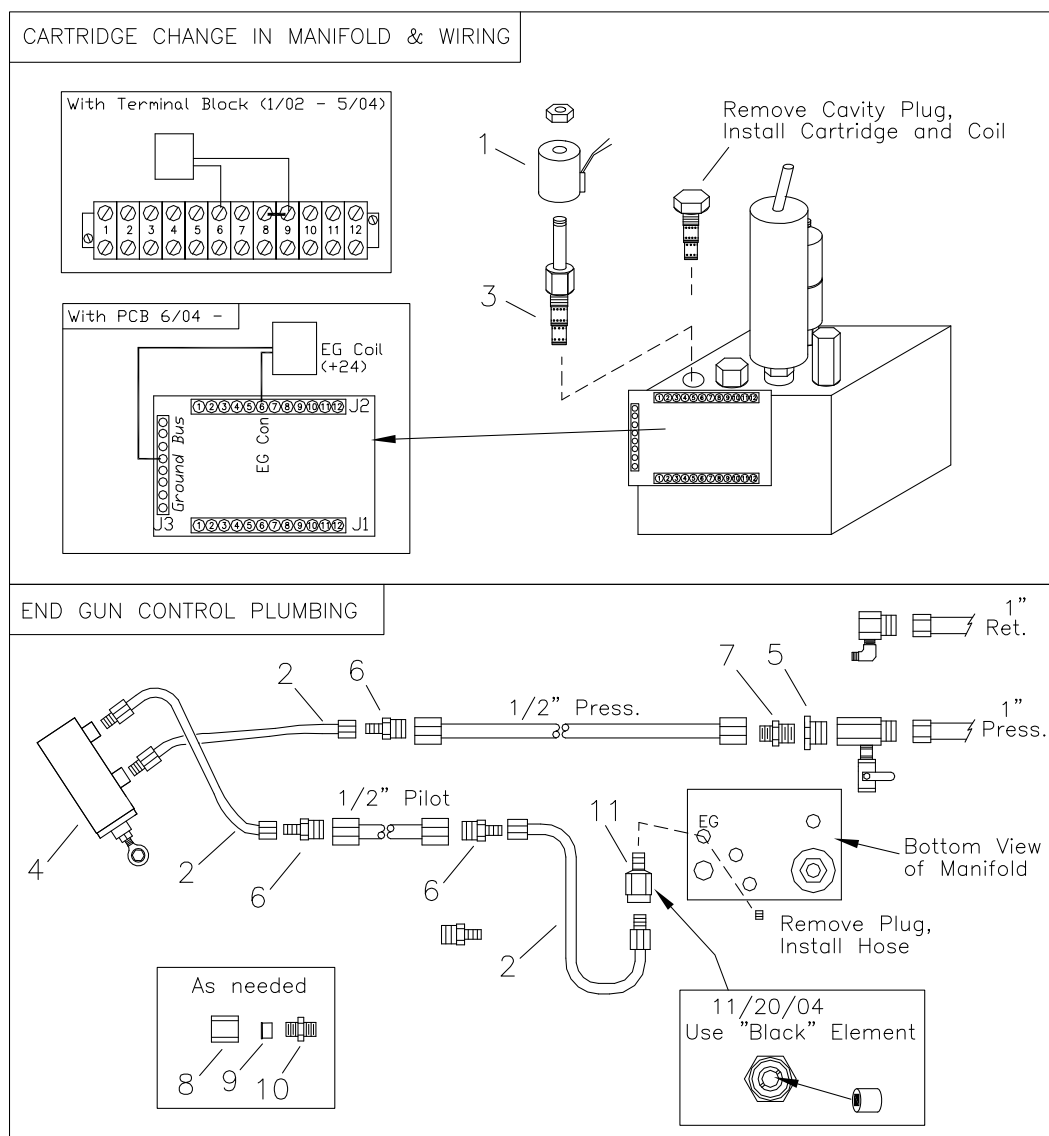
1. Install two PS-EG Control Rings (2) around the Telescope Pipe above the water inlet. Use eight 3/8 x 2" bolts w/nuts (6,7).
2. Attach Mount Plate-Micro Switch Assy (3) to the Tractor Base bracket using two carriage bolts 1/4 x 3/4 with wing nuts (5,8).
3. Attach Activation Lever (4) to Micro Switch. Adjust wand so Trigger Assy (1) will activate the Micro Switch.
4. Attach Position Stop Trigger Assemblies (1) to Control Rings (2) so Micro Switch will be activated each time End Gun is to turn on and each time End Gun is to turn off.



<u>Ref.</u>	<u>Part No.</u>	<u>Description</u>	<u># Req'd.</u>
1	AA2812	Position Stop Trigger Assy	8
2	AW2828	PS-EG Control Ring Half	2
3	EA53267	Point QT EG 24 VDC Latching Assy	1
4	EH5268	LSZ54N Activation Lever Rod SS	1
5	FB5432	Carriage Bolt 1/4 x 3/4	2
6	FB5444	HHCS 3/8 x 2	8
7	FN5401	Hex Nut 3/8	8
8	FN5411	Wing Nut 1/4	2

Figure 26. End Gun Control Switch – Quick Tow Pivot Point

End Gun Control Without Booster Installation – Last Tower MB45247 (Figure 27)



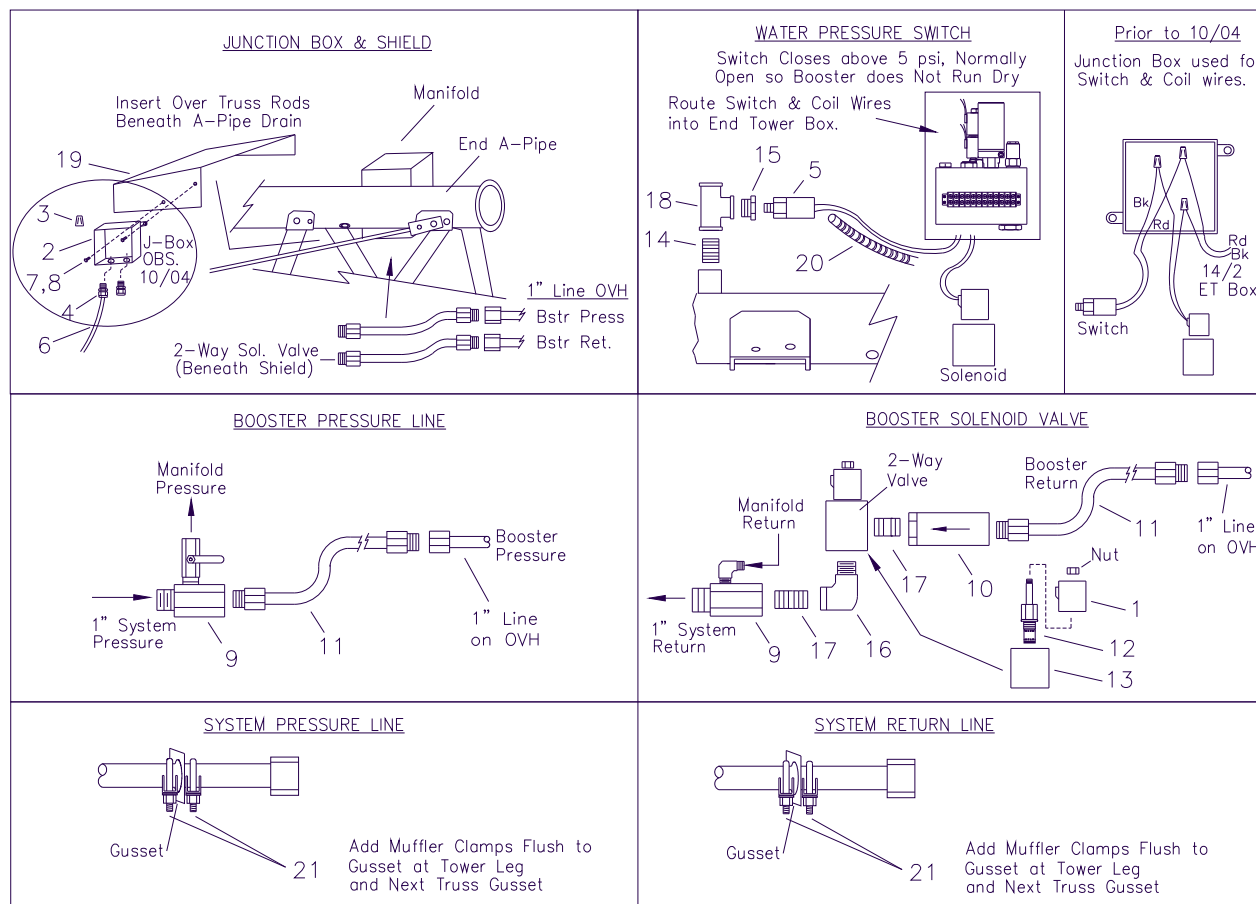
Ref.	Part No.	Description	# Req'd.
1	EC50539	Coil 24vdc 14watt Watertight (Non-Eff 4/08)	1
	EC98370	Lead Wires & EC98371 Coil (Eff 4/08)	
2	HH61205	Hose 4CIT-4FP-4MPX-30"	3
3	HV98043	3-Way Cartridge for EG Control	1
4	IV97080	2" Ball Valve W/Hyd Actuator	1
5	PB7998	Bushing 3/4" x 1/2"	1
6	TC9613	Conn 1/4"MPT x 1/2"Erm	3
7	TC9614	Conn 1/2"MPT x 1/2"Erm	1
8	TN9604	Tube Nut 1/2" Erm	2
9	TS9608	Tube Sleeve 1/2" Erm	4
10	TU9622	Union Conn. 1/2" Erm	2
11	HF57602	Brass Filter, 1/4" Male X Female (11/04 -)	1

Figure 27. Installation of End Gun Control without Booster

HARDWARE INSTALLATION



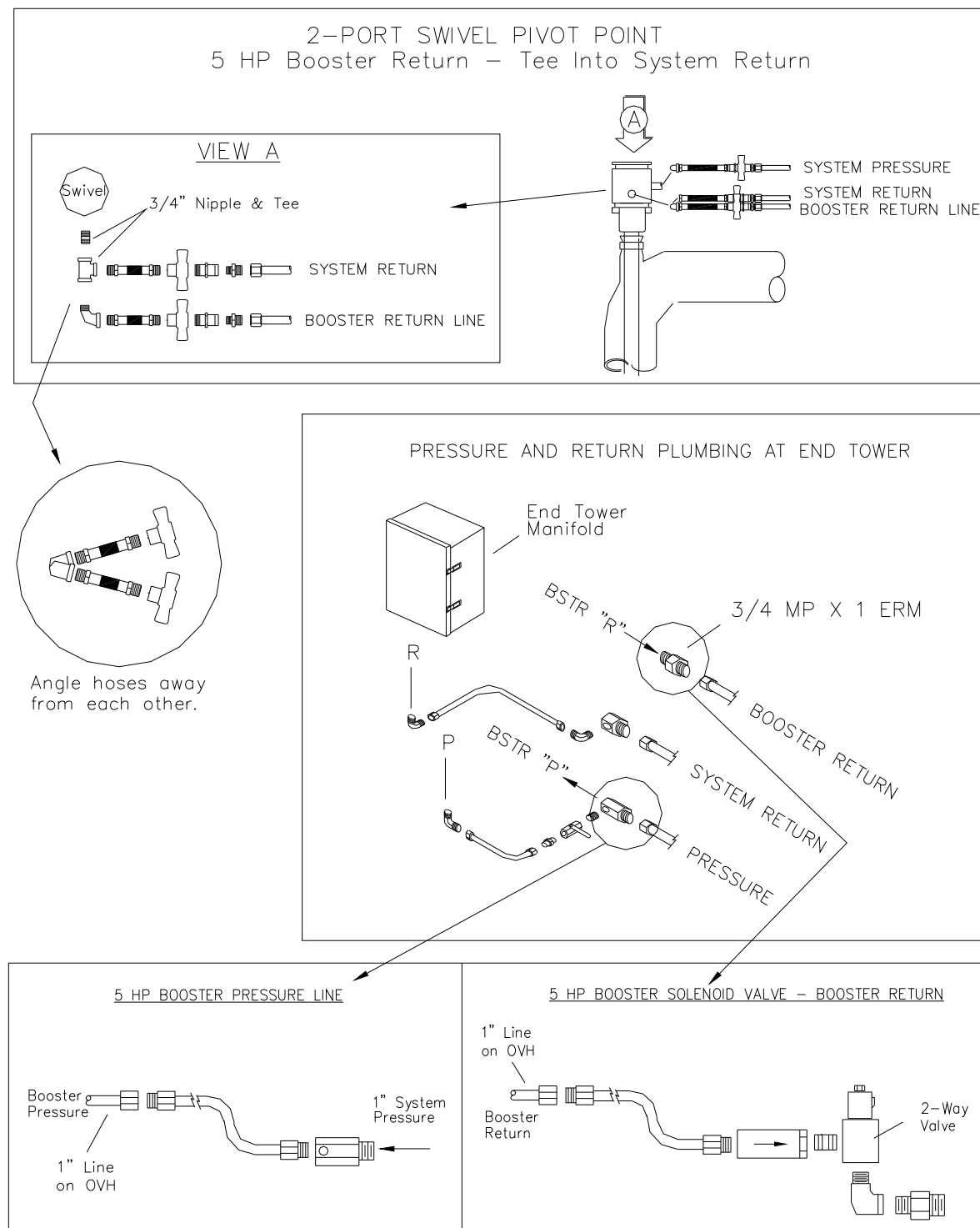
2 HP End Gun Booster with Electric Control – Last Tower MP56909 (Figure 28)



Ref.	Part No.	Description	# Req'd.
1	EC98389	Sterling Coil CCS024L (Non-Eff 4/08)	1
	EC98370	Lead Wires & EC98371 Coil (Eff 4/08)	
2	EH50808	Point Junction Box	1
3	EH52354	Wire Connector Brown	3
4	EH5244	½" Strain Relief Connector	3
5	ES52975	Pressure Switch 5 PSI 48" Leads	1
6	EW5356	14/2 TC Cable	18 ft.
7	FN5424	Nut 10-24	2
8	FS5481	SRHMS 10-24 x ½	2
9	HA83356	Tee Assy 1"Erm x ¾"FP x ½" Erm (remove conn.)	1
10	HF57775	¾" In-Line Filter	1
11	HH6130	Hose ¾" – ¾"MP x 1"Erm x 23"	1
(Note: One HH6130 is also included in MP56803 2HP Bstr End Span Box)			
12	HV98382	Sterling 2-Way NC Valve Cartridge	1
13	HV98386	Sterling Body for 2-Way Valve	1
14	MD8208	Nipple ¾ x Cl Galv	1
15	PB80412	Bush ¾" x ¼" Galv	1
16	PE8113	Street Elbow ¾" x 90 Plated	1
17	PN8206	Nipple ¾ x Cl BEH	2
18	PT8154	Tee ¾" Galv	1
19	AS16923	End Gun Solenoid Valve Shield	1
	(Shipped Loose in Component Box)		
20	EH52495	Black Poly Loom ½"	2 ft.
21	FC4924	Muffler Clamp	8

Figure 28. 2 HP End Gun Booster with Electric Control

5 HP EG Booster - Electric Control, 3-Line (Separate Booster Return Line) (Figure 29)



(See Figure 28 for solenoid valve installation.)

Figure 29. 5 HP End Gun Booster with Electric Control, 3-Line



Booster with Flood Drop (Figure 30 & 31)

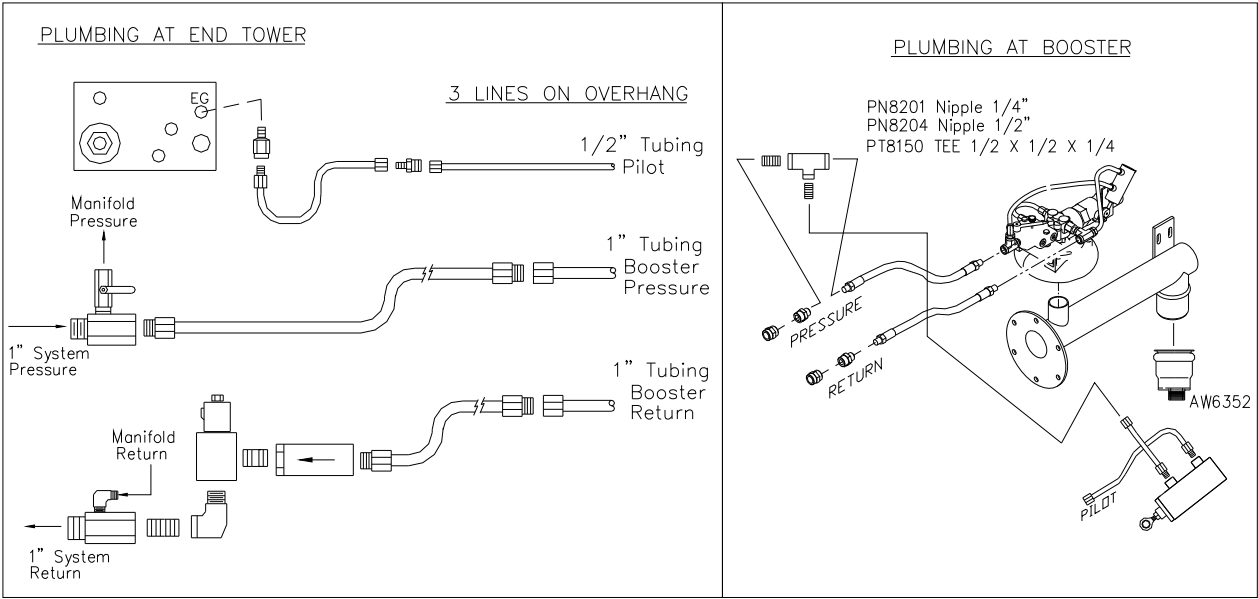


Figure 30. Booster with Flood Drop Plumbing

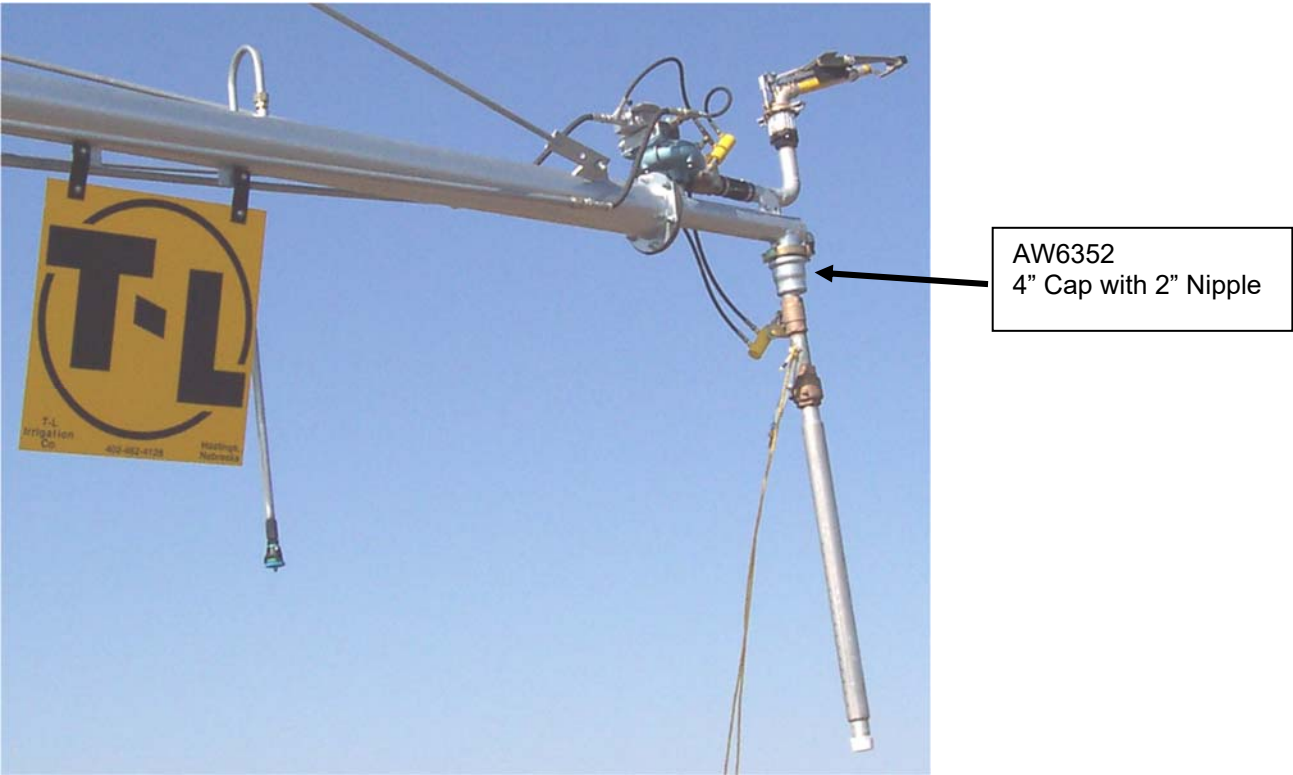


Figure 31. Picture of Booster with Flood Valve

Auto Reverse Switch at Pivot Point MB46446 (6") MB46448 (8") MB46440 (10") (Figure 32)

The Auto Reverse Switch can be installed at the Pivot Point to provide a reversing point in the field. This can be installed by itself or in addition to the existing Auto Reverse Switch at the End Tower. In any case, this should NOT take the place of the Reversing points required at each end of the field on part circle machines where Physical Barricades should be installed. (See Operation – Reversing Systems later in this manual.)

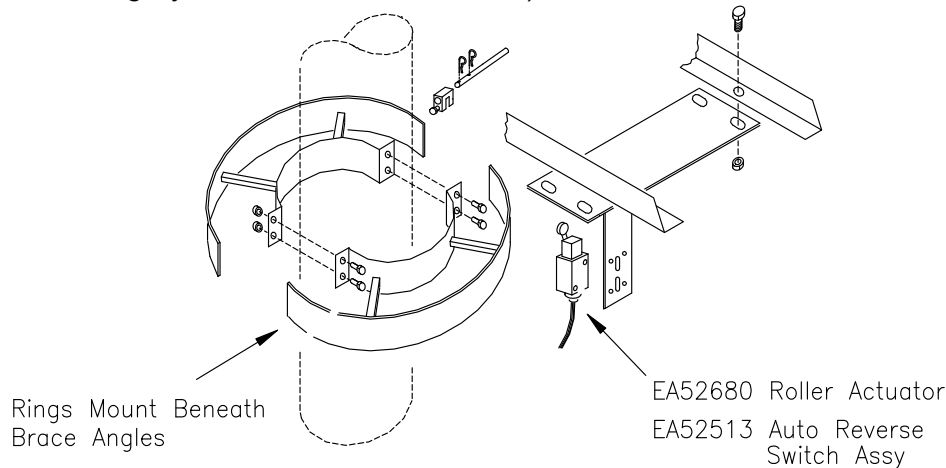
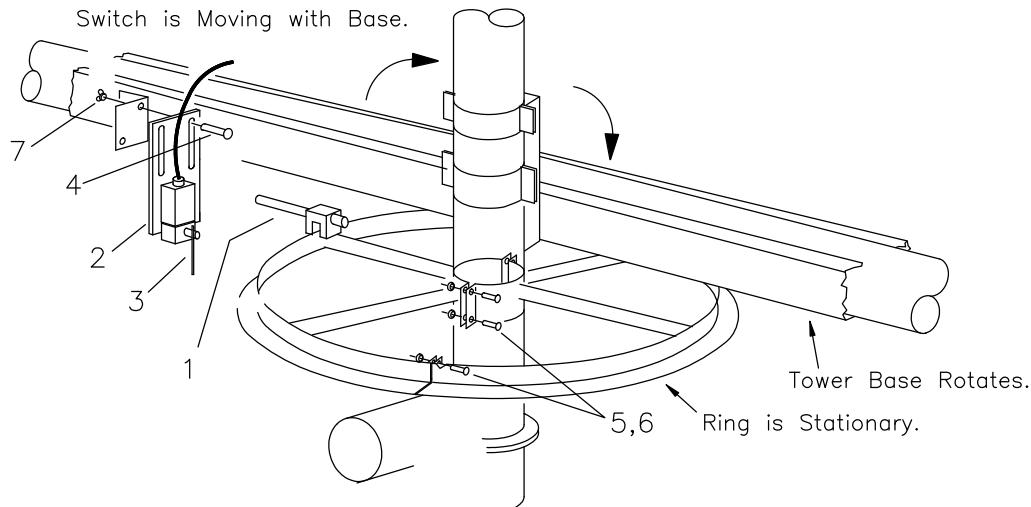


Figure 32. Reversing Switch at Pivot Point.

Auto Reverse Switch at QUICK TOW Pivot Point - MB46449



Ref.	Part No.	Description	# Req'd.
1	AA2812	PS Trigger Assy / QT	2
2	EA52699	PC QT AR SW ASSY (Change to Prewired Switch Dec. 2011)	1
3	EH5267	LSZ68 ACT Lvr Rod /Spring	1
4	FB5432	Carriage Bolt 1/4 X 3/4 GR5	2
5	FB5444	HHCS 3/8 X 2 GR5	8
6	FN5401	Nut Hex 3/8-16	8
7	FN5411	Wing Nut 1/4"-20	2

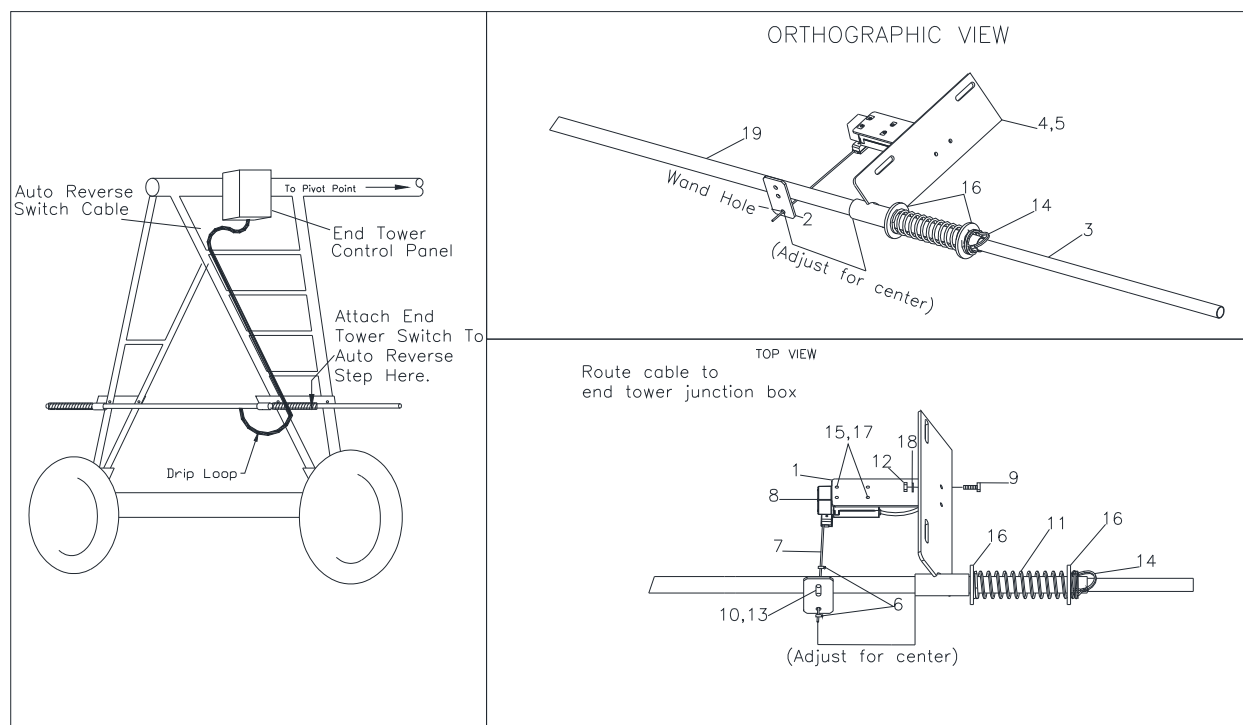
Figure 32.1 Reversing Switch at QUICK TOW Pivot Point.



End Tower Auto Reverse Switch MB45108 (Figure 33)

1. Attach Left and Right Auto Reverse Steps (4,5) to End Tower Legs using 3/8 x 1 bolts provided with system. Insert Slide Tube (21) through Steps. (NOTE: The Left Auto Reverse Step may need to be field drilled. It may be easier to drill the holes before installing.)
2. Insert Telescoping Tubes (3) into each side of the Slide Tube. Slide on a Large Washer (18), Compression Spring (13), Large Washer (18) and attach Telescoping Tube to Slide Tube with Lynch Pin (16).
3. Attach Wand Stop Strap (2) to Slide Tube (21) with U-Bolt (12) in approximate location shown.
4. Attach Switch Mount Plate (1) to Auto Reverse Step using 1/4" x 1" bolts, washers and nuts (11,20,14).
5. Attach Reversing Switch (9) to Switch Mount Plate (1) with two #10-32 x 1/2" screws and washers (17,19).
6. Attach Activation Lever (8) to MicroSwitch (9). Adjust length so wand extends at least 1" through Stop Strap (2). Attach Blackburn Wire Clamps (6) to the Activation Lever (8) on both sides of the Stop Strap (2) so lever is held captive.
7. Adjust position of Wand Stop Strap and MicroSwitch so switch is in neutral position when slide tube is centered with Springs. **IMPORTANT:** Check that slide tube will slide freely and return to spring centered position after activation.

HARDWARE INSTALLATION



NOTE: Reversing Switch should have Potting Sealant fill inside to seal cable entrance from moisture.

Ref.	Part No.	Description	# Req'd.
1	AP28610	Tall AR / PS Switch Bracket	1
2	AP28615	Wand Stop Strap	1
3	AT2990	Auto Reverse Telescope Tube	2
4	AW13590	LT Step Weldment AR / POS Stop	1
5	AW13610	RT Step WELDMENT AR / POS Stop	1
6	EH5103	Blackburn Wire Connector	2
7	EH5268	LSZ54N Activation Lever Rod SS	1
8	EA52513	Rev Switch Prewired and Sealed (12/11 -)	1
9	FB5435	HHCS 1/4 x 1	2
10	FB5725	U-Bolt 5/16 x 1 3/8 x 1 1/2	1
11	FD4864	Compression Spring 1 3/4 ID x 8" long	2
12	FN5399	Hex Nut 1/4	2
13	FN5400	Hex Nut 5/16	2
14	FPP5729	Locking Pin with wire retainer 1/4" x 1 3/4"	2
15	FS5537	SRHMS 10-32 x 1/2	2
16	FW50	Washer 1 5/16 ID	4
17	FW5646	Flat Washer 3/16	2
18	FW5648	LW 1/4"	4
19	TZ449	Auto Reverse Slide Tube	1

Figure 33. Installation of Auto Reverse Switch.

Auxiliary Speed Switch MB52018 (see previous Figure 24 and Figure 25)

- Not used with a Position Sensor.

The Precision Point Control III has the ability to switch the pivot to an Auxiliary Speed that is programmed by the Operator in the Set-up of the Precision Point Control III. This Auxiliary Speed is triggered any time +24 VDC is present on terminal 10 in the Control Panel.

The Auxiliary Speed can be wired from an End Gun Control Switch, a Flood Drop Control Switch or a separate Auxiliary Speed Switch. If both an End Gun Control Switch and an Auxiliary Speed Switch are required, a double actuator mount can be installed so the switches and rings can be stacked. This way the settings can be independent of each other.

NOTE: Logic can be reversed to reduce the number of black wafers installed. Wire the switch NO (Normally Open), terminals 3 and 4. Then install wafers at set points where the Auxiliary Speed is to be active.

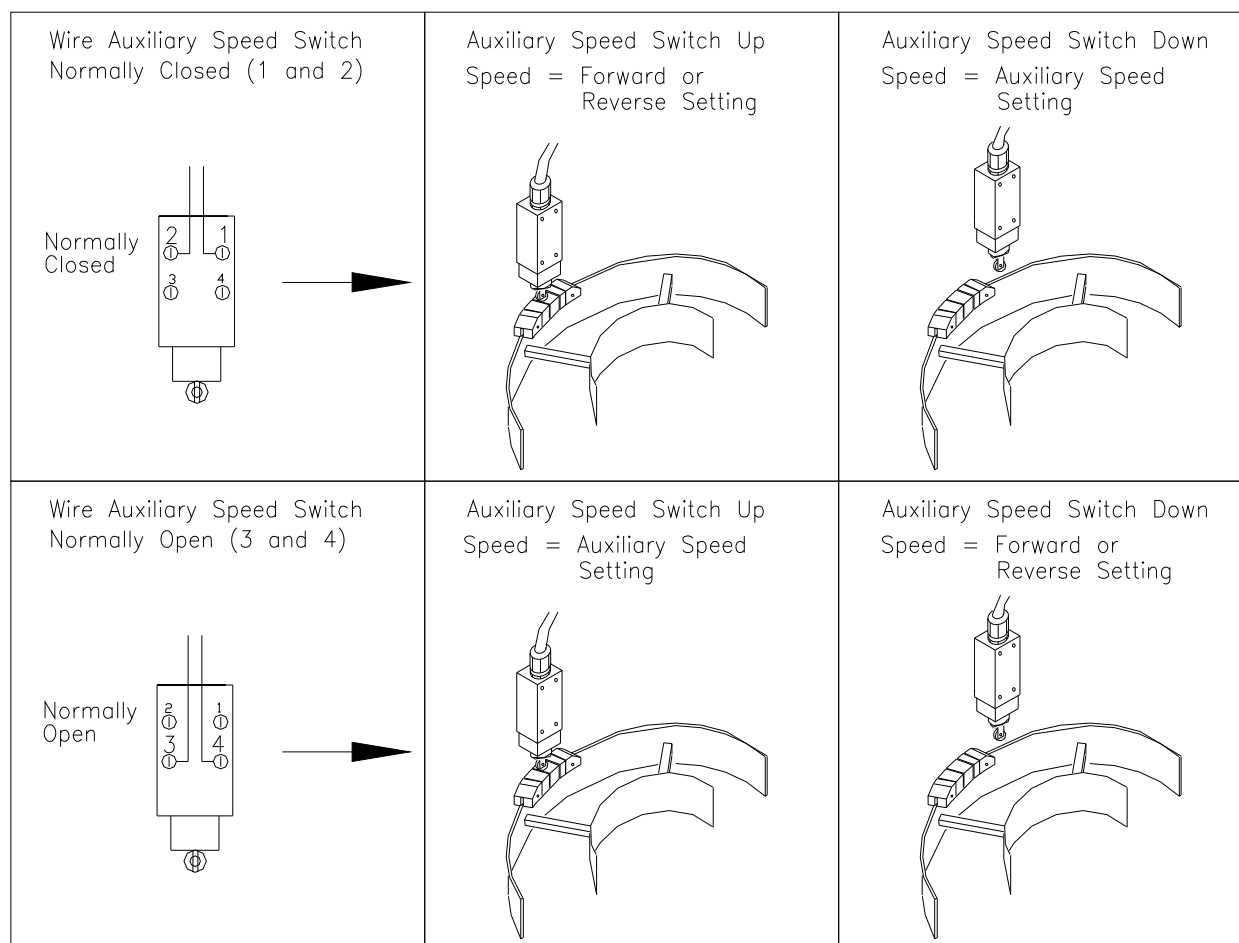
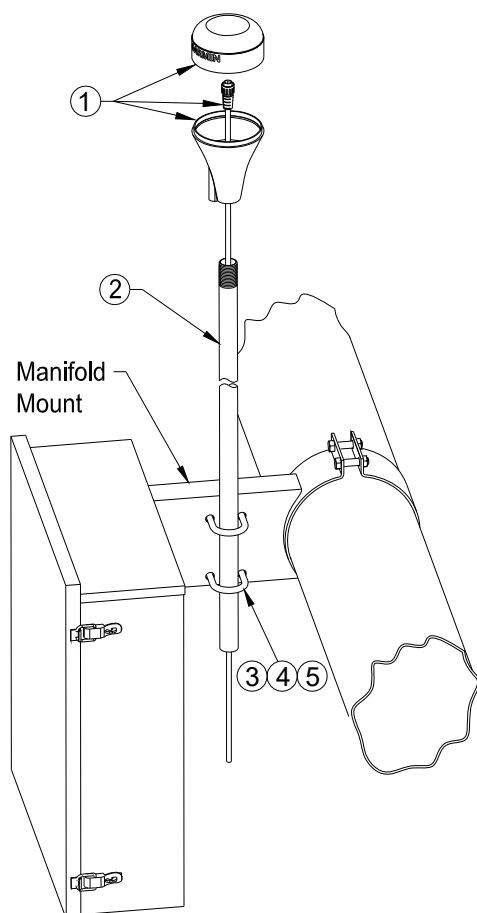


Figure 34. Auxiliary Speed Switch

GPS Antenna Installation at End Tower. MB45216 & MB45219



Ref.	Part No.	Description	Qty
Shipped in MB45219:			
1	EC53231	Garmin GPS Antenna	1
3	FB5725	U-Bolt 5/16" x 1 3/8" x 2 3/16"	2
4	FW5649	Flat Washer 5/16"	4
5	FN5400	Nut, Hex 5/16"-18	4
	FT5702	Wire Tie (not shown)	3
	CD91174	I.S. GPS Mounting and Wiring (not shown)	1
	EH52350	Wire Nut - Orange (not shown)	2
	EH52394	1/2" Conduit Locknut (not shown)	1
	EH52413	1/2" Cord Conn. (not shown)	1
Shipped Loose:			
2	TY10183	TB 1" X 36" GPS	1

Figure 34.1 GPS Antenna Mounting at End Tower.

NOTE: Clip and discard Fuse Holder from GPS antenna cord.

Underground Control Cable Installation (Figure 35)

VERY IMPORTANT NOTES:

1. Independent Ground Rods required at Control Panel location and Pivot Point.
2. Underground Cable Requirement:
 - a. 16 ga. Buried wire max. of 800 ft (w/ qty 1 of 12 ga. gnd) + 16 ga. Span = 2640 ft. max. combined length.
 - b. 16 ga. Buried length 800 -1300 ft (w/ qty 2 of 12 ga. gnd) + 16 ga. Span = 2640 ft. max. combined length
 - c. 14 ga. Buried, 1300 to 1800 ft (w/ qty 2 of 12 ga. gnd) + 16 ga. Span = 3200 ft. max. combined length
3. Communication Twisted Pair required for GPS or Encoder use.
4. **Dealer Help:** see the Pivot Design Guide - Cable Selection or consult T-L Engineering Dept.

Sleeve the cable in PVC conduit on each end where it exits the ground to protect against rodents. If the area is known to have rodent problems, the entire length of the buried wire may need to be sleeved.

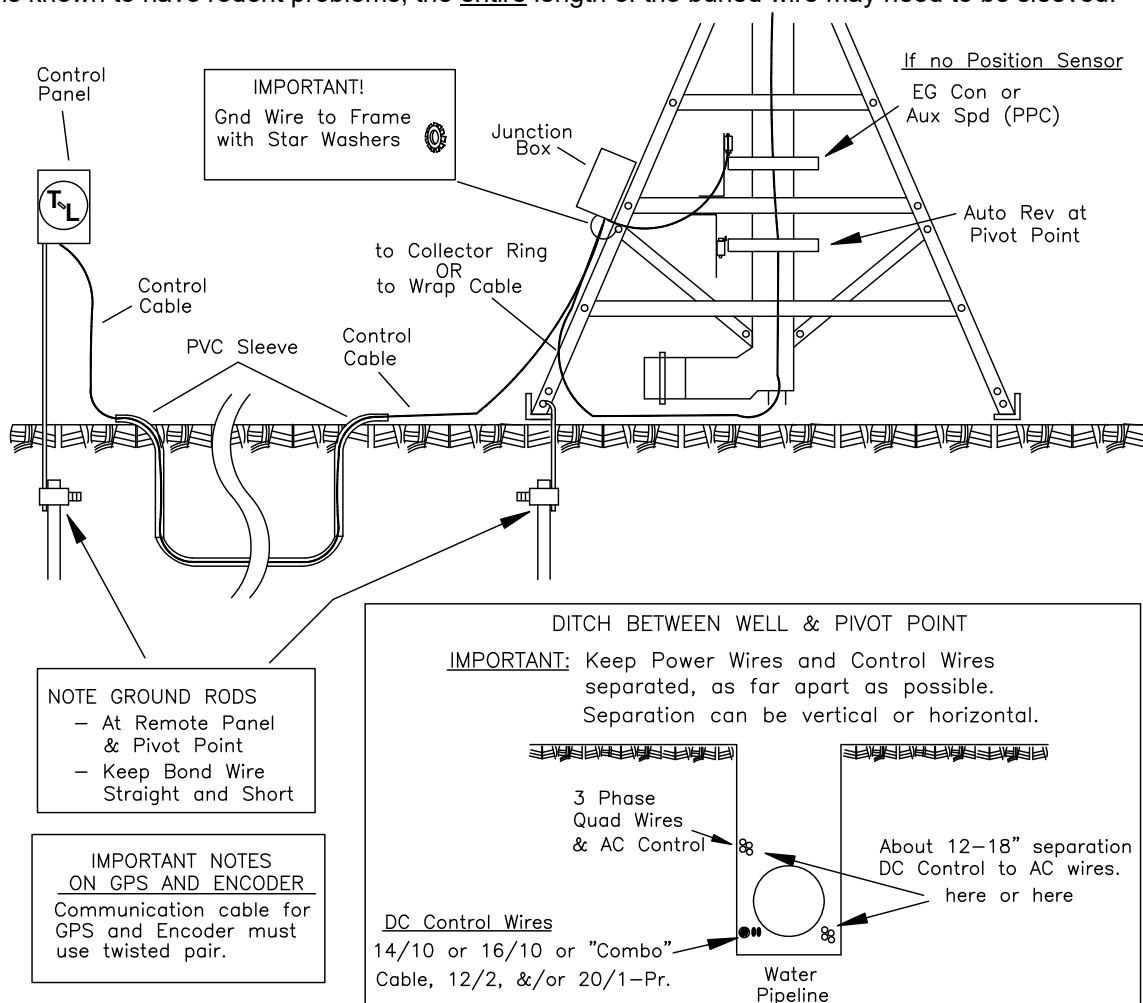


Figure 35. Underground Control Cable Installation

Special Note On Run Lights:

Run Lights at the Pivot Point CANNOT be used with a Remote Mounted Control Panel UNLESS a separate wire is used.

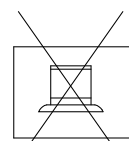


Figure 36. Run Lights need Separate Wire.

Remote Mounted Panel: GPS at End Tower, Using “Combo” Cable Underground.

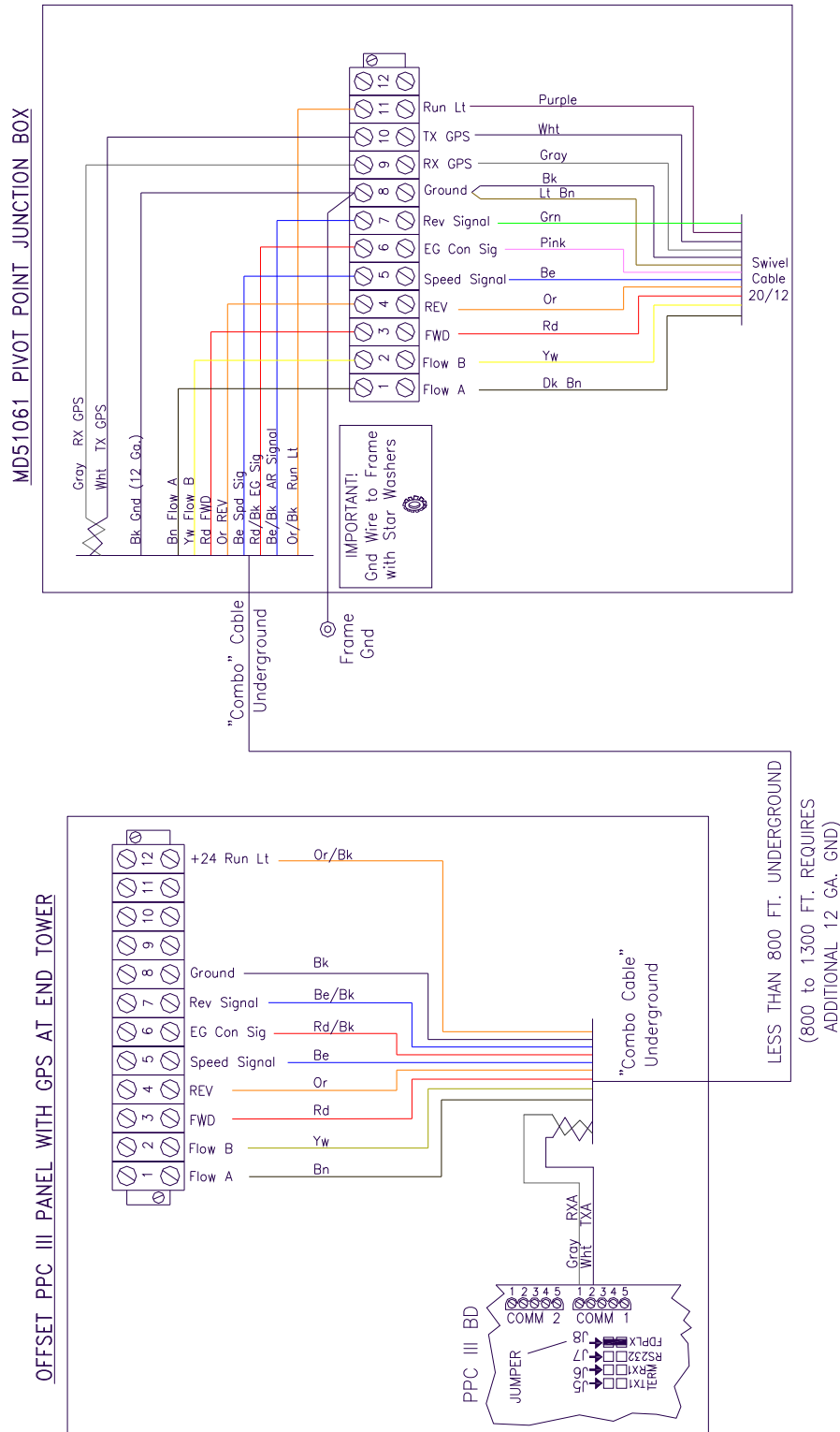


Figure 36a. Wiring with Remote Mounted Panel, Underground “Combo” Cable.

Engine Drive Power Supply Wiring (Figure 37)

- DC-DC Converter mounted in Precision Point Panel converts 12 to 28 VDC (green bd.) +12 VDC into board is fused, +28 VDC out is fused.
- Engine alternator **MUST** be capable of producing 15 amps for the Point Control Power.
- **VERY IMPORTANT:** If engine more than 30 ft. from pivot point, mount the Control Panel at the Engine OR use larger gauge wire to run power from the Engine to the Control Panel.

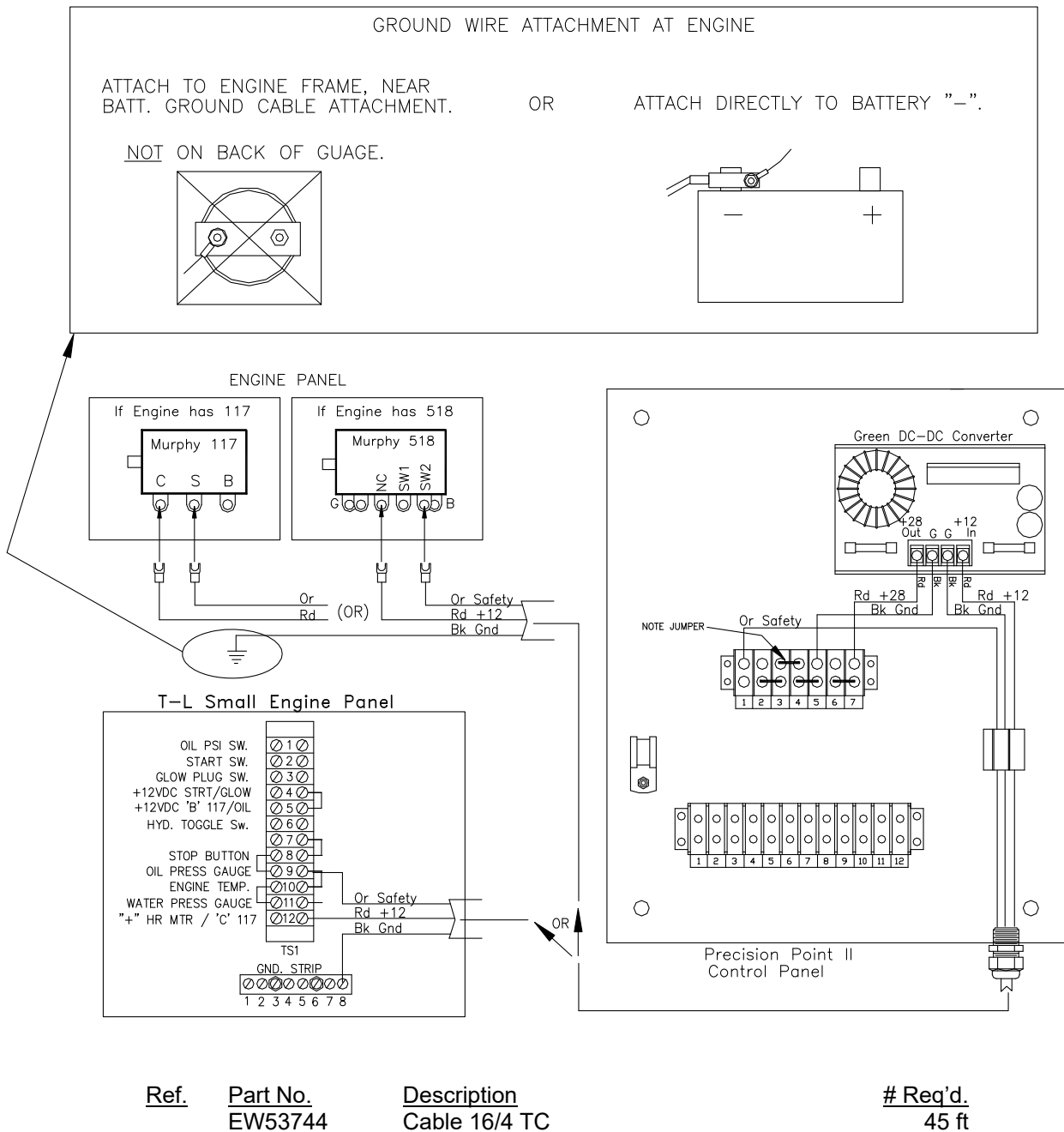


Figure 37. Engine Drive Power Supply Installation.

Electric Drive Power Supply Installation (Figure 38)

With an Electric Driven Hydraulic Pump, the DC power for *The Precision Point Control III* system is drawn from the T-L Electric Panel. A Transformer takes the line voltage to 24 VAC and the Rectifier takes the 24 VAC to approximately 30 VDC, and it is then regulated to 28 VDC.

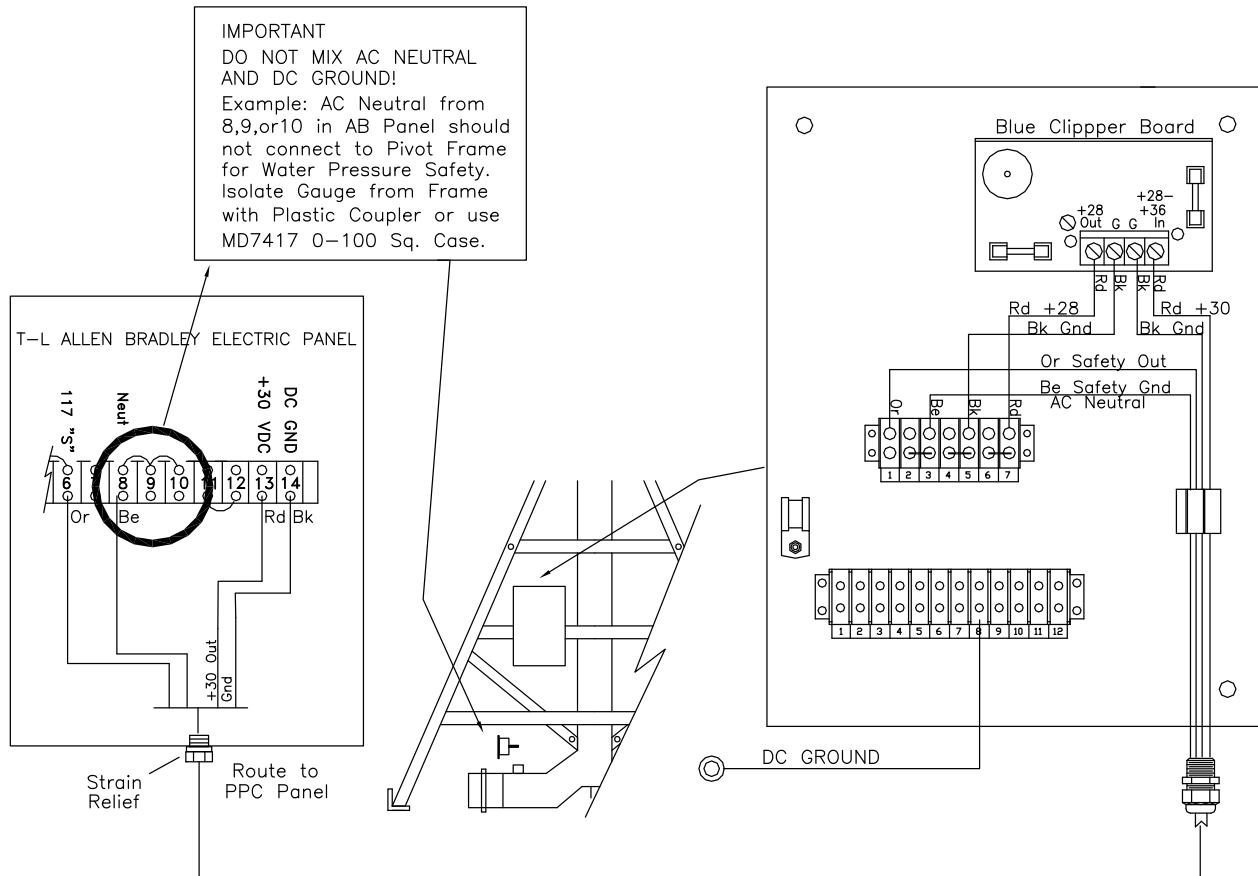
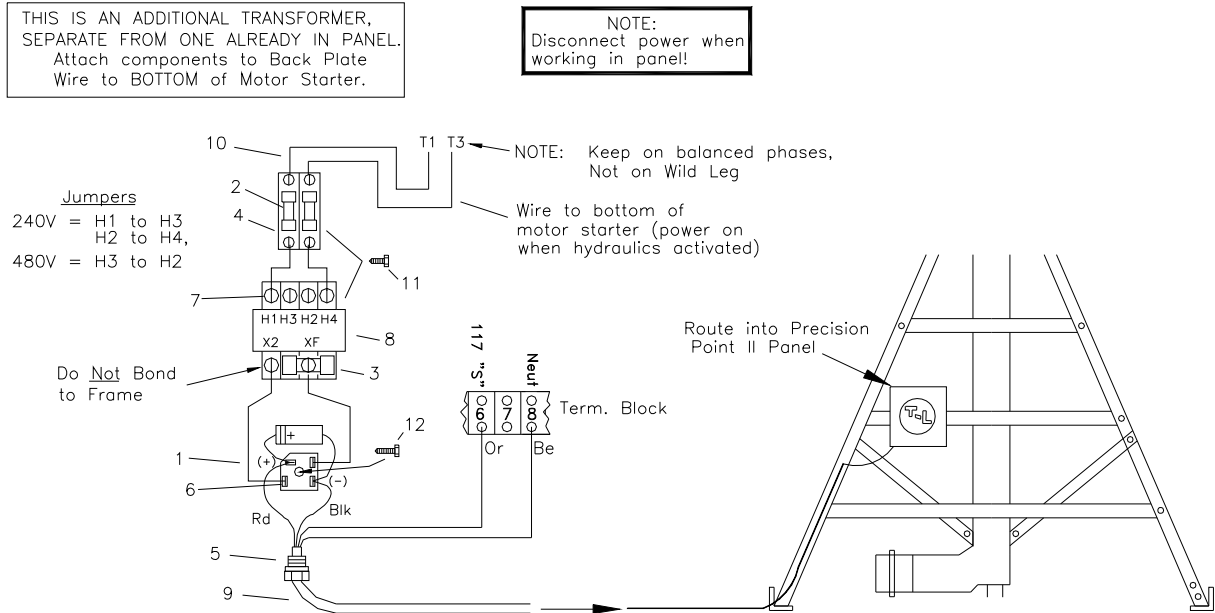


Figure 38. Electric Drive Power Supply Installation

RETROFIT Electric Drive Power Supply Installation EA51051 (Figure 39)

These components are field installed inside the T-L Electric Panel.

NOTE: This is an additional Transformer and Fuse Block added to the Panel, separate from the Control Transformer already in the Panel.



Ref.	Part No.	Description	# Req'd.
1	EC50450	Rectifier Assy	1
2	EC50885	Fuse FNQ 0.4 amp	2
3	EC5104	Fuse SNM 3.2 amp	1
4	EH5090	Fuse Block	1
5	EH5244	Strain Relief Connector	1
6	EH53065	Terminal Slip On Spade	4
7	EH53234	Transformer Jumper	2
8	EH5323	Transformer 240/480 to 24 VAC	1
9	EW53744	Cable 16/4 TC	45 ft
OR	EW53745	Cable 16/6 TC	45 ft
	(16 ga. wire found in MB45974 Pivot Point Box)		
10	EW5383	Wire Single #16	8 ft
11	FS5425	Self Tap 8 x ½	6
12	FS5426	Self Tap 8 x ¾	1

Figure 39. Electric Drive Power Supply Installation for Retrofit Precision Point Panel

Wiring to 7-Terminal Block - Precision Point Control III Panel (Figure 40)

1. Wiring to the 7 Terminal Block in the Precision Point Control III Panel is as shown below.

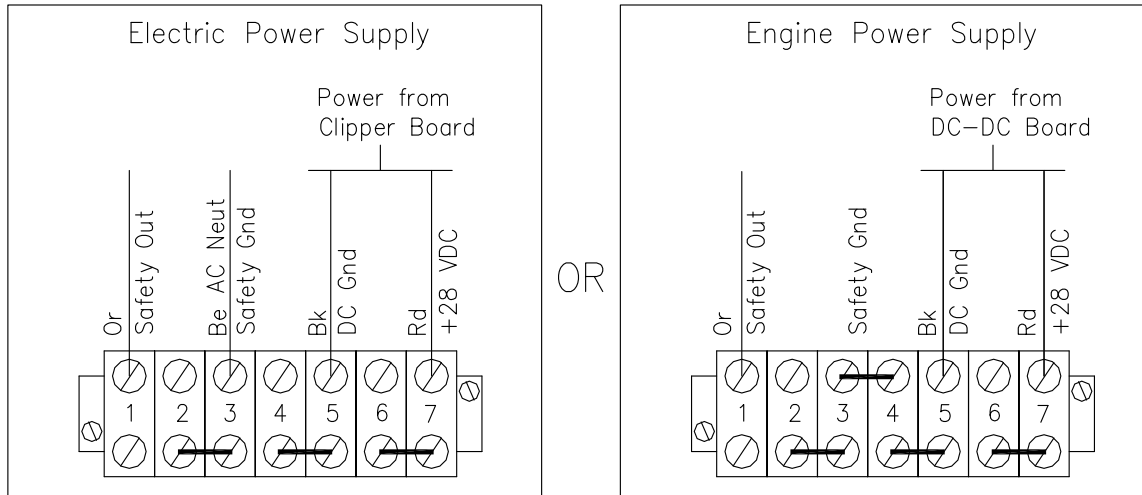


Figure 40. Wiring for Electric and Engine Installation.

Wiring Precision Point Control III Panel (Figure 41 and 42)

- Figure 41: Precision Point Control III on Swivel Pivot Point.
If the pivot has a Hydraulic Swivel Pivot Point, a Collector Ring will be installed on top of the Swivel. A gray Swivel Cable (20 ga./12 wire) or 20 ga./15 with Encoder option will route through the middle of the swivel pivot point and up into the Collector Ring.
- Figure 42: Precision Point Control III on Hose Wrap Pivot Point.
With a Hose Wrap Pivot Point, there are 2 options of wiring in the span cable:
- a. for part circle systems where the pivot will never make a full circle
 - Wire span cable directly into Precision Point Control III Panel
 - b. for full circle systems where the hydraulic hoses will be re-wrapped
 - Wire span cable into Socket, use excess cable to wire plug to Panel

Wiring Precision Point Panel – Swivel Cable combo/11 (June 2018)

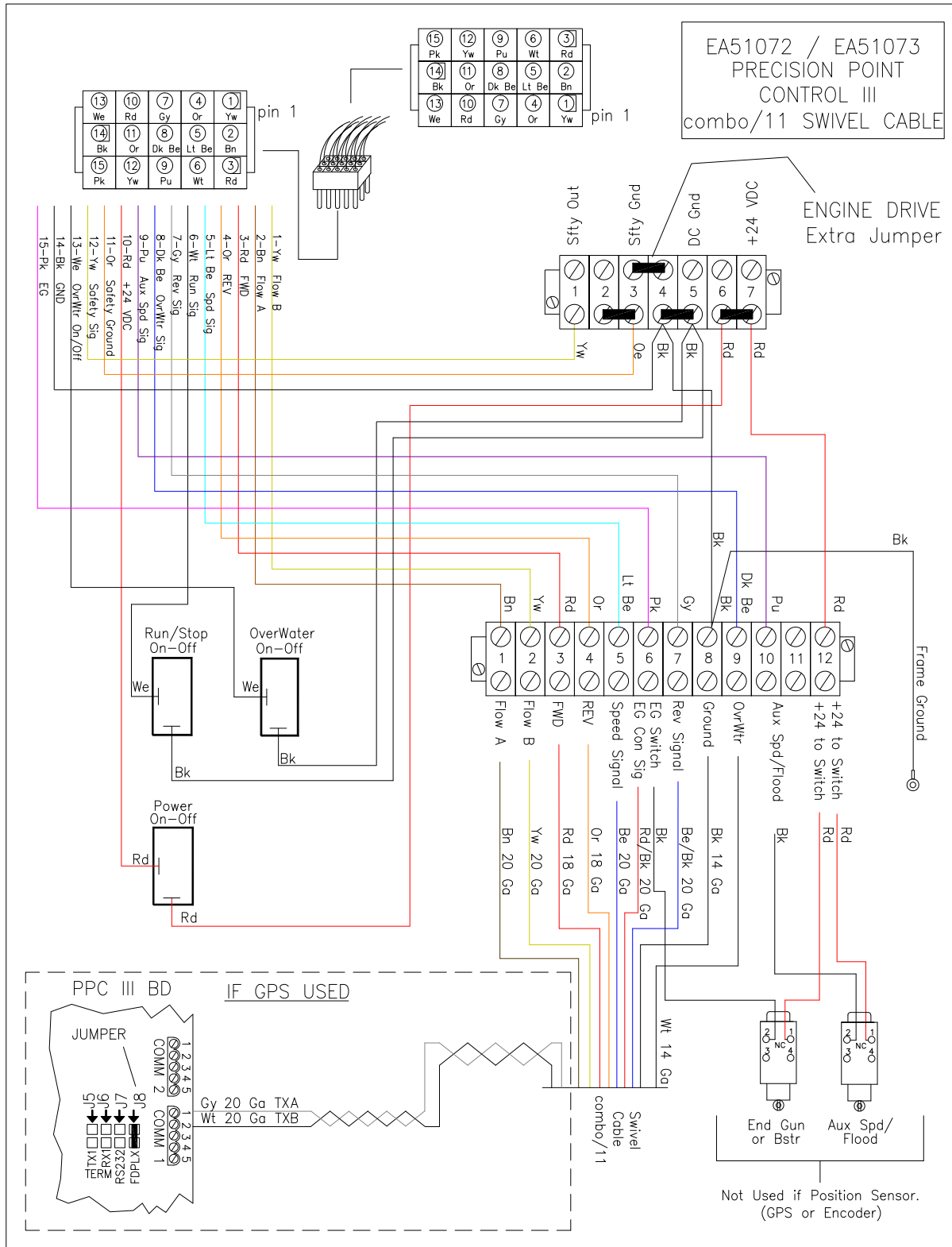


Figure 41. Wiring of Precision Point Control III Panel to Swivel Cable.

NOTE: Connect all wires needed. Maintain continuity from source to destination.

Wiring Precision Point Panel – Swivel Cable 20/12 (Pre-June 2018)

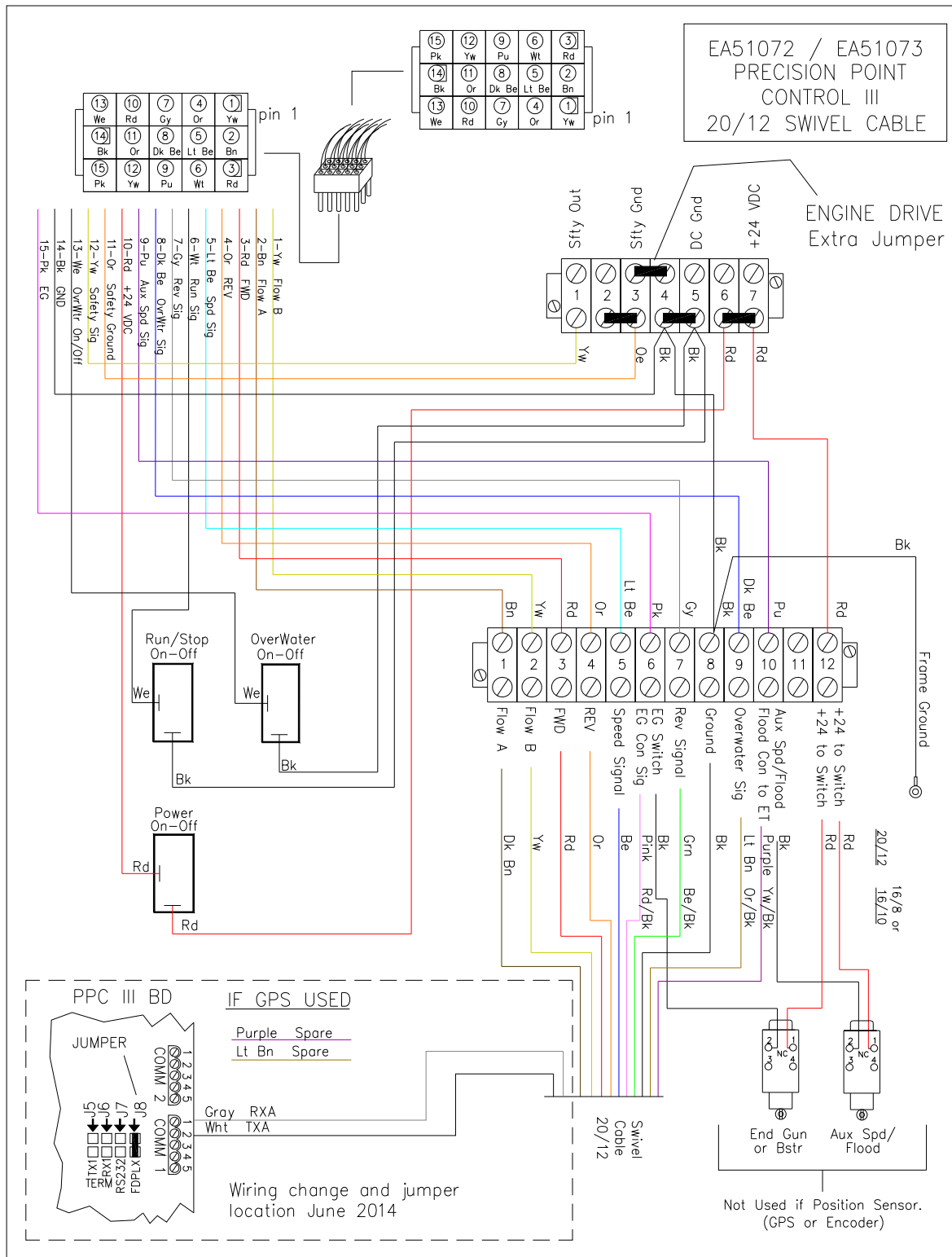


Figure 41.1 Wiring of Precision Point Control III Panel to 20/12 Swivel Cable.
NOTE: Connect all wires needed. Maintain continuity from source to destination.

Wiring Precision Point Panel – Swivel Cable with Encoder 20/15 Cable

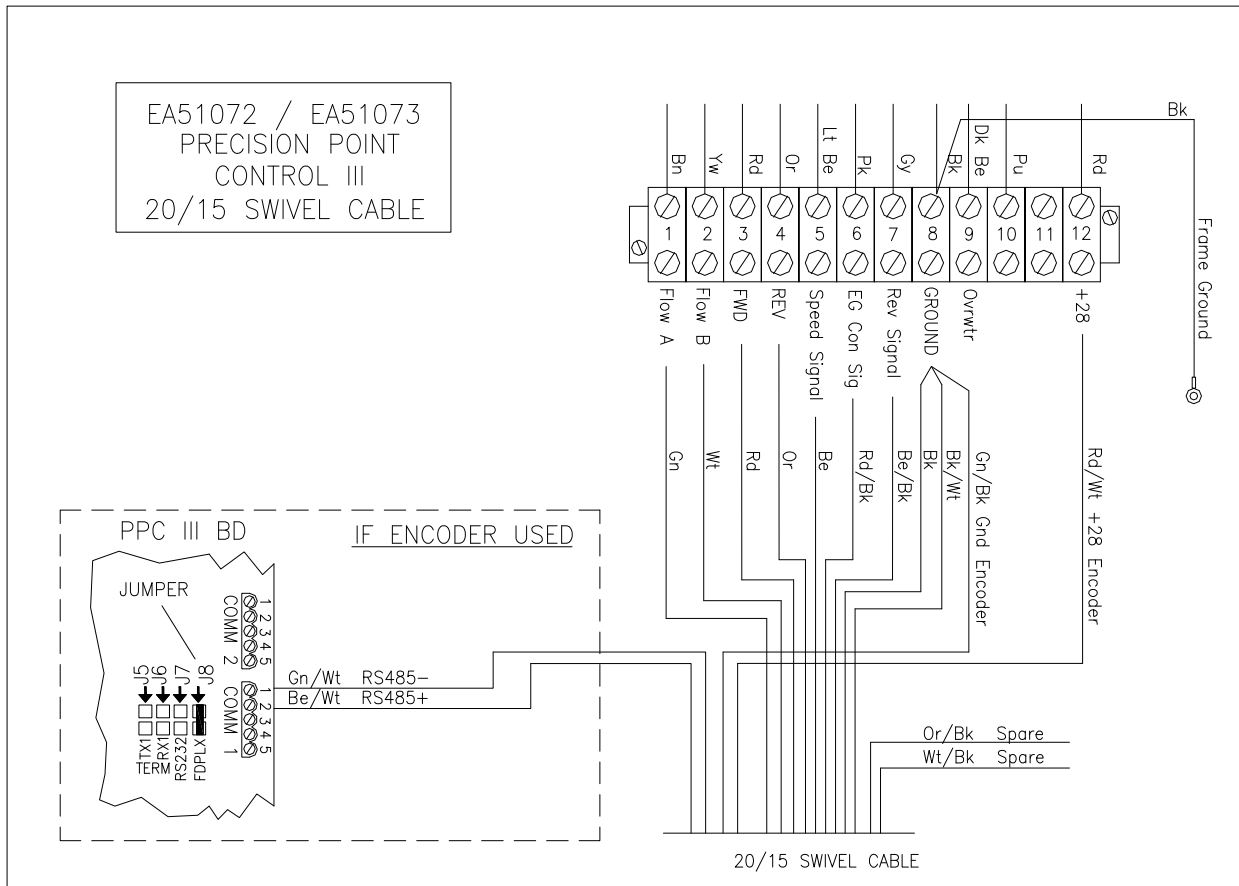


Figure 41.2 Wiring of Precision Point Control III Panel to 20/15 Gray Swivel Cable with Encoder.

Wiring Precision Point Panel – Hose Wrap Pivot Point

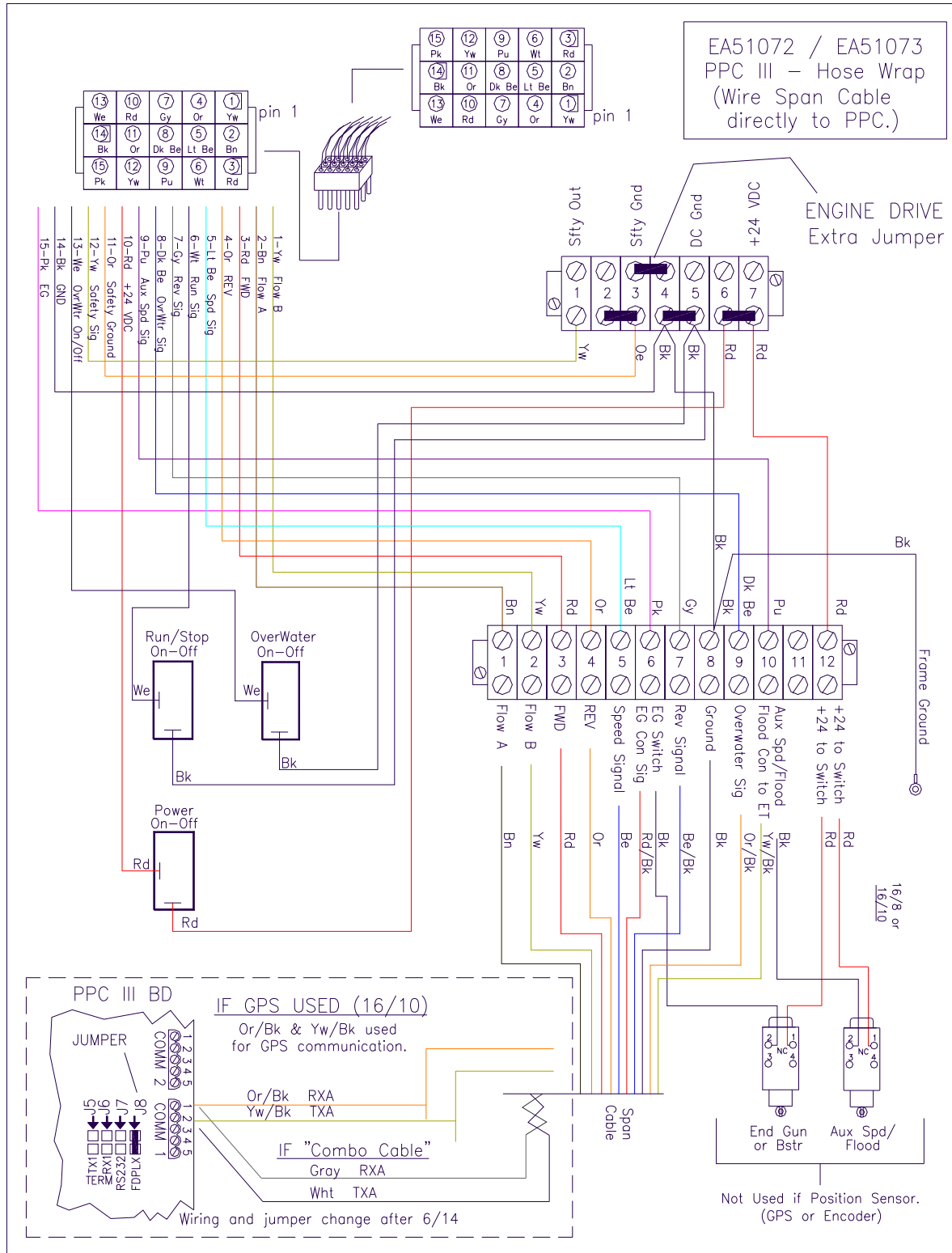
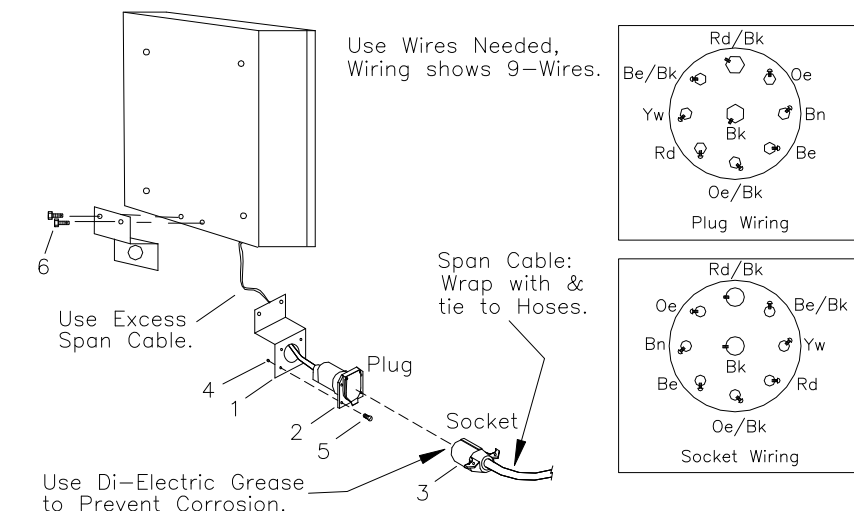


Figure 42. Wiring of Precision Point Control III Panel to Wrap Cable.

NOTE: Connect all wires needed. Maintain continuity from source to destination..

Wrap Cable Plug and Receptacle Wiring MB45978 (Figure 43)

On full circle Hose Wrap Pivot Points, a Plug and Socket is used to quickly disconnect and reconnect the span cable from the Control Panel when the hoses are rewrapped. If the system is an auto reverse and will never turn full-circle, the span cable can be wired directly into the control panel.



Ref.	Part No.	Description	# Req'd.
1	AB15724	Point Cable Wrap Plug Mount	1
2	EH52901	Point Cable Wrap Plug	1
3	EH52902	Point Cable Wrap Socket	1
4	FN5414	Nut 8-32	4
5	FS5485	SRHMS 8-32 x 1/2	4
6	FB5435	HHCS 1/4 x 1	2
7	FN5399	Nut Hex 1/4-20	2

Figure 43. Wiring of Wrap Plug and Socket

End Gun Roller Switch Wiring (Figure 44)

NOTE: Wire Roller Switch for minimum number of Plastic Wafers. Override switch may also be added.

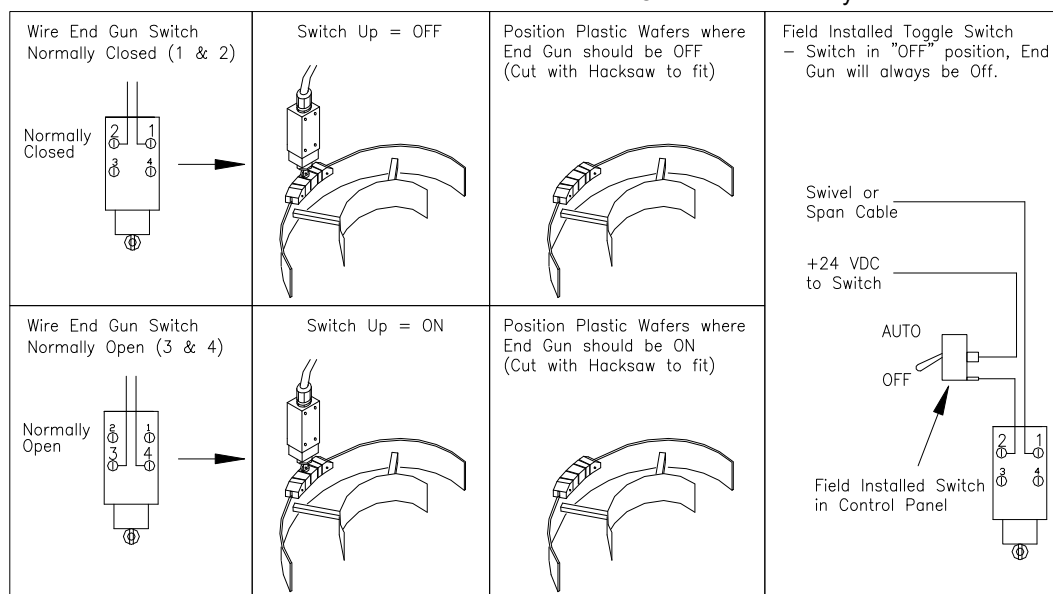


Figure 44. End Gun Control Switch Wiring.

Auto Reverse at Pivot Point Wiring MB46448 or MB46440 (Figure 44a) (If No Position Sensor)

The Auto Reverse at Pivot Point option can be used to eliminate field backstops on full circle machines (if a Position Sensor is not used). This will allow the pivot to reverse at a different cropping location, etc. On systems that must reverse because of obstacles, a Physical Barricade and End Tower Auto Reverse Switch **MUST BE USED**.

Auto Reverse at the Pivot Point can be used in conjunction with Auto Reverse at the End Tower. With this, the system may reverse off the End Tower Switch at one side and then off the Pivot Point Switch at some location in the field.

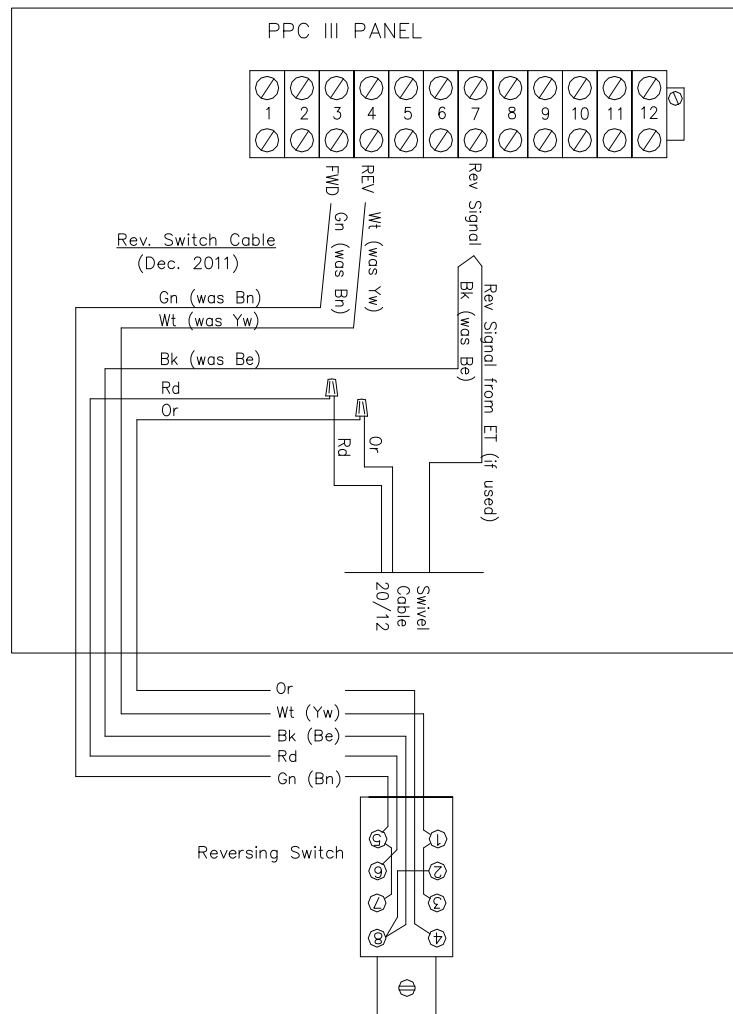


Figure 44a. Auto Reverse at Pivot Point Wiring.
(Shown with Yellow Cable prewired switch, Dec. 2011 -)

Collector Ring Wiring – Swivel Pivot Point without Position Sensor (Figure 47)

If there is a swivel pivot point on the system, a Collector Ring for the Precision Point Control III will be shipped. The wiring in the Collector Ring consists of a) wiring the Swivel Cable to the Ring Extension Wires and b) wiring the Span Cable to the Ring Brushes. The Swivel Cable shipped with the system will be a combo/11.

Place a Strain Relief on the combo/11 to prevent the cable from sliding back into the swivel. NOTE: Strip the insulation back from each span cable wire so it can be doubled up on the brush.

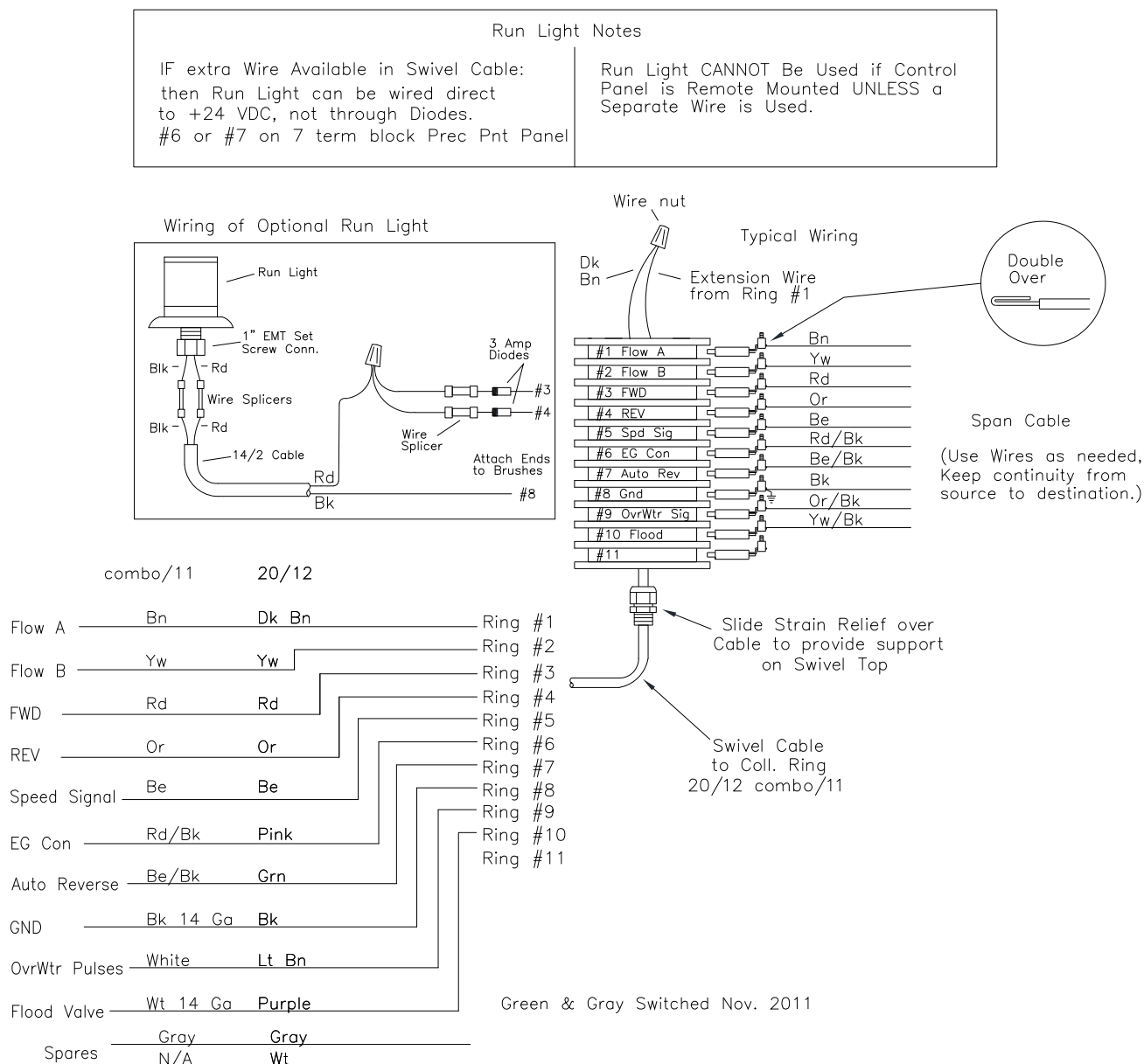


Figure 47. Collector Ring Wiring – Swivel Pivot Point without Position Sensor.

Collector Ring Wiring – Swivel Pivot Point with GPS (Figure 47.1)

Fig 47.1 PPC III COLLECTOR RING WITH GPS

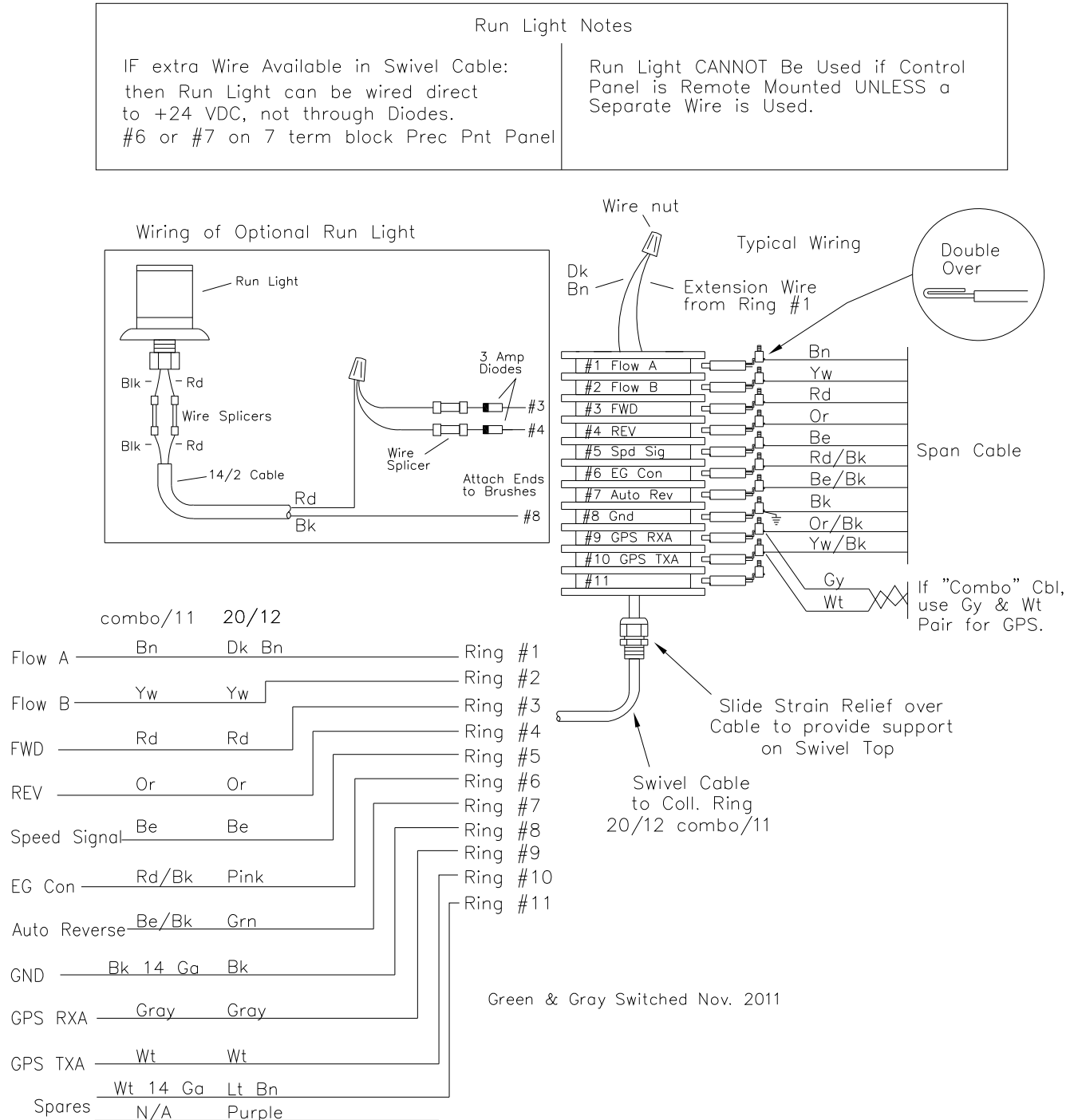


Figure 47.1 Collector Ring Wiring – Swivel Pivot Point with GPS at End Tower.

Wiring of Optional Run Light

Run Light

Blk - Rd

Wire Splicers

Blk - Rd

14/2 Cable

Rd

Bk

3 Amp Diodes

#3

#4

Wire Splicer

Attach Ends to Brushes

#8

Serial Encoder

Blue Wire Nuts

Red

Wt/Bk

Gnd

485+ Be/Wt

485- Gn/Wt

#1 Flow A

#2 Flow B

#3 FWD

#4 REV

#5 Spd Sig

#6 EG Con

#7 Rev Sig

#8 Gnd

#9 Spare

#10 Spare

#11

Bn

Yw

Rd

Or

Be

Rd/Bk

Be/Bk

Bk

Or/Bk Spare

Yw/Bk Spare

Span Cable

20/15 Swivel Cable to Coll. Ring

20/15 Cable

Gn = Flow A

Wt = Flow B

Rd = FWD

Or = REV

Be = Speed Sig

Rd/Bk = End Gun

Be/Bk = Rev Sig

Bk

Bk/Wt

Gnd

Or/Bk = Spare

Wt/Bk = Spare

Rd/Wt = Encoder +28 DC

Gn/Bk = Encoder Gnd

Be/Wt = RS485+

Gn/Wt = RS485-

Encoder

47

Encoder Wiring (Figure 47.3)

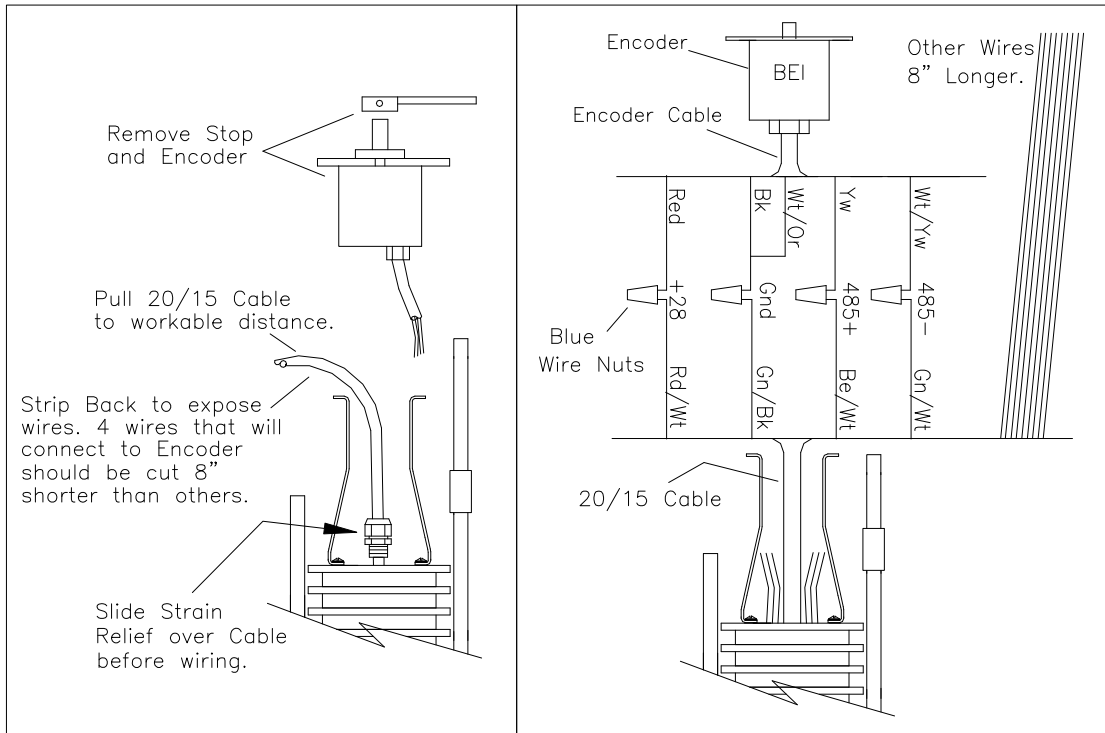


Figure 47.3. Encoder Wiring

Collector Ring Finish Work (Figure 47.4)

After the Encoder (Figure 20) and the Collector Ring wires (Figure 21) have been connected to the Swivel Cable, finish the installation as shown in Figure 20a. Note that the degree angle of the pivot will be set later in PIVOT SETUP (See Operators Manual).

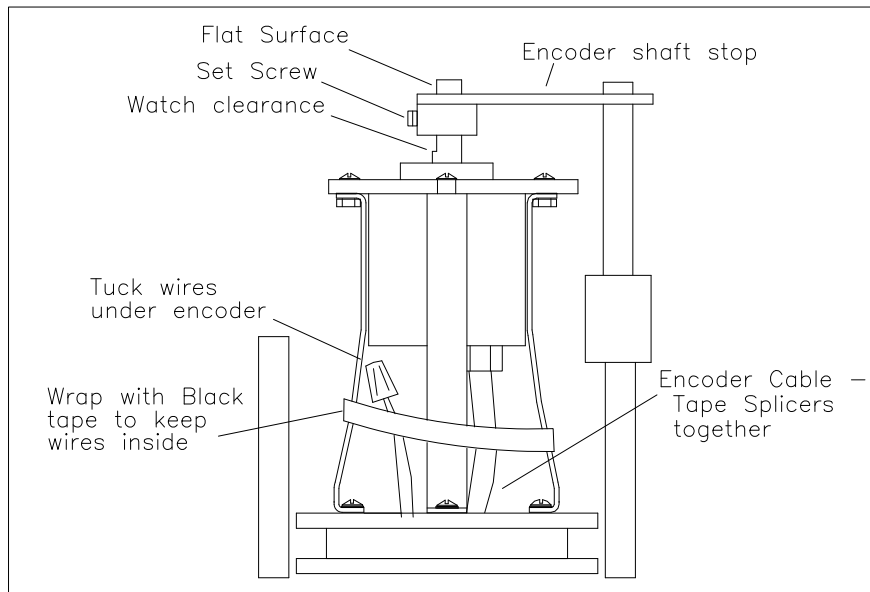


Figure 47.4. Collector Ring Finish Work

OverWater Junction Box Wiring (Figure 48, 6/04 -)

A Speed Sensor Motor located on the next to last tower provides the signal for the Overwater Safety Shutdown. The Overwater pulses must be brought into the Precision Point Control III Panel. To wire this, the Span Cable is spliced at the next to last tower as shown in Figure 10.

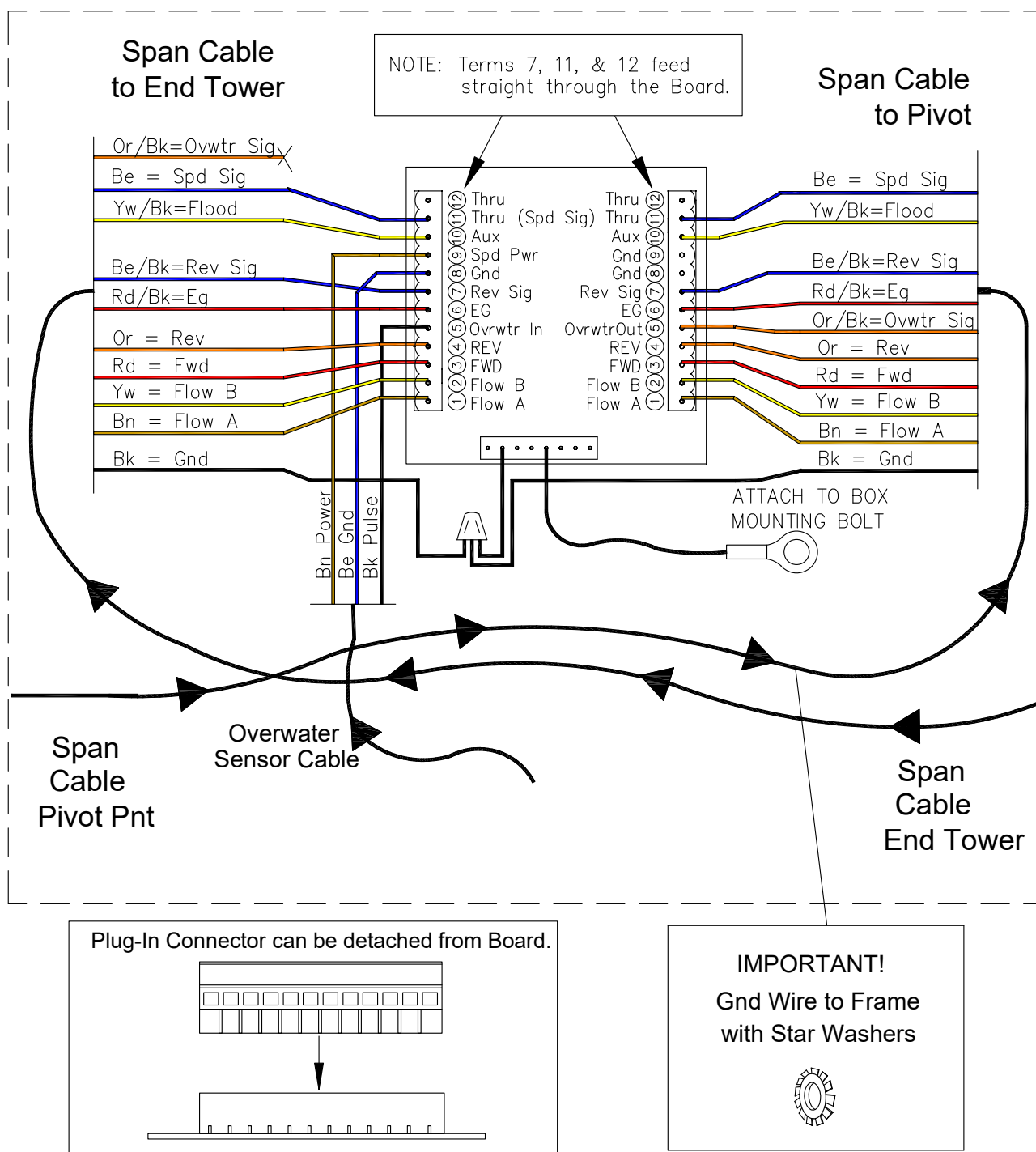


Figure 48. Wiring of Overwater Junction Box with PCB (6/04 -)

OverWater Junction Box Wiring (Figure 49, prior to 6/04)

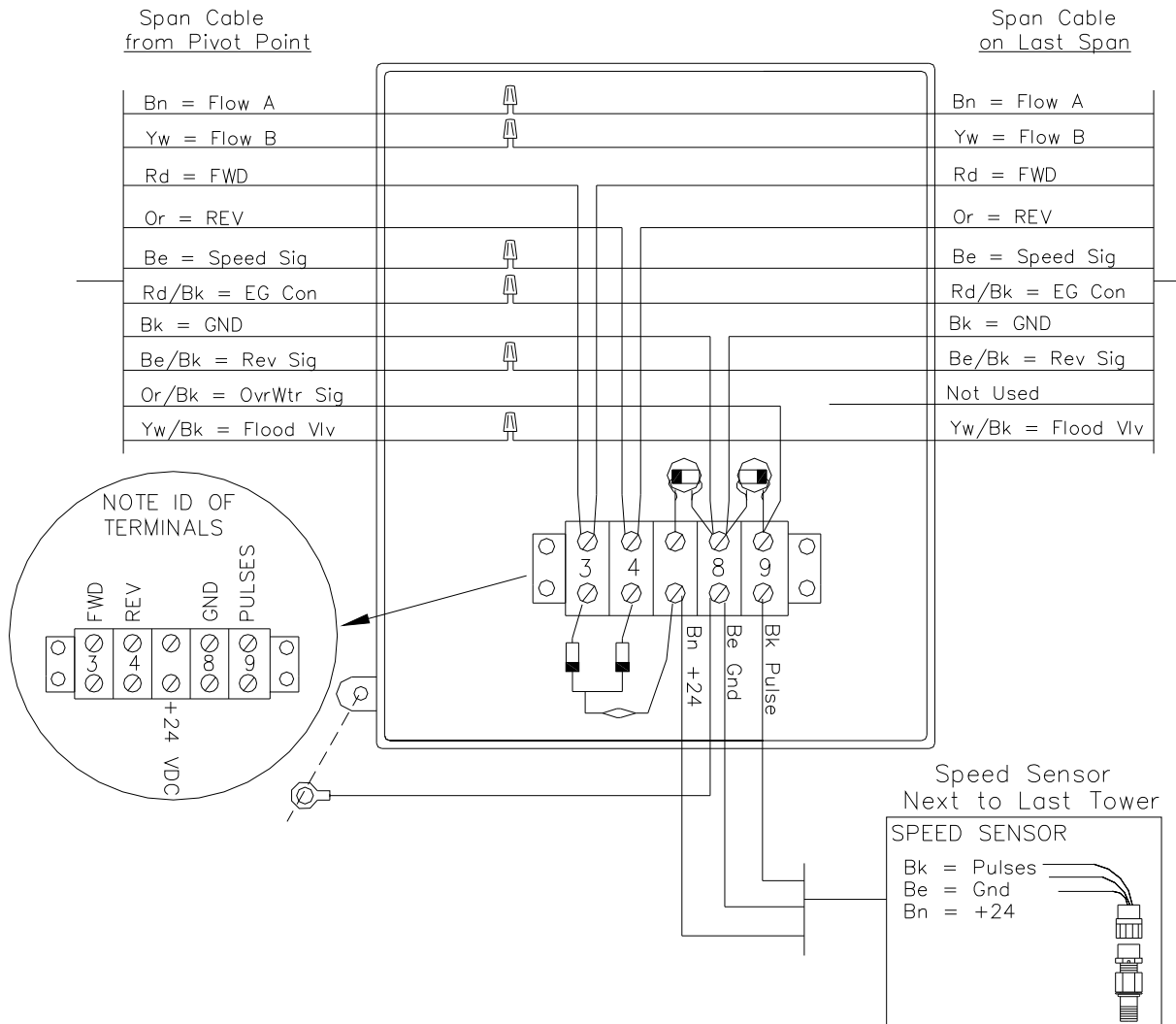


Figure 49. Wiring of Overwater Junction Box

Intermediate Auto Reverse Wiring (Figure 49a) (See CD91100)

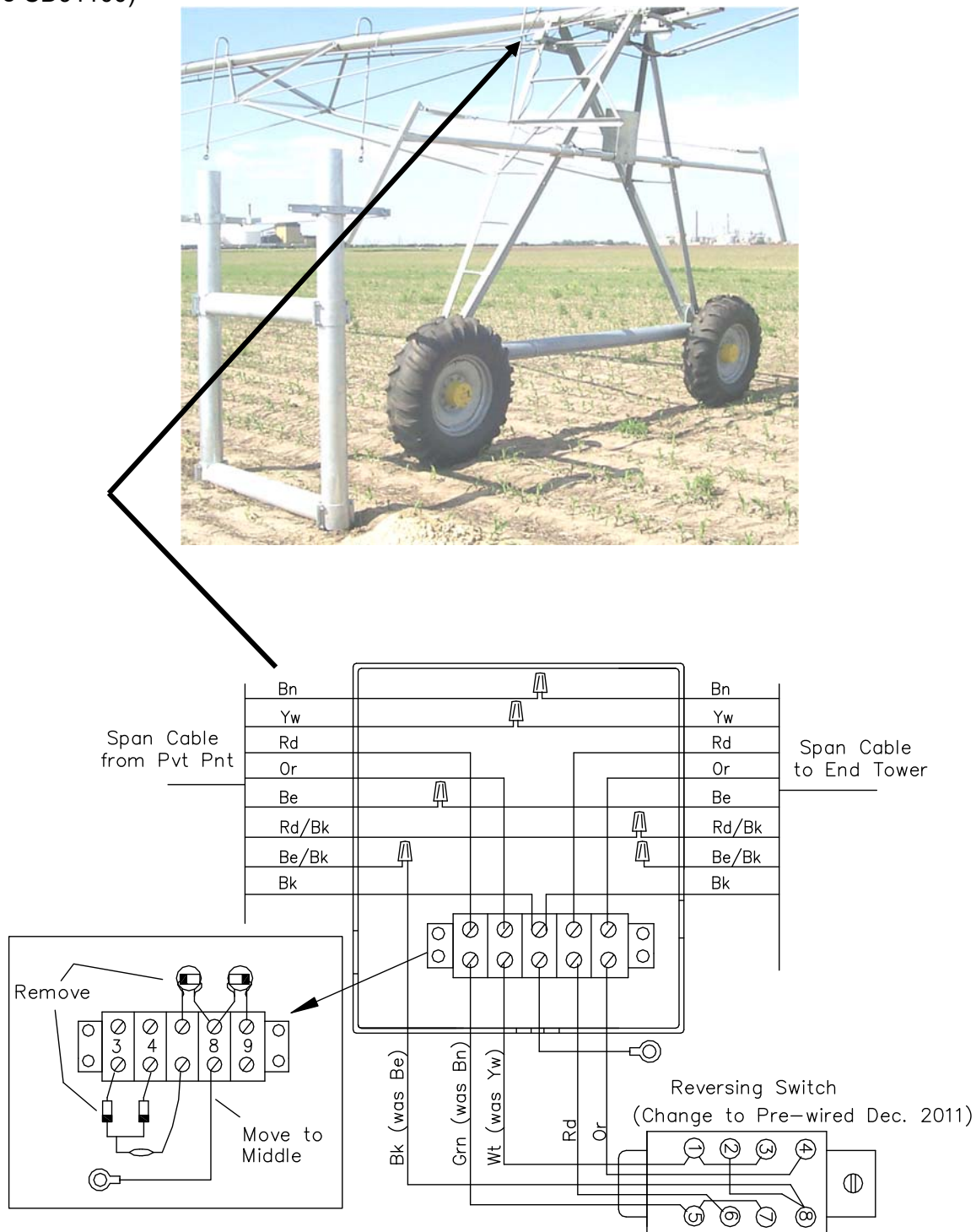


Figure 49a. Wiring Intermediate Auto Reverse Junction Box

- Use Overwater Junction Box, drill one more hole.



END TOWER CONTROL WIRING - FULL CIRCLE

Dir. Con. 1/08 -

Rev (+24)

Fwd (+24)

NOTE:

Silver Cartridge = Top REV, Bott FWD. (1/08 -)

Gold Cartridge = Top FWD, Bott REV. (1/99 - 12/07)

Gn Wire Disconnected (June 2020)

To Mounting Bolt with Star Washers

END GUN 1

END GUN 2

END GUN 2 Same as End Gun 1

END GUN BOOSTER

Wtr. Press. Sw. (5 PSI)

EG Con

EG Coil (+24)

EG Bstr Coil (+24)

Bk = Pulses

Be = Gnd

Bn = +24

SPEED SENSOR

GPS ANTENNA

Red 20 VDC

Bk Gnd

Be Gnd

OR/WT Gnd

Wt Gnd

Gray TXA

Wht/Rd TXB

Span AntL RX TX

Span AntL RX TX

Wht TXB

Gray TXA

SPAN CABLE

END TOWER CONTROL WIRING WITH AUTO REVERSE

Dir. Con. 1/08 -

Rev (+24)

Fwd (+24)

NOTE:

Silver Cartridge = Top REV, Bott FWD. (1/08 -)

Gold Cartridge = Top FWD, Bott REV. (1/99 - 12/07)

Gn Wire Disconnected (June 2020)

To Mounting Bolt with Star Washers

EG 1

EG 2

EG WIRING SEE ABOVE

SPEED SENSOR

Bk = Pulses

Be = Gnd

Bn = +24

Rev Switch (Yellow Cord)

Wt (was Yw)

Gn (was Bn)

Bk (was Be)

Or

EG 2

Yw/Bk

SPAN CABLE

CD90344

See previous diagrams

6/16/14 - MFC Adj/GPS Comm

3/07/18 - Added EG 2 for PPT

6/8/20 - Disconnected Green Wire

53

End Tower Wiring with Manifold System (Figure 51, 1/02 to 6/04)

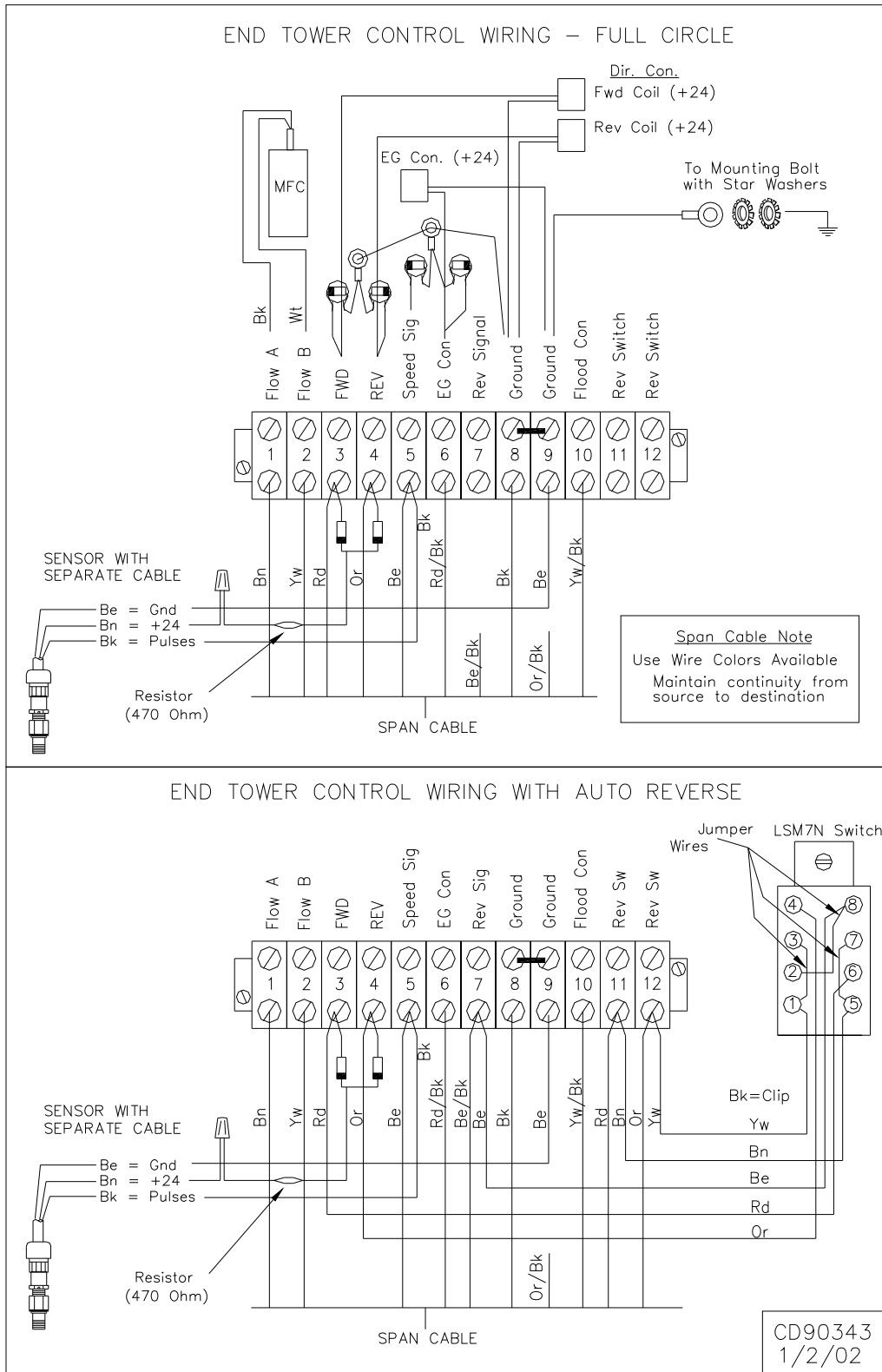


Figure 51. End Tower Control Wiring (1/02 to 6/04).

End Tower Wiring with Small Junction Box (Figure 51a, prior to 1/02)

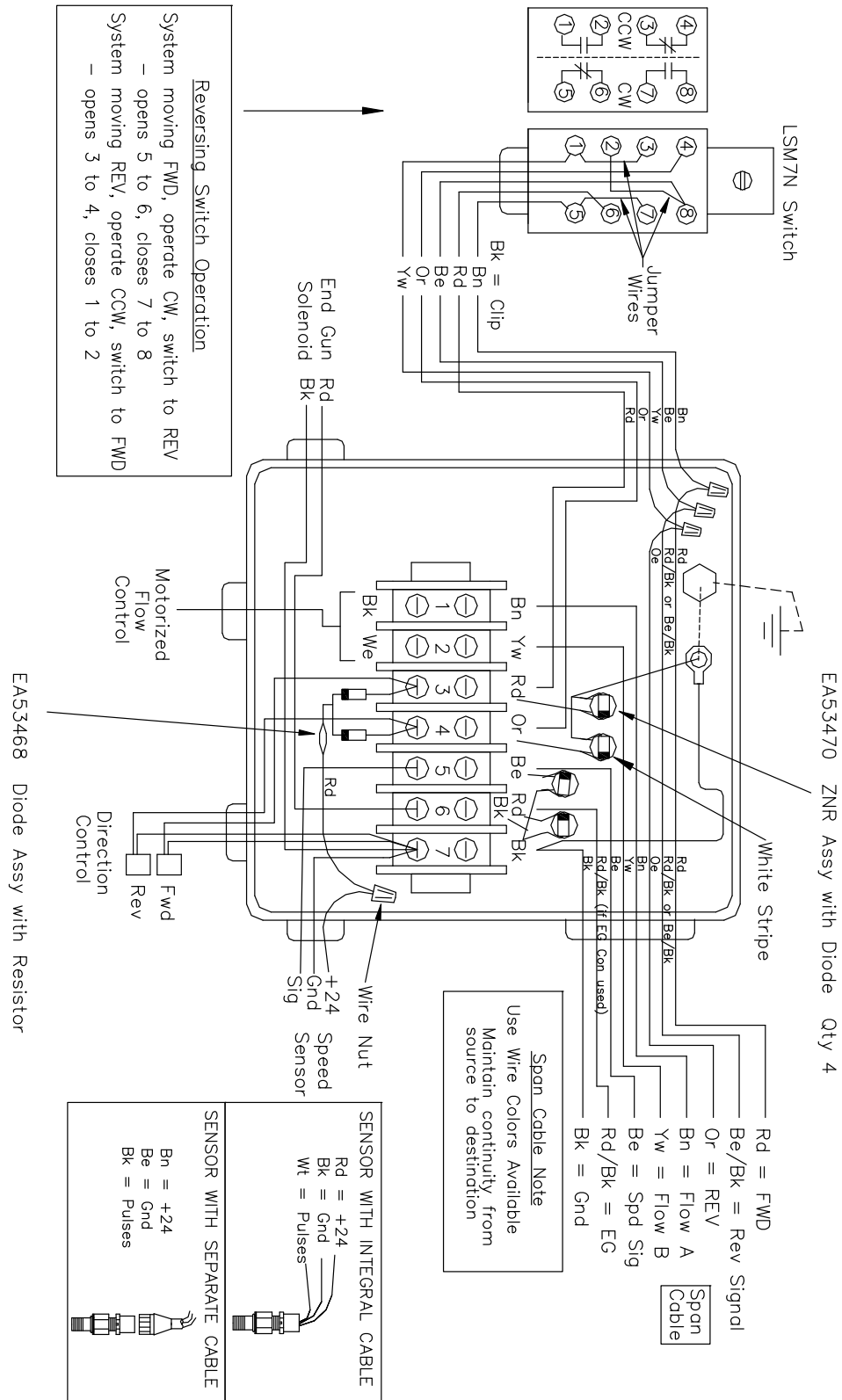


Figure 51a. End Tower Control Wiring prior to 1/02

Reversing Switch Schematic

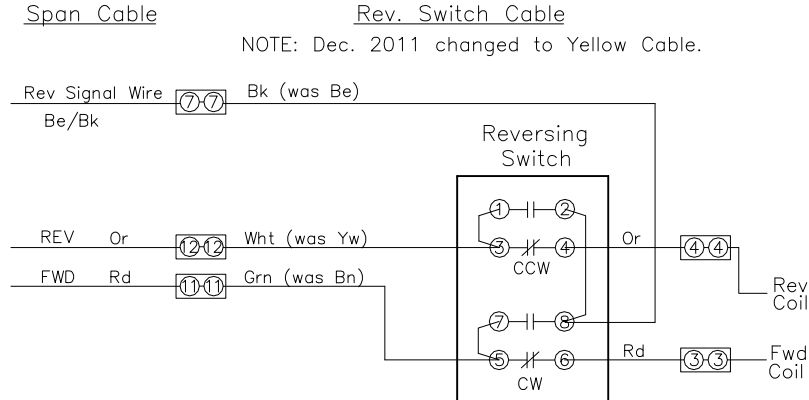


Figure 52. Auto Reverse Switch Schematic.

End Gun Booster Control Wiring - Last Tower (Figure 53)

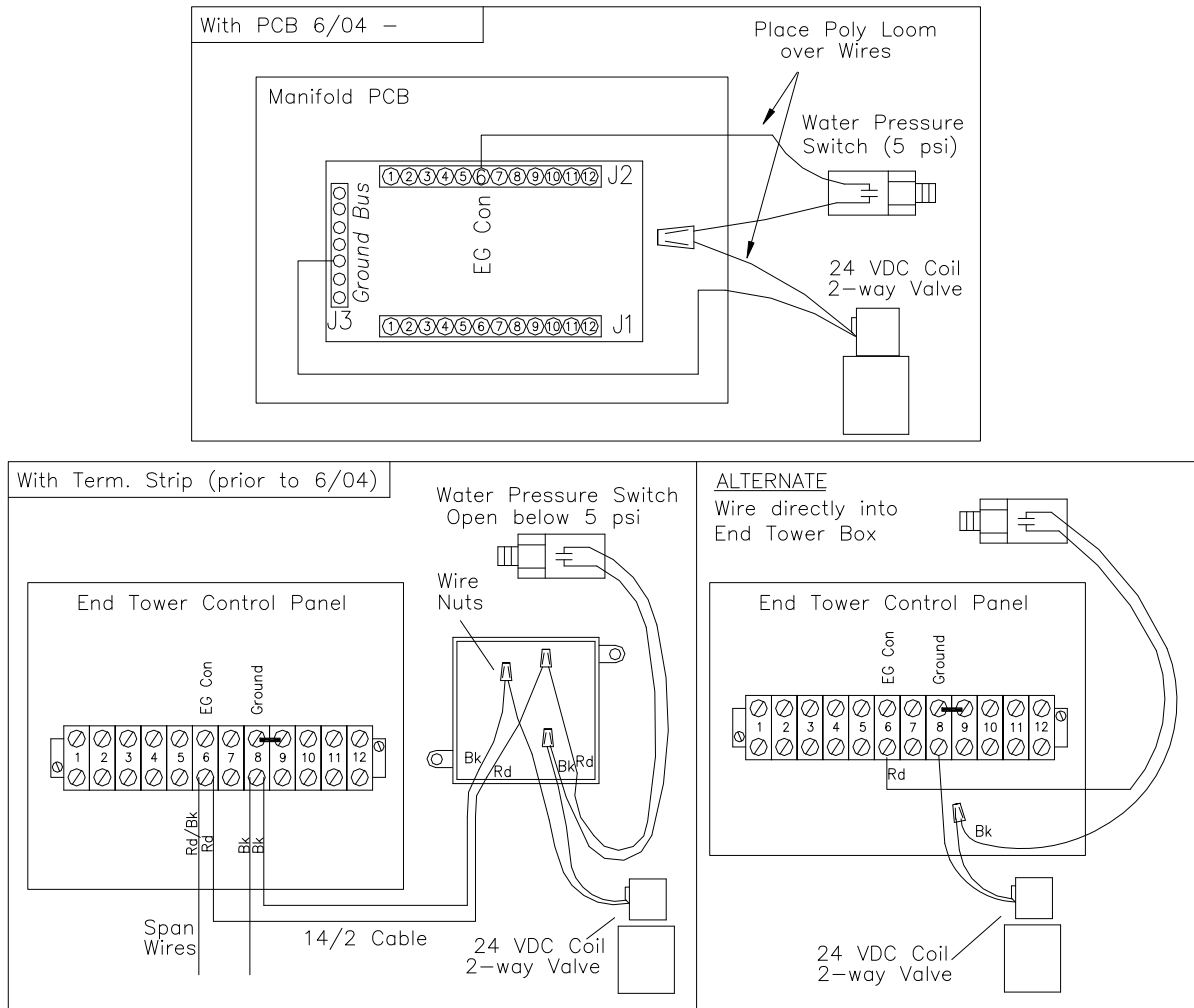


Figure 53. Wiring of End Gun Booster Control Solenoid

Spray Protective Coating on Wiring Connections

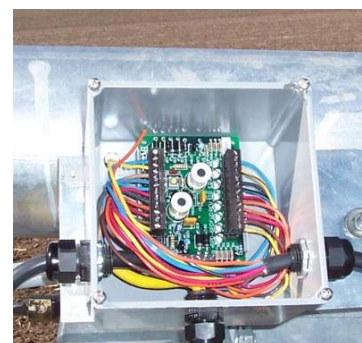
To protect against corrosion, spray coating on all wiring connections.

- Do NOT spray on Rings/Brushes in Collector Ring.

T-L Part No. EH53613 Weather Guard Spray

Wiring Connections that should be sprayed with Weather Guard:

- Wiring Connections & Battery Terminals on engine.
- Wiring inside of the Precision Point Control Panel.
- Wiring to Brushes in Collector Ring (Do NOT Spray the Rings.)
- Wrap Cable Socket and Plug connections.
- Overwater Junction Box Circuit Board Wiring.
- End Tower Manifold Circuit Board Wiring.



Initial Check to Precision Point Control III Panel Before Switching Power On

1st **Before** powering on the PPC III, check Terminals 5 (gnd) and 7 (+28) of 7 term block in PPC III panel. You should be measuring Volts DC.

Engine drives - with 117 pushed in, +28 VDC

Electric drives - with electric motor running, approx. +28 VDC

2nd Jumper terminals 1 to 2 of the 7 term. block in PPC III, pumping unit should safety out and stop.

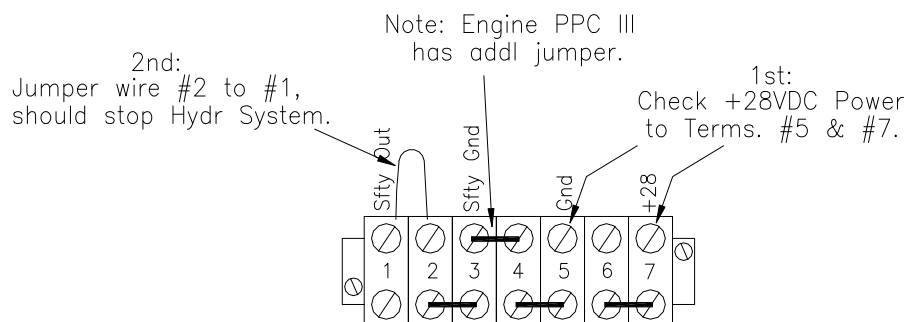


Figure 53.1 Initial Checks Before Switching Power "On" to PPC III.

PPC III Control Panel Keys and Switches Overview

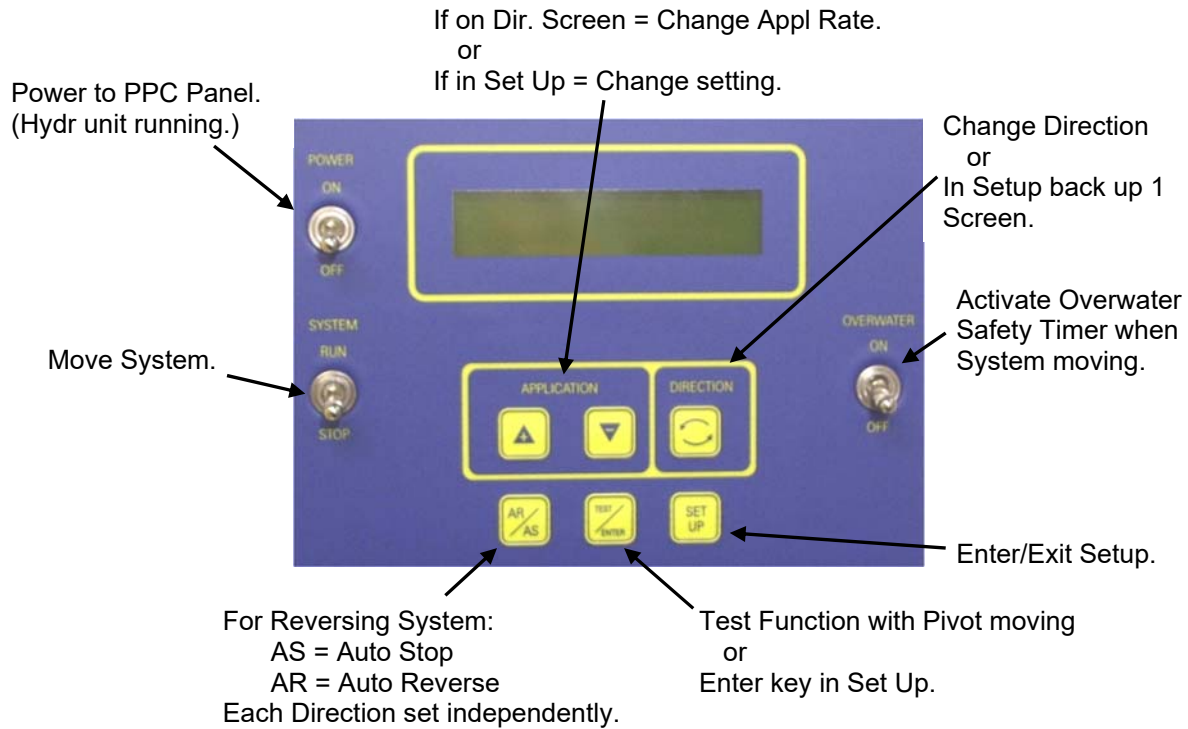


Figure 54. Display, Keypad and Toggle Switch Operation

Main Direction Screen Overview

The "Direction Screen" is considered the "Home" screen. The upper left will show either FWD or REV.
- Direction, Application and AR/AS can be changed on this screen.

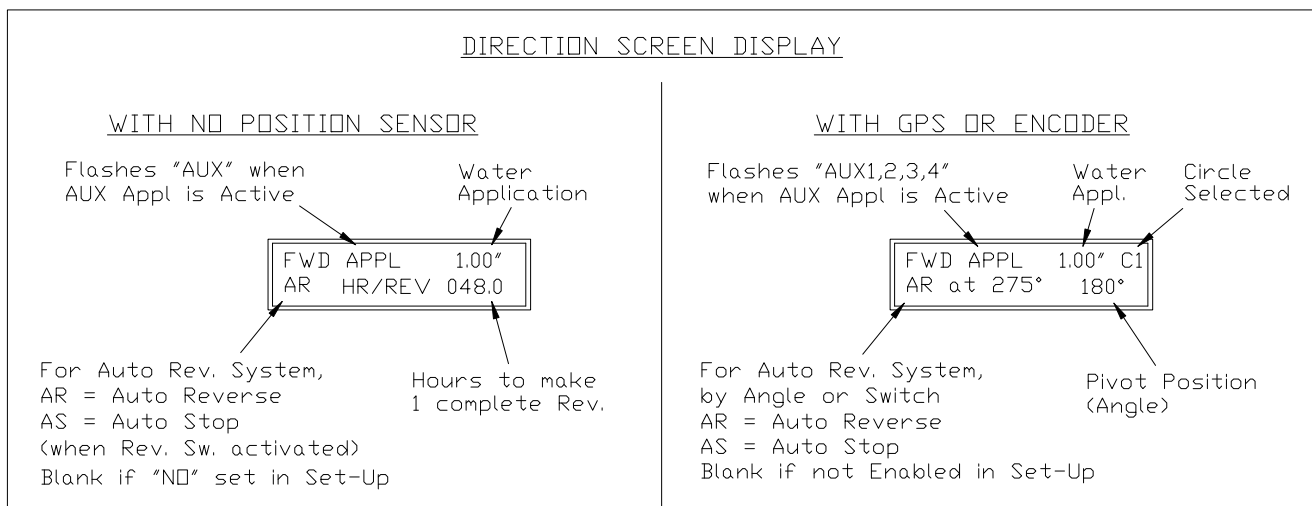


Figure 54.1 Main Direction Screen

Board Power-Up (Figure 55)

When the Precision Point Control III is powered up, the Title Screen will be displayed with firmware version. The screen displayed after the Title Screen will depend upon the following:

- Is this the first time power up after board programming? If yes, go directly to Set-Up.
- Was there a safety?
 - If Yes, display the Safety message and <Press Enter>, then continue below.
 - If No, continue below.
- The board will then display the speed and direction last saved in the system memory.
 - If the Run/Stop switch is in the "Run" position on power up, the system will begin moving the direction and speed saved last.
- From the Direction Screen, the Set-up and Test Screens can be accessed. If an Auto Reverse Delay has been set, the Direction Screen will switch to a timer if AR is selected for that direction and the Auto Reverse Switch is activated.

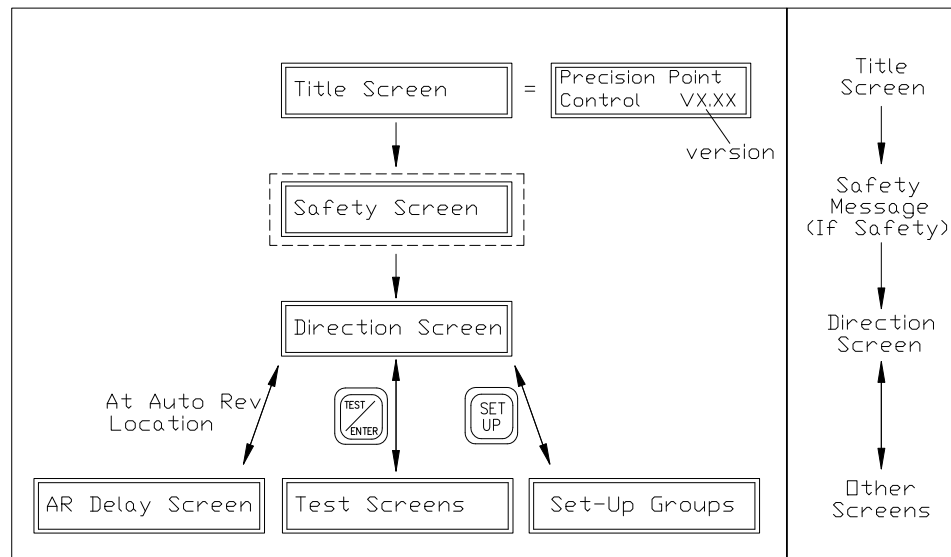


Figure 55. Precision Point Panel Operation.

Firmware Version & Features

The "Title Screen" showing the Firmware Version is displayed for roughly 3 seconds at power up.

V 6.58 = 2015-2021



- Application set in 0.05" increments
- GPS Com Lost Initial Delay = 20 secs
- Speed checked every 20 secs for 2 minutes, then every 10 minutes.

V 7.04 = 2021 & after



- Application set in 0.01" increments
- GPS Com Lost Initial Delay = 40 secs
- Set vs. Actual Speed continually checked every 20 secs.
- Com Port 2 can be used for Position Sensor if Com Port 1 has failed.

Backlit Display

The display will be backlit and remain backlit for 2 minutes with no keypad activity. Press any key to turn on backlighting after it has timed out.



Entering Set-up Information (Figure 56, 57)

SYSTEM Run/Stop Switch:

- Must be in STOP to change "Position Sensor" and "System Set Up" settings.
- If in RUN, "Position Sensor" and "System Set Up" information can be reviewed, not changed.
- Can be in STOP or RUN to change "End Gun Areas", "Auto Stop/Reverse" or "Aux Appl" settings.

"SET UP" Key:

- Press once to access Setup Groups. Press "Set-up" again to exit Set Up.

"DIRECTION" Key:

- When in a Setup Group, press once to back up one screen.

"Test/ENTER" Key: Press key to accept the change. If not, the item will not be changed.

Set Up Keys

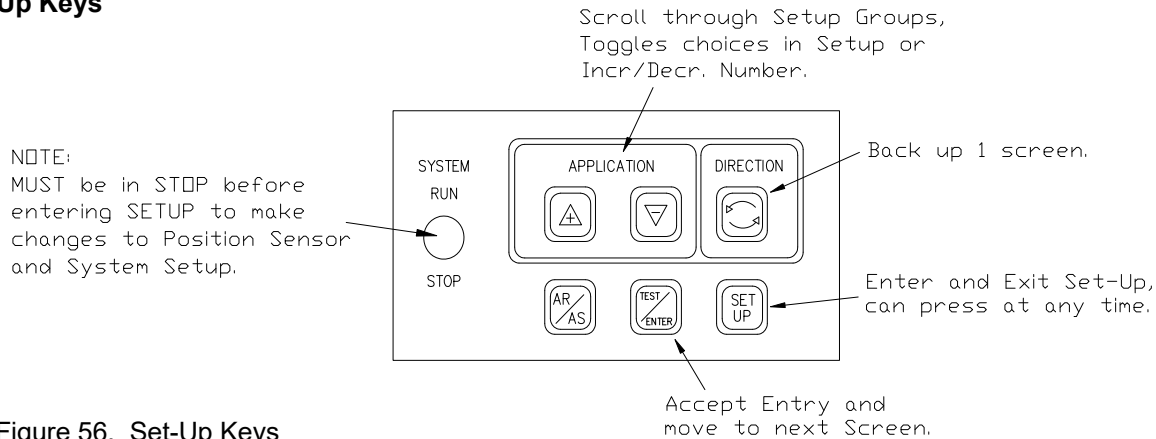


Figure 56. Set-Up Keys

Setup Groups

- Press Setup Key to enter/exit Set Up Groups.



- Use + / - Keys to change to a different Set Up Group.



- (Order of Setup Groups for 6.58 & 7.04 on)

SETUP GROUP System Setup	SETUP GROUP End Gun Areas	SETUP GROUP Auto Stop/Reverse	SETUP GROUP Aux Appl	SETUP GROUP Position Sensor	SETUP GROUP Safety History
-----------------------------	------------------------------	----------------------------------	-------------------------	--------------------------------	-------------------------------

- Press Enter to access the Set Up Group settings.



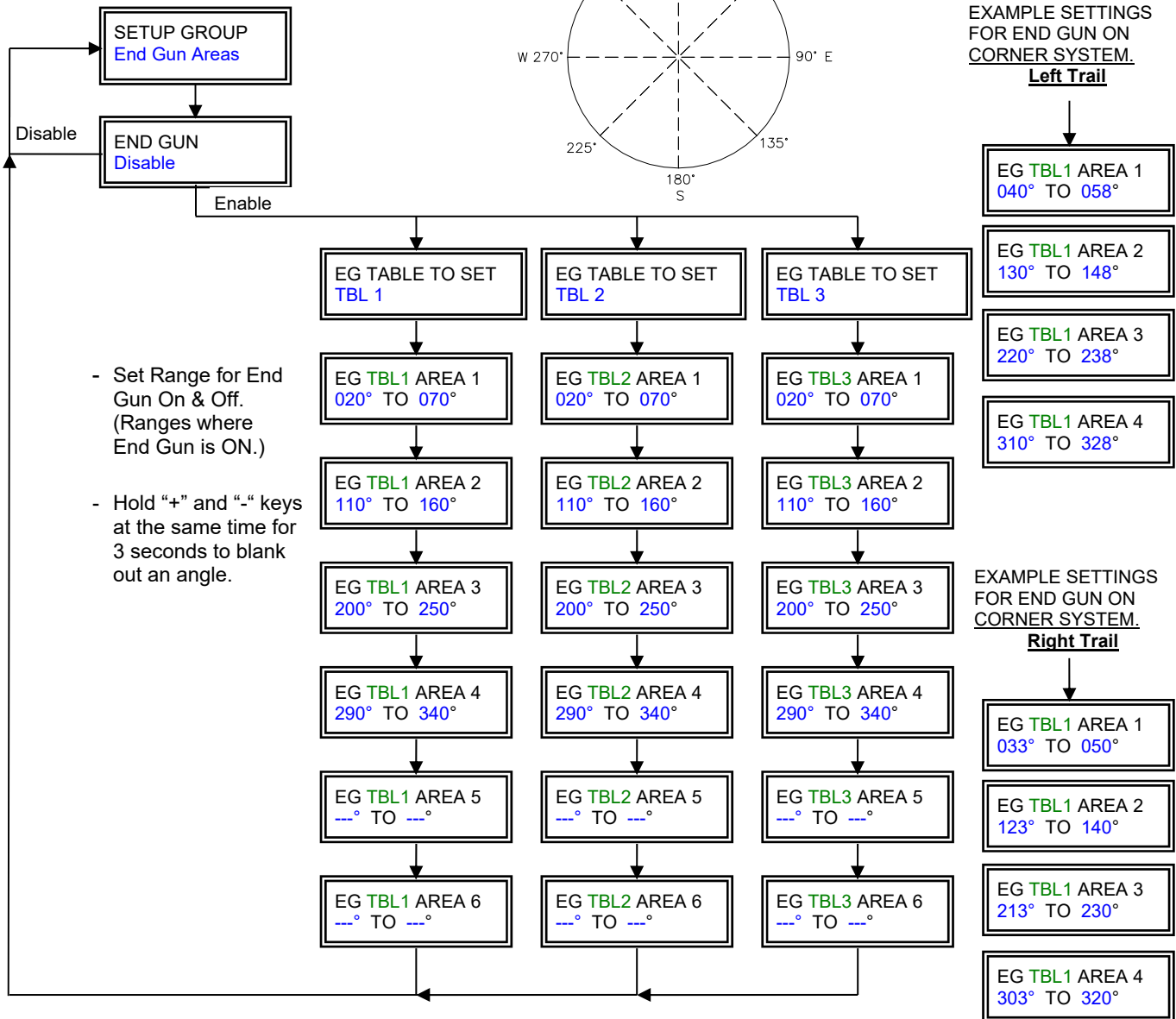


System SETUP Screens

		Description	Choices
If GPS Or Enc	CIRCLE TO USE Circle 1	Select Circle (corresponds to PP1, 2 or 3 and EG TBL1, 2 or 3)	1, 2, or 3 (1 is default, use 2 or 3 for towable pivots.)
No Pos Sensor, start here	UNITS English	Choose Units of display for System.	English Or Metric
	LAST WHEEL TRACK 1264 Ft	Distance to Last Tower Wheel Track.	101 to 3000 Ft 30 to 1000 Meters
	IRRIGATED RADIUS 1300 Ft	Distance to End of Overhang + EG Throw.	101 to 3200 Ft 30 to 1000 Meters
	WATER FLOW 0800 GPM	Water Flow through Center Pivot.	0 to 4000 GPM 1 to 252 litres/Sec
	LAST TOWER GEAR Planetary	Gear Type used On Last Tower	Worm or Planetary 68:1 or 24:1
	LAST TOWER TIRE 023.5 In	Rolling Radius of Last Tower Tire.	20.0"/50.8cm = Recap/11.2 24.0"/61.0cm = 16.9 x 24 23.5"/59.7cm = 14.9 x 24 27.0"/68.6cm = 11.2 x 38 (12.0 to 45.0")
If GPS or Enc	OVERWATER TIMER GPS	Select sensor to use for overwater timer.	If Pos. Sensor = None, this screen does not appear, and uses "Sensor Motor". GPS or Encoder if have Pos. Sensor
	OVERWATER TIME 10 Min	Time to wait for pulses or move 1 degree before safety.	1 to 60 min Defaults: Sensor Motor = 20 Min. (1-20) GPS = 60 Min., Encoder = 60 Min. (1-90)
	MAX SPD 0.20" 060 IPM	Max Speed Setting. (Adjust in 1 IPM increments)	6 to 300 IPM 15 to 761 cm/min
	SPD ADJ RATE MFC 0.018 sec/IPM	Multiplier to determine time for change when adjusting MFC	0.001 to 0.050 (default is 0.015)
If GPS or Enc	WRONG DIR SAFETY ENABLE	If Pos. Sensor used, the "Wrong Direction" Safety can be Disabled.	Enable or Disable
	ET SENSOR SAFETY 060 Sec	Set time with no ET Sensor pulses before safety.	30 to 120 sec (60 sec. default)
	ET SEN SPD CON Enable	ET Sensor Safety (also disables Speed Control). (If set to "Disable", then MFC will adjust for 45 secs, to slightly open.)	Enable or Disable
If GPS or Enc	COM LOOP TEST No	Test Communications? (If Position Sensor used.)	No or Yes (see Troubleshooting section.)
	RUN TIME 000000:00:00 HRS	Hours with "System" Toggle Sw. in Run Position	NO ADJUSTMENT ALLOWED Read out is HRS:MIN:SEC
	AgSense Link No	Is AgSense Precision Link wired into PPC III Panel?	No or Yes (Yes Enables Comm. with PPC III.)



End Gun Areas SETUP Flowchart



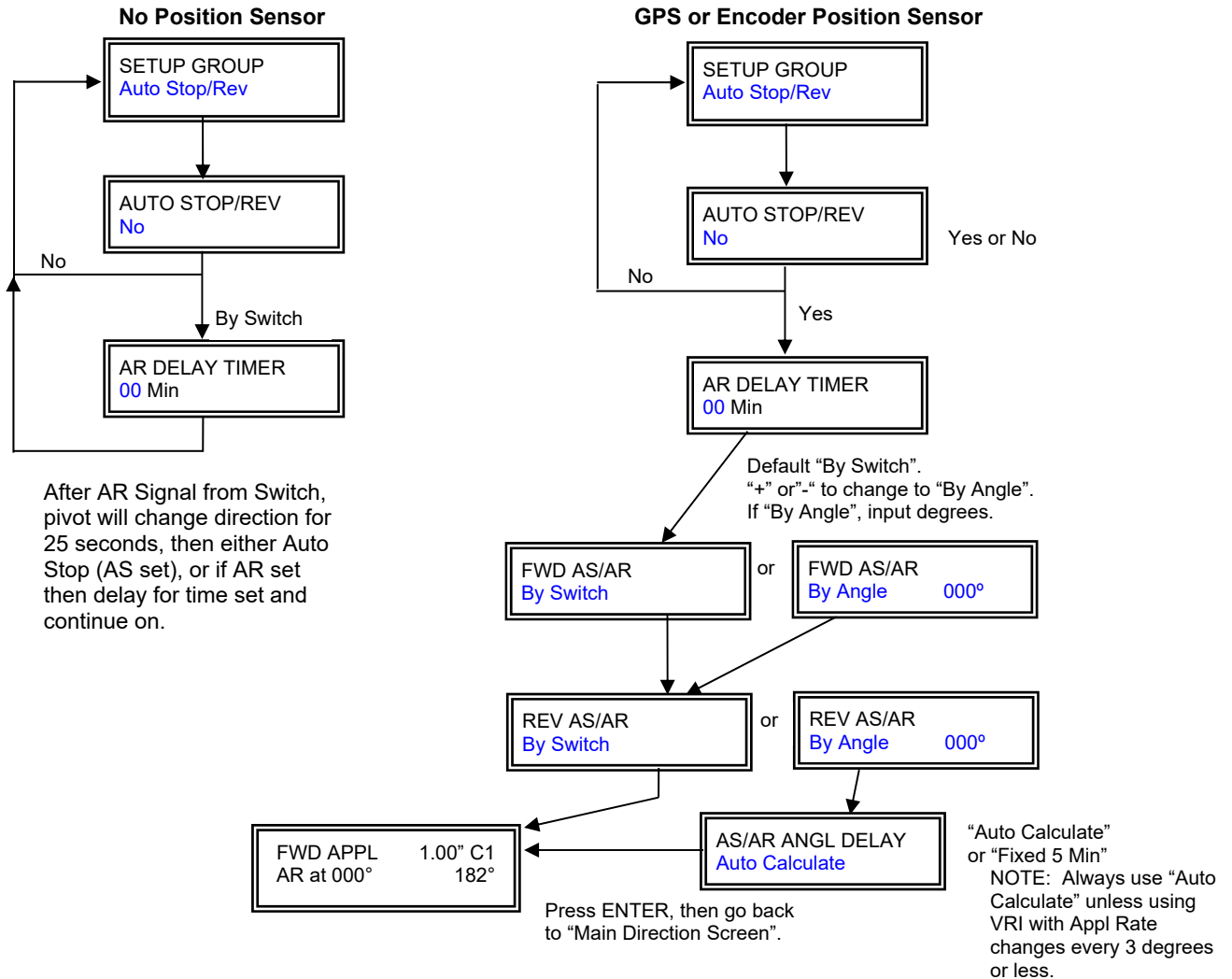
Important Notes about EG Tables:

- Use the + or – keys to change the areas when they are flashing.
- Use the Test/Enter key to advance. Use Direction Key to back up.
- Areas must be entered in “ascending” order.
- Area 1 can start at 000.
- Last Area can end at 000, however, and area cannot be programmed 000 to 000.
- Do NOT program an area across 000, example 290 to 100.
- Any areas not used should be blanked out (---°) by using the + or – keys. The dashes are located between 359 and 000.
 - Hold both “+” and “-” keys at the same time for 3 secs to blank the area.

Position Sensor “None”, can use FieldWise Remote GPS with Computer Controller for End Gun Control:

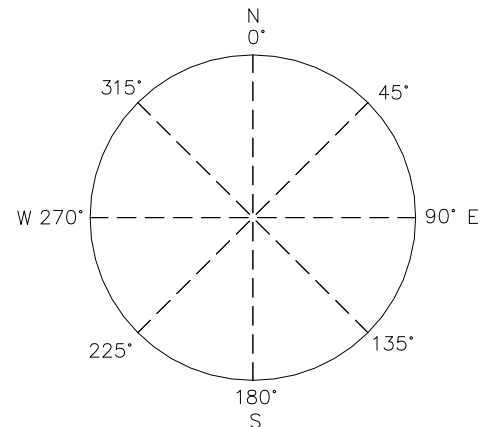
- With Position Sensor “None”, Enable the End Gun Areas to control the End Gun with the FieldWise Computer Controller with Remote GPS. EG Angles are set on the FieldWise Portal.

Auto Stop/Reverse SETUP Flowchart



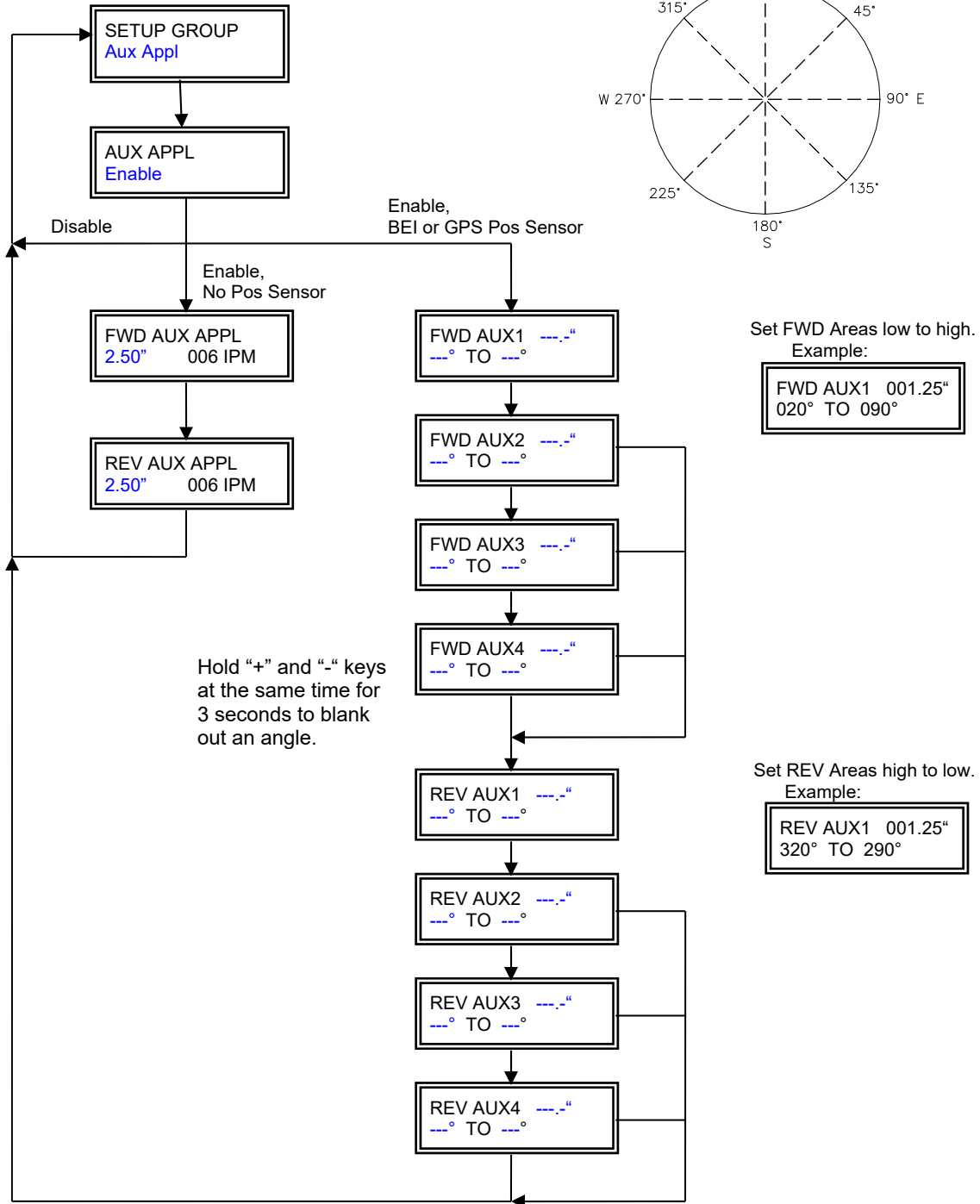
IMPORTANT NOTES for GPS & Encoder Pos. Sensor:

- **Ignore Angle Startup delay** on AS / AR by angle is calculated using distance to Last Wheel Track and SET IPM when System Toggle place into RUN.
- AS/AR angle ignored for **1st XX minutes calculated up to a max of 90.**
- "AS/AR Angle Delay" timer shown on Test Screen.
- Do **NOT** program the system to AS or AR by an Angle **within 3 degrees** of the current position **when starting.**
- Set AS/AR ANGL DELAY to "Auto Calculate" unless using VRI with Appl Rate changes every 3 degrees or less.





Aux Appl SETUP Flowchart



Important Notes about Auxiliary Application Tables:

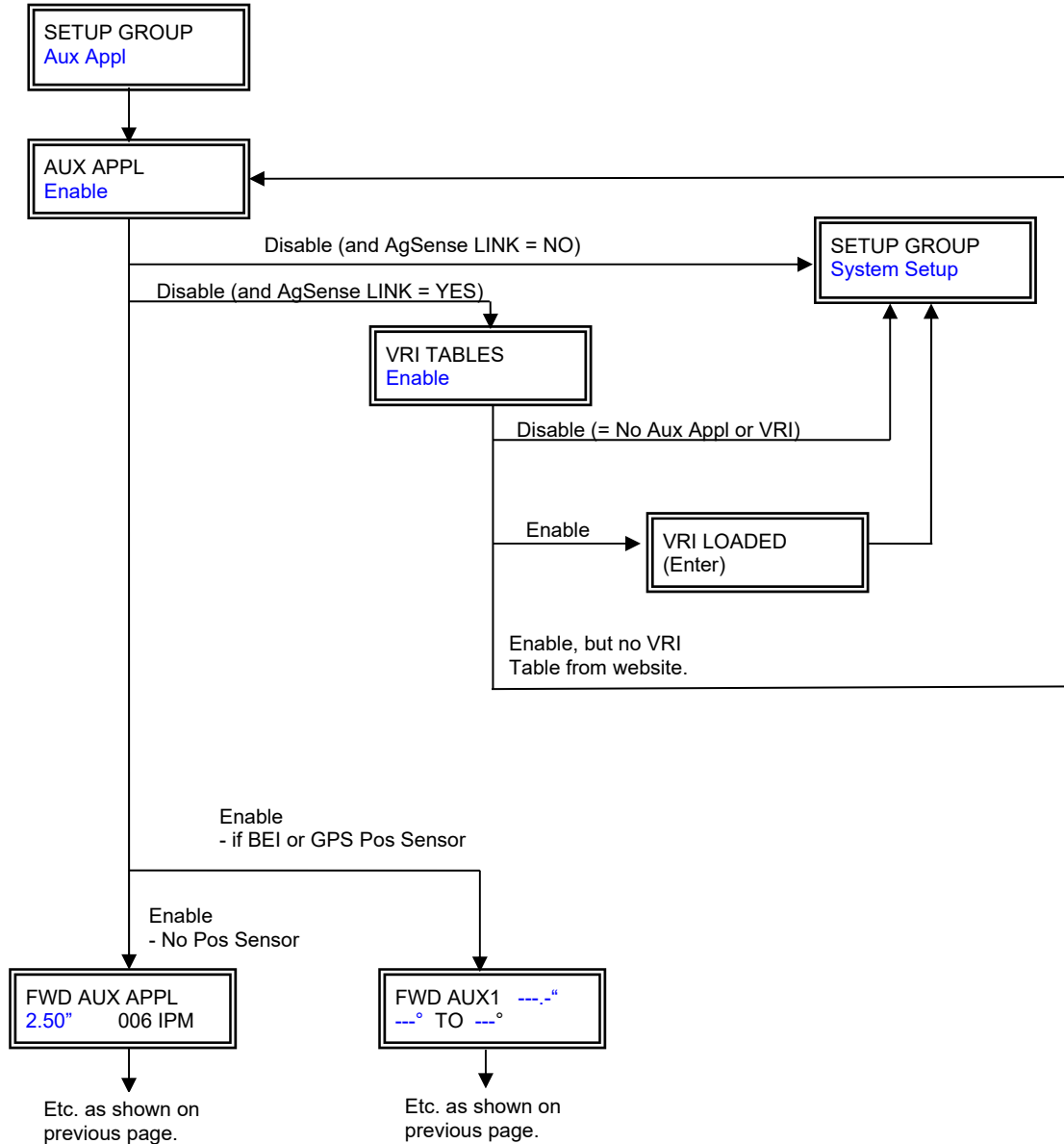
- FWD Areas must be entered in "ascending" order. (start at 000 and work CW)
- REV Areas must be entered in "descending" order. (start at 000, then 359 and work CCW)
- Do NOT program an area across 000, example 290 to 100.
- Areas following a blank area will not be shown for that direction:
Example: FWD AUX1 and AUX2 have entries, 3 does not, so 4 will not be shown.
- Blank out areas not used. Hold both "+" and "-" keys at the same time for 3 secs.
or scroll angle with "+" or "-" key to blank located between 359 and 000.



Aux Appl SETUP Flowchart – VRI with Precision Link

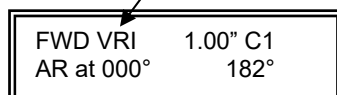
Applies only if AgSense Link set to YES and there is a Precision Link wired into the PPC III Panel.

The Precision Link adds the capability of running a VRI (Variable Rate Irrigation) Table. The VRI Table is set on the AgSense website, with anywhere from 5 to 120 steps. Each step is an area where the application rate is different.



Main Direction Screen if VRI Table being used:

- VRI Table set at AgSense Website.
- Flash VRI with APPL when VRI TABLE is Enabled by PPC III.
- whatever angle it is at, use the application set in the VRI Table for that step.

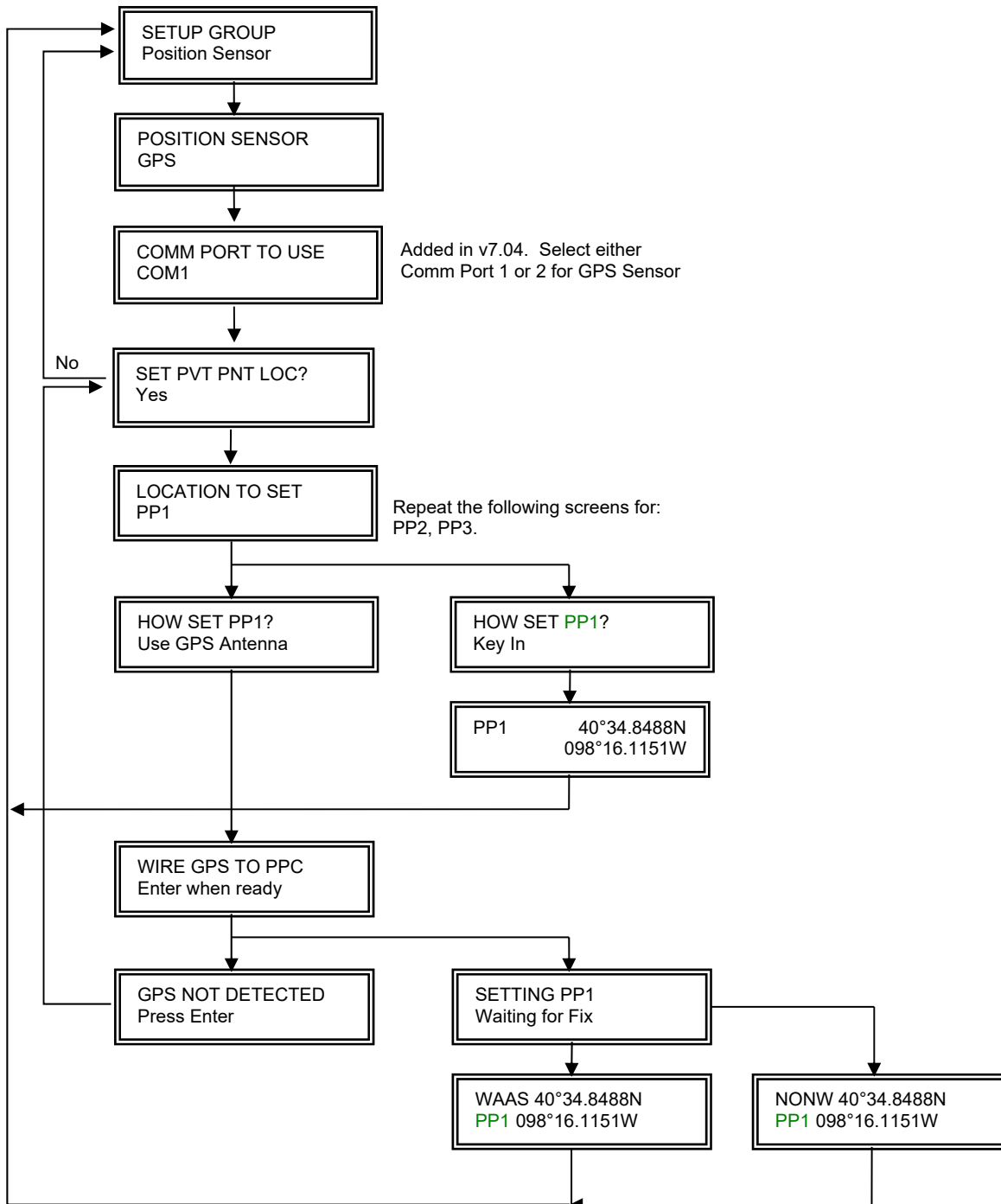




Position Sensor SETUP Flowchart

Position Sensor = GPS

(Position Sensor = None or Encoder, see next page)





Position Sensor SETUP Flowchart

Position Sensor = None
Position Sensor = BEI Encoder

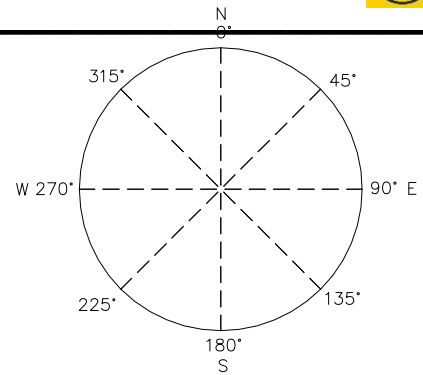
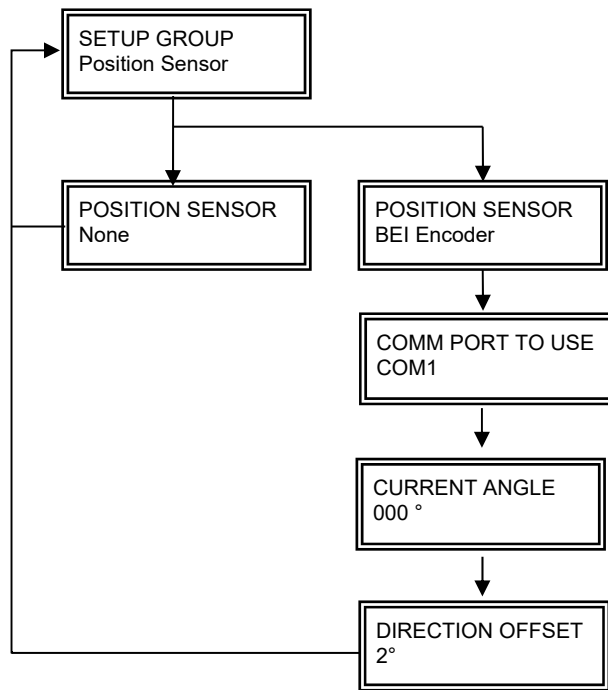


Figure 56.1. Compass Degrees.

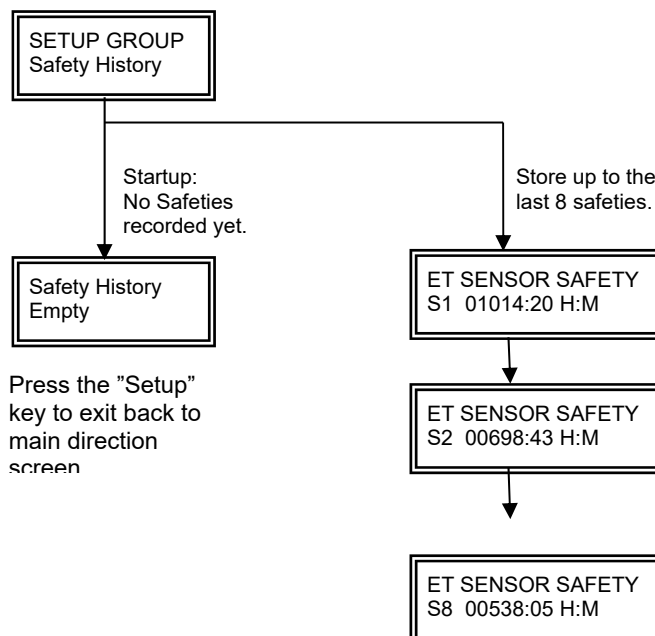
Added in v7.04. Select either Comm Port 1 or 2 for BEI Encoder.

Enter actual angle of pivot spans

Set Pivot Direction Offset - Compensates for the difference of the end tower position between FWD and REV directions. When the system changes direction and moves 2 degrees in that direction, the angle is changed by the value of the offset.

Safety History Flowchart

Review last 8 safety shutdowns. Total "Run Time" from System Setup is logged with the Safety.



Press the "Setup" key to exit back to main direction screen

Use "+" or "-" keys to scroll back and forth through the screens.

Press the "Setup" key to exit back to Main Direction screen.



TEST Screens Flowchart

(Use “+” or “-” keys to scroll back and forth through screens.)

No Position Sensor

SET IPM	038	+
ACT IPM	036	

OVRWTR SENSOR	*
ET SENSOR	*

CURRENT RUN TIME	000:00:00
------------------	-----------

CURRENT RUN TIME:
The time since the Run-Stop System
Switch has been placed in RUN.

Encoder Position Sensor

SET IPM	038	+
ACT IPM	036	

OVERWATER	60 min
ET SENSOR	*

HR/REV 128.2	
EG TBL1-3 ON	

CURRENT RUN TIME	000:00:00
------------------	-----------

GPS Position Sensor

SET IPM	038	+
ACT IPM	036	

OVERWATER	60 min
ET SENSOR	*

HR/REV 128.2	
EG TBL1-3 ON	

CURRENT RUN TIME	000:00:00
------------------	-----------

If Auto Stop or Auto Reverse by Angle set, show ignore angle delay
time that the Auto Stop or Auto Reverse action will be ignored.

AS/AR ANGL DELAY	13:22 mm:ss
------------------	-------------

AS/AR ANGL DELAY	13:22 mm:ss
------------------	-------------

Show communication to Position Sensor. If AgSense Precision Link
connected, show communication to it. Asterisk flashing = good
comm. Asterisks may not flash at the same time.

COM TO POS SNS	*
COM TO PL	*

COM TO POS SNS	*
COM TO PL	*

PP1	40°34.8488N
	098°16.1151W

ET1	40°34.8488N
	098°16.1151W

No. of Satellites	10
WAAS	Yes

Est. Speed from GPS:
Current Time = 3:00 min, 1st display
- then every 2 minutes after that.

ESTIMATED SPEED	
FROM GPS	000 IPM

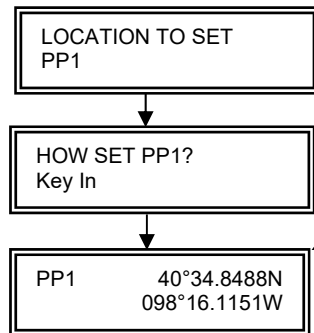
GPS Distance
- Calculated distance of Pivot Point to
End Tower.
- Helpful in determining if Pivot Point
lat-long is set correctly.
- Will show “ERR” if greater than 9999.

GPS DISTANCE	
PP1 ET1	1264ft

GPS Position Sensor, Set Pivot Point Location (PP1 or PP2 or PP3)

Key In

If the operator has access to a handheld GPS unit, then stand at the Pivot Point with it and determine the GPS coordinates. Manually enter these for either PP1 (or PP2 or PP3).



IMPORTANT: GPS Format:

DD°MM.MMMM

(Degrees, Minutes, Decimal Minutes)

Use the “+” and “-” keys to change each number or direction, press Enter to advance to the next. Press Enter to the next screen with all entries complete to Save.

Use the GPS Antenna supplied with the system:

1. Assemble the GPS Mounting and Antenna. Temporarily clamp it to the Pivot Point as shown and wire it into the PPC III Panel as shown.

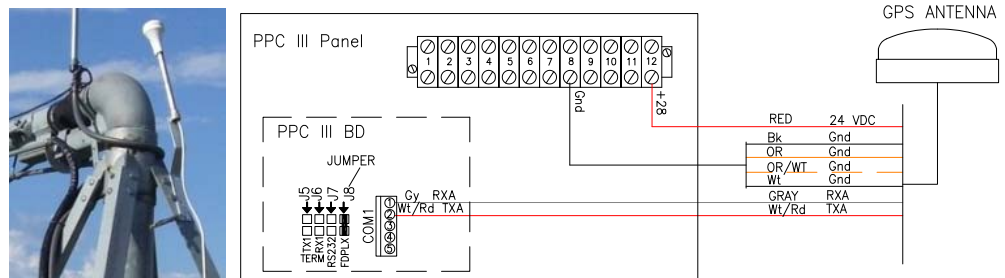
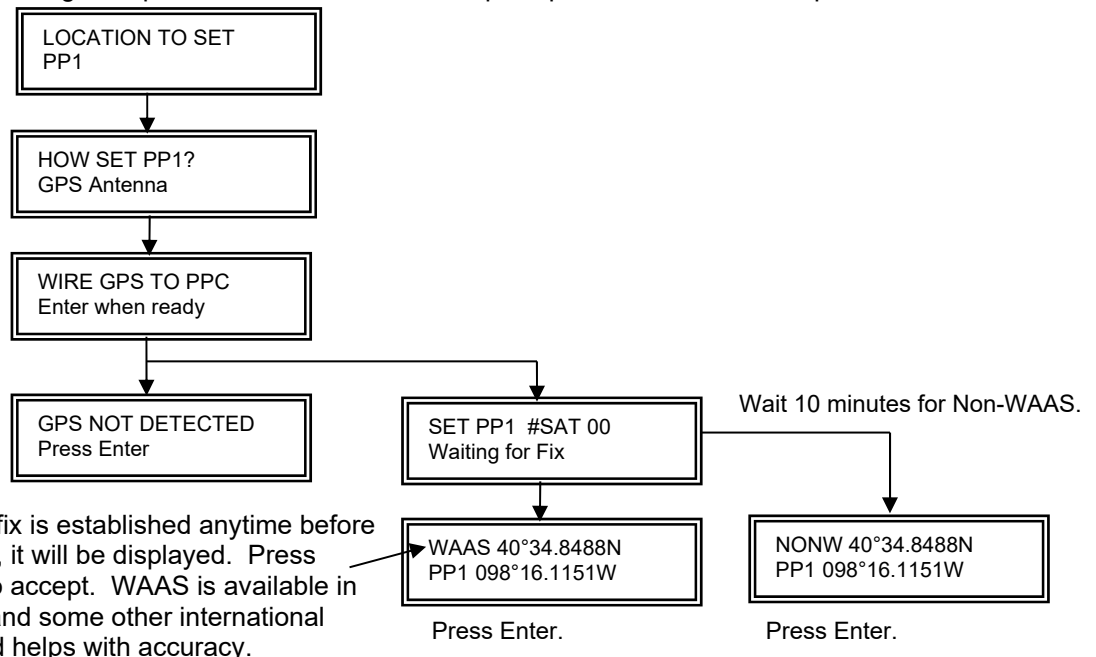


Figure 57. Wire GPS Antenna to PPC III Panel.

2. Step through the procedure, Press Enter after pivot point location comes up on the screen.





Setting Direction

Use the "DIRECTION" key to switch between Forward and Reverse.

DIRECTION



FWD APPL

REV APPL

Figure 58. Setting Direction.

Setting Application

Use the "+" and "-" keys to change Appl.
 - from Min. (6 in/min) to the Max. in Setup
 - See information at right for firmware v6.58 & v7.04.
 Application can be set for each Direction.

APPLICATION



V6.58 = 0.05" increments.

FWD 1.00"

REV 0.50"

V7.04 = 0.01" increments.

FWD 0.97"

REV 0.54"

Figure 58.1 Setting Application.

Determine Hours per Revolution for Set Application

No Position Sensor

Hr/Rev shown on Main Direction screen.
 - No angle shown.

FWD APPL	1.00"
AR	HR/REV 073.0

With Position Sensor

Main Direction screen shows angle.
 - Test Screen #3 shows HR/REV.

FWD APPL	1.00"C1
AR at 000°	182°



HR/REV 128.2
EG OFF

Automatic Speed Control

System Toggle to RUN, Speed Control activated:

- Checks "Actual Speed" for 20 seconds.
- Compares "Actual" to "Set".
- Makes Adjustment if needed (increase or decrease).
 - amount of adjustment time determined by "Spd Adj Rate MFC" in System Setup.
- Checks "Actual Speed" for 20 seconds again, etc.

6.58 - Continues for 2 minutes, then every 10 mins.

7.04 - Continually checks every 20 seconds.

Anytime Speed or Direction is changed while in "RUN", it will reset the Automatic Speed Control and start again.

- Checks "Actual Speed" for 20 seconds, etc.

1st Test Screen

SET IPM	038	+
ACT IPM	036	

+ or - displayed at 20 seconds.

2nd Test Screen

OVRWATER	60 min
ET SENSOR	*

Flashes with Sensor Pulses.



Overwater Timer Operation

The Overwater Timer is used to shut down the system if the End Tower gets stuck and does not move.
- System Toggle to RUN and Overwater Toggle to RUN, begin timing.

No Position Sensor

Uses Sensor Motor pulses from next to last tower.

- Overwater Safety occurs if set time exceeded without pulses from Sensor Motor.
- The board must receive 3 pulses in a 30 sec. time frame to reset the Overwater Timer.



Flashes with Sensor Pulses.

With Position Sensor

Uses GPS or Encoder Angle

- Overwater Safety occurs if set time exceeded without moving 1 degree.
- If the system moves 1 degree, then the timer is reset.



Auto Stop/Reverse Operation by Switch

The Pivot can be automatically reversed or stopped using an Auto Reverse Switch at the End Tower or at the Pivot Point. In Auto Stop/Reverse Setup, select "By Switch" for the Auto Stop/Reverse option.

Use the AR/AS key when on Main Direction screen:

AS = set to Auto Stop in that direction.

AR = set to Auto Reverse in that direction.

Auto Reverse Switch activated, end tower direction changes and no changes can be made for 25 seconds immediately following.

AR (Auto Reverse) = Once 25 seconds has elapsed the speed in the new direction will be displayed, and 20 seconds later, a speed change will occur.

AS (Auto Stop) = will cause the system to reverse for 25 seconds then shut down.

A delay time for Auto Reverse can be set, see "AR Delay Timer Operation" for explanation.

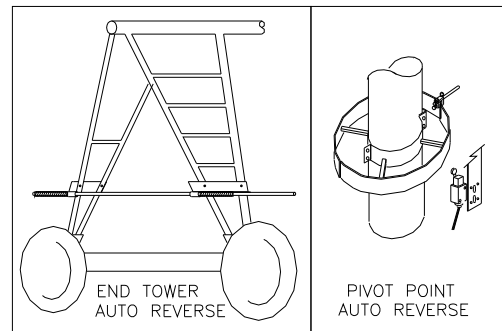


Fig. 61. AR Switch.



Auto Stop/Reverse Operation by Angle (with Position Sensor)

Use the Auto Stop/Reverse Setup Group, to set the Auto Stop/Reverse options.

A Delay Time can be set for end tower Auto Reverse.

- See "AR Delay Timer Operation" for explanation.

The Auto Stop/Reverse location (angle) can be specified in each direction or a Reverse Switch can be used for either direction.

- when Auto Stop or Auto Reverse is activated "by Angle", the action will occur immediately. (Will NOT move opposite direction for 25 seconds like "by Switch.")

Auto Stop/Reverse SETUP with Pos. Sensor:



Set time to delay if Auto Reversing.



Set activation method for each direction.





Use the Main Direction Screen to set whether the system will AS (Auto Stop) or AR (Auto Reverse) when activated.



FWD APPL	1.00" C1
AR by Switch	000°

REV APPL	0.75" C1
AS at 180°	000°

Main Direction Screen SHORTCUT:

- To quickly change the angle set for AS/AR, press and hold the AR/AS button from the main direction screen on which the angle appears.

Change the angle and press Enter to return back to the Main Direction Screen.

At Main Direction Screen.

REV APPL	0.75" C1
AS at 180°	000°

Press & Hold AR/AS 3 secs.



REV AS/AR	
By Angle	180°

Will NOT AS or AR while timing.

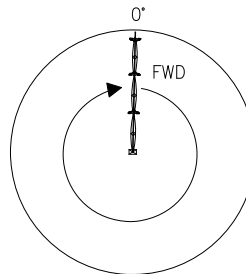
CHECK TIMER: View Test Screen. See below.

AS/AR ANGL DELAY
13:22 mm:ss

Auto Stop by Angle – Full Circle System

When setting Auto Stop by Angle on a Full Circle System, the same angle can be set as what appears on the Main Direction Screen, allowing a full circle operation.

The AS/AR Ignore Angle Delay Timer is used to keep the Auto Stop or Auto Reverse from functioning when first starting the system moving.



FWD APPL	0.75" C1
AS at 000°	000°

AS/AR ANGL DELAY
13:22 mm:ss

IMPORTANT NOTE for Auto Stop / Reverse by Angle – Ignore Angle Delay Timer:

- **Ignore Angle Startup Delay Time** on AS/AR by Angle is calculated using distance to Last Wheel Track and SET IPM when System Toggle place into RUN.
- The set AS/AR angle will be ignored (not activated) for the **1st XX minutes calculated up to a max of 90 minutes**. Auto Stop or Auto Reverse will **NOT** occur when the timer is active. Once timed down to 00:00, then AS or AR can occur.
- AS/AR Ignore Angle Delay timer shown on Test Screen.
- Do **NOT** program the AS or AR by an Angle **within 3 degrees** of the current position **when starting**.
 - The action will be ignored and will not happen.

Test Screen, use “+” to scroll to AS/AR ANGL DELAY timer screen.

- Countdown timer during which Auto Stop or Auto Reverse will be ignored. After the timer times out, Auto Stop or Auto Reverse by Angle will be active.

AS/AR ANGL DELAY
13:22 mm:ss



AR Delay Timer Operation

Allows the pivot to stop and water at Auto Reverse point.

- not used for "Auto Stop".

Auto Reverse delay time is entered in "Set-Up" from 0 to 45 minutes.

- With 0 selected, the pivot will auto reverse as described previously.

AR By Switch

- With an AR Delay Time of 1 to 45 minutes, when the Reversing Switch is activated, the pivot will move the opposite direction for 25 seconds, then stop with the AR Delay Timer Screen showing. After the delay time has elapsed, the pivot will begin moving again.

AR by Angle

- When the set angle is reached, the end tower will stop and the AR Delay Timer Screen will be showing. After the delay time has elapsed, the pivot will begin moving again the opposite direction.

If power is lost during timing, the system will immediately begin moving the other direction at the next power up, bypassing the rest of the delay time (with the Run/Stop in the Run position). Moving the Run/Stop switch into the "Stop" position during timing will also bypass the timer.

Auto Stop/Reverse Setup.



Delay Timer Screen.



NOTE: Do not press any keys while Delay Timer is active.

To bypass Delay Timer:

- System Toggle to Stop
- Power Off

AR by Switch Example:

- pivot moving Forward, set at 1.00" application
- AR switch is triggered,
- Pivot moves in Reverse for 25 seconds (at Forward Speed).
- stop and water in place for 15 minutes.
- After delay timing is finished, pivot will move Reverse, set at 0.75" application.
- AR switch is triggered on other side of field, the pivot will change direction.
- Pivot moves in Forward for 25 seconds (at Reverse Speed).
- pivot will "Auto Stop" and shut-down.

Example



System moving FWD, AR Switch activated



Reverse for 25 secs. at FWD Spd



Stop and Water in Place for 15 mins.



System moving REV, at REV Speed



AR Switch activated FWD for 20 secs. then shut-down.

Fig. 61.1. AR Example.

IMPORTANT NOTES About Auto Reverse Delay Timer Operation:

- With the Delay Timer activated, do not press any keys.
- To by-pass the Delay Timer, either toggle the Power to Off and back On, or toggle the Run/Stop switch to Stop.
- If the Precision Point Control III is a retrofit on an older system with OPL Style Gauges, the Auto Reverse Delay Timer will cause the left hand OPL Gauge to safety because flow is "0" when the system is stopped during the delay.
 - Replace OPL Gauges with 1500 psi Switch for bypass.
 - Overwater Timer on PPC will monitor end tower stuck.

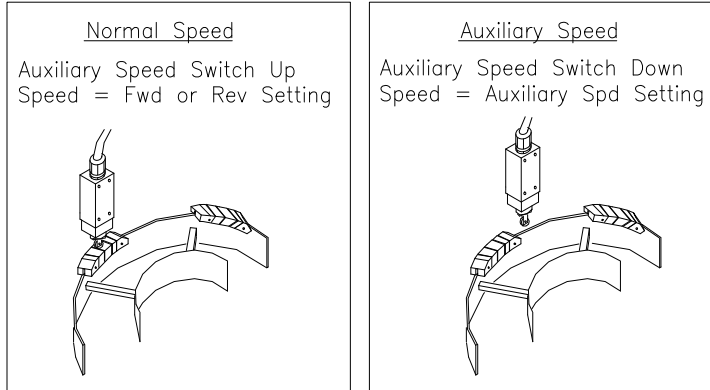


Auxiliary Application – No Position Sensor, Use Roller Switch

Set one Forward Aux. Appl. and one Reverse Aux. Appl.
 Use Roller Switch at Pivot Point.
 Can set several areas, but same Aux. Appl.
 Changes can be made while running.
 If “Disabled” in Setup, Aux. Application will be ignored.

Can set Fwd & Rev differently.

FWD AUX APPL 1.25" 018 IPM	REV AUX APPL 0.75" 030 IPM
-------------------------------	-------------------------------



Can set several areas, but same application rate in each area in the same direction.

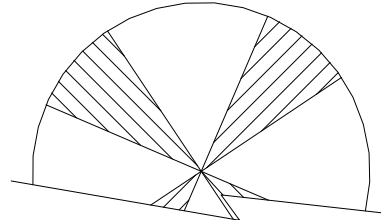


Fig. 62. AR Example Using Roller Switch and No Position Sensor.

Auxiliary Application With Position Sensor, Use GPS or Encoder

Set Forward Aux. Appl. and Reverse Aux. Appl.
 Set by location (degrees).
 Set up to 4 different areas in each direction.
 Changes can be made while running.
IMPORTANT:

- Fwd Application Areas, set low to high
 - start at lowest degree range, from 000
- Rev Application Areas, set high to low
 - start at highest degree range

If “Disabled” in Setup, Aux. Application will be ignored.

Can set Fwd & Rev differently.
 Up to 4 Aux. Appl Areas each direction.

FWD AUX1 001.25" 90° TO 180°	REV AUX1 001.00" 350° TO 320°
FWD AUX2 000.75" 270° TO 320°	REV AUX2 001.25" 270° TO 180°

AUX3 & AUX4 not shown.

Auxiliary Application With AgSense Precision Link – VRI Table

From the AgSense website, a VRI Table can be created with 5 to 120 steps.
 This table can then be loaded wirelessly into the PPC III and Enabled through the Aux Appl SETUP in the PPC III.

See previous “Aux Appl SETUP Flowchart – VRI with Precision Link.”
 Also see CD90441 – Precision Link Manual.

Step	Start	Stop	App Rate
1	0	45	1.00
2	45	90	0.85
3	90	135	0.95
4	135	180	0.90
5	180	190	0.95
6	190	200	0.80
7	200	210	1.00
8			



“ET SEN SPD CON” in System Set-Up (Set to Disable if ET Sensor Safety)

Set to “DISABLE” in System Setup.

This also disables the Automatic Speed Control.

Auto Stop/Reverse still functional.

Direction Screen will display “SPD DSABL”.

System Setup:

ET SEN SPD CON
DISABLED

MFC ADJUSTING
WAIT 45 SECONDS

Closes MFC, then slightly opens.

Manual Speed Adjustments can be made, use “+” or “-” keys on the Main Direction Screen.

System Toggle Switch must be in “RUN”.

FWD SPD DSABL
AR



FWD SPD INCR
AR



FWD SPD DECR
AR

Test Screen will show Actual Speed if pulses are present.

SET IPM DSABL
ACT IPM 036

Safety Messages

If a Safety from the Precision Point Control III (PPC) Panel has shut the system down, it will be displayed at start up the next time the Precision Point Control III Panel is powered up. Possible safety messages are:

Message:	Initial Delay	Run Delay	Description
Encodr Comm Lost	6 seconds	6 seconds	Encoder Position Sensor cannot be detected or communicated with (6 second delay).
GPS Comm Lost:	20 seconds 40 secs v7.04	5 minutes	GPS Antenna cannot be detected or communicated with. NOTE: Initial delay changed to 40 seconds in v7.04.
GPS Error:	10 minutes	6 minutes	Number of Satellites detected less than 3.
Wrong Direction (GPS):	10 minutes	7 minutes	System has moved 5 degrees in the opposite direction set.
Wrong Direction (Enc):	7 minutes	7 minutes	System has moved 5 degrees in the opposite direction set.
ET Sensor Safety:	user set in System Setup		PPC Board not receiving pulses from End Tower, reading speed as 0 IPM for time set (30 to 120 secs). System has stopped moving, speed sensor or cable defective, or wiring problem, or other.
Overwater Safety:	user set in System Setup		Used if the End Tower gets stuck: No Position Sensor - Pulses not received from Sensor Motor at next to end tower in time set GPS or Encoder - system has not moved 1 degree in the time set.
Auto Stop Safety:			AS Angle reached or End Tower Switch activated. If set “By Angle”, AS ignored for first XX minutes as calculated when system 1 st started moving.
GPS Not Set:			GPS Pivot Point location not set for selected Circle.

Operational Checks

Operate System and check the following:

Direction of system vs. display.

Auto Reverse and Auto Stop Operation.

Speed of system vs. display.

- use "Test" key on Keypad with system running.
- the "Last Tower Tire" rolling radius may need to be adjusted in "Set-Up".

Check Optional Auxiliary Speed Switch:

Roller Switch in UP position, on cam wafers, System is at FWD or REV Spd.

Roller Switch in DOWN position, off cam wafers, AUX SPD is activated.

Check Optional End Gun Switch Control Operation:

Roller Switch in UP position, on cam wafers, cylinder extends and ball valve closes.

Roller Switch in DOWN position, off cam wafers, cylinder retracts and ball valve opens.

Check Optional End Gun Booster Operation:

With Water

Roller Switch in UP position, on cam wafers, Booster Valve Closed, Motor not turning.

Roller Switch in DOWN position, Booster Valve Open, Motor Turns.

Without Water

Roller Switch in DOWN position, Booster Valve Closed, Motor not turning.

Set Physical Barricades at Reverse Points:

Physical Barricades must be installed at reversing points on part circle machines. This insures that the pivot will stop should the Auto Reverse circuit fail. As with any man-made machine, a failure can occur.

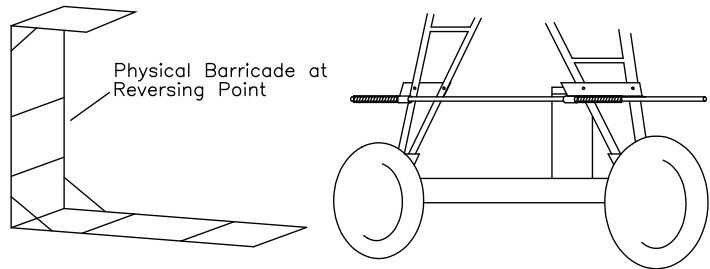


Figure 64. Physical Barricade

OPERATION



Table 1. Maximum End Tower Speed, Inches per Minute

Gear Type	Planetary				**
Pumping Unit	7 1/2 HP Electric				
	End Tower Tire Size				
	11.2 X 24	14.9 X 24	16.9 X 24	11.2 X 38	
4 SPANS	90	104	95	108	
8 SPANS	51	58	54	61	
12 SPANS	36	42	36	42	
16 SPANS	28	32	28	31	
20 SPANS	22	26	22	26	

End Gun Boosters N/A w/ 7½ HP

Gear Type	Worm				**
Pumping Unit	7 1/2 HP Electric				
	End Tower Tire Size				
	11.2 X 24	14.9 X 24	16.9 X 24	11.2 X 38	
4 SPANS	50	58	61	68	
8 SPANS	32	36	37	42	
12 SPANS	22	24	25	28	
16 SPANS	17	19	19	21	
20 SPANS	14	16	16	18	

End Gun Boosters N/A w/ 7½ HP

Gear Type	Planetary				**
Pumping Unit	10 HP Electric, Isuzu 3CB1				
	End Tower Tire Size				
	11.2 X 24	14.9 X 24	16.9 X 24	11.2 X 38	
4 SPANS	102	117	121	137	
8 SPANS	70	80	72	81	
12 SPANS	49	56	49	56	
16 SPANS	38	43	38	42	
20 SPANS	30	35	31	35	

End Gun Boosters N/R w/ 10 HP

Gear Type	Worm				**
Pumping Unit	10 HP Electric, Isuzu 3CB1				
	End Tower Tire Size				
	11.2 X 24	14.9 X 24	16.9 X 24	11.2 X 38	
4 SPANS	71	82	85	96	
8 SPANS	43	49	50	57	
12 SPANS	29	33	34	39	
16 SPANS	23	25	26	29	
20 SPANS	19	21	22	25	

End Gun Boosters N/R w/ 10 HP

Gear Type	Planetary				**
Pumping Unit	15 HP Electric, Isuzu 3CE1				
	End Tower Tire Size				
	11.2 X 24	14.9 X 24	16.9 X 24	11.2 X 38	
4 SPANS	102	117	121	137	
8 SPANS	102	117	108	122	
12 SPANS	70	80	77	85	
16 SPANS	56	65	57	64	
20 SPANS	47	54	46	53	

With 2 HP End Gun Booster = 66% of Table

Gear Type	Worm				**
Pumping Unit	15 HP Electric, Isuzu 3CE1				
	End Tower Tire Size				
	11.2 X 24	14.9 X 24	16.9 X 24	11.2 X 38	
4 SPANS	71	82	85	96	
8 SPANS	63	72	75	84	
12 SPANS	44	50	51	58	
16 SPANS	34	38	39	44	
20 SPANS	28	31	32	36	

With 2 HP End Gun Booster = 66% of Table

Gear Type	Planetary				**
Pumping Unit	20 HP Electric, Belt Drive 1800 RPM				
	End Tower Tire Size				
	11.2 X 24	14.9 X 24	16.9 X 24	11.2 X 38	
4 SPANS	102	117	121	137	
8 SPANS	102	117	121	137	
12 SPANS	90*	104*	94*	105*	
16 SPANS	70*	80*	72*	81*	
20 SPANS	56*	65*	57*	65*	

With 2 HP End Gun Booster = 75% of Table
With 5 HP End Gun Booster = 55% of Table

Gear Type	Worm				**
Pumping Unit	20 HP Electric, Belt Drive 1800 RPM				
	End Tower Tire Size				
	11.2 X 24	14.9 X 24	16.9 X 24	11.2 X 38	
4 SPANS	71	82	85	96	
8 SPANS	71	82	85	96	
12 SPANS	54	63	66	74	
16 SPANS	43	49	50	57	
20 SPANS	34	38	39	44	

With 2 HP End Gun Booster = 75% of Table
With 5 HP End Gun Booster = 55% of Table

(Table continued on next page)

Table 1. Maximum End Tower Speed, Inches per Minute (continued)

OPERATION



Gear Type	Planetary				**
Pumping Unit	25 HP Elec, Belt 2200 RPM, Isuzu 4LE				
	End Tower Tire Size				
	11.2 X 24	14.9 X 24	16.9 X 24	11.2 X 38	
4 SPANS	102	117	121	137	
8 SPANS	102	117	121	137	
12 SPANS	102*	117*	112*	127*	
16 SPANS	80*	93*	86*	97*	
20 SPANS	68*	78*	69*	78*	

With 2 HP End Gun Booster = 80% of Table

With 5 HP End Gun Booster = 65% of Table

Gear Type	Worm				**
Pumping Unit	25 HP Elec, Belt 2200 RPM, Isuzu 4LE				
	End Tower Tire Size				
	11.2 X 24	14.9 X 24	16.9 X 24	11.2 X 38	
4 SPANS	71	82	85	96	
8 SPANS	71	82	85	96	
12 SPANS	66	76	79	89	
16 SPANS	50	58	61	65	
20 SPANS	40	46	47	54	

With 2 HP End Gun Booster = 80% of Table

With 5 HP End Gun Booster = 65% of Table

*Speed may need to be reduced if operating at temperatures below 60 degrees F

**If the system uses an End Gun Booster, the maximum speed will be less than listed in the tables. Use notes at bottom of each table to calculate maximum speed.

NOTE: Maximum speeds calculated as follows: Worm gear systems = All P Motors

Planetary with 16.9 x 24 or 11.2 x 38 tires = All G Motors

Planetary with 11.2 x 24 or 14.9 x 24 tires = G Motors on End Tower, all others = E's

Engine Drive Note:

The alternator must be capable of 15 amps outputs, with a maintained battery voltage of at least 12 VDC. Below this voltage, the DC-DC Converter may not work properly.

SPECIAL NOTE FOR GAS POWERED ENGINES (SPARK IGNITION):

To reduce electronic noise to the Precision Point Control III Panel, the engine should have:

- suppresser plug wires
- resistor spark plugs

TROUBLESHOOTING



If you need help with a problem, please have the following information ready for your dealer:

- A. Serial Number, Type of Power Supply, Full Circle or Auto Reverse, Collector Ring Or Cable Wrap, what is the system Hydraulic Pressure, what Safety Message is being generated if any.

Troubleshooting Table

Symptom	Possible Cause	Test Procedure	Possible Remedy
1. System will not move in one direction, moves ok in other.	A. Direction coil defective at end tower B. Open Connection C. Direction Solenoid Contaminated	A.1. Figure 67 - Check coil ohms B.1. Check Voltage C.1. Remove Cartridge Fig 68	A.1. Replace coil, T-L Part No. EC98389 C.1. Blow Cartridge out.
2. System will not move in either direction.	A. No Power from Power Supply B. End Tower Filter plugged C. Motorized Flow Control either fully closed or plugged D. Direction Solenoid Contaminated	A.1. Figures 69 & 70. Check Power Supply B.1. Remove filter element and check C.1. Figures 73, 74, 75 - Check MFC D.1. Remove Cartridge Fig 68	A.1. Replace fuse, see Figure 69 or 70 for size B.1. Replace element, Part HF57778 C.1. Replace MFC if it will not respond. Part HV98384 D.1. Blow Cartridge out.
3. Speed will not adjust.	A. Fuse Blown on Circuit Board B. Continuity lost to MFC at End Tower C. Speed will not slow down, runs wide open	A.1. Figure 65 check Speed control fuses. B.1. Figures 73 and 74. C.1. Figure 77 to 77.6- Check Sensor	A.1. Replace Fuse, see Figure 65 B.1. Check Connections C.1. Replace sensor.
4. ET Sensor Safety or Overwater Safety using Sensor Motor occurs continuously.	A. Speed sensor defective on End Twr or Next to Last Twr B. Collector Ring Brushes dirty or not making contact C. Overwater Speed Sensor is defective D. Pulses not getting to Circuit Board E. High Voltage Spike	A.1. Figure 77 to 77.6 - Check Speed Sensor B.1. Check continuity from brush to ring See Figure 80 C.1. Test Sensor, see Fig. 77 to 77.6 D.1. Check pulses, Fig. 77 to 77.6 E.1. Check ET Manifold Bd	A.1. Replace sensor, ES53218 (M12) B.1. Clean collector ring and brush See Figure 81 C.1. Replace Sensor D.1. Check connection in wiring harness E.1. Replace ET PCB, EH53240.
5. Optional End Gun Control will not switch End Gun On or Off	A. Solenoid coil shorted open at end tower B. Open Connection C. End Gun Solenoid Contaminated	A.1. Check coil ohms Fig. 82 B.1. Check Voltage C.1. Remove Cart. Fig 82.2	A.1. Replace coil, EC98389 B.1. 24 VDC C.1. Blow Cartridge out.
6. Auto Reverse Switch will not reverse pivot.	A. Loose or broken wire connections B. Wired improperly C. Reversing Switch at ET not working D. Direction Solenoid Contaminated	A.1. Check wiring from point to end B.1. Review wiring diagrams B.2. Test reversing in panel, Figure 83 C.1. Check operation of switch D.1. Remove Cartridge Fig 68	A.1. Repair, tighten as needed B.1. Rewire as needed C.1. Replace, T-L P/N EA52513 D.1. Clean Cartridge



Precision Point III Circuit Board (2011 -)

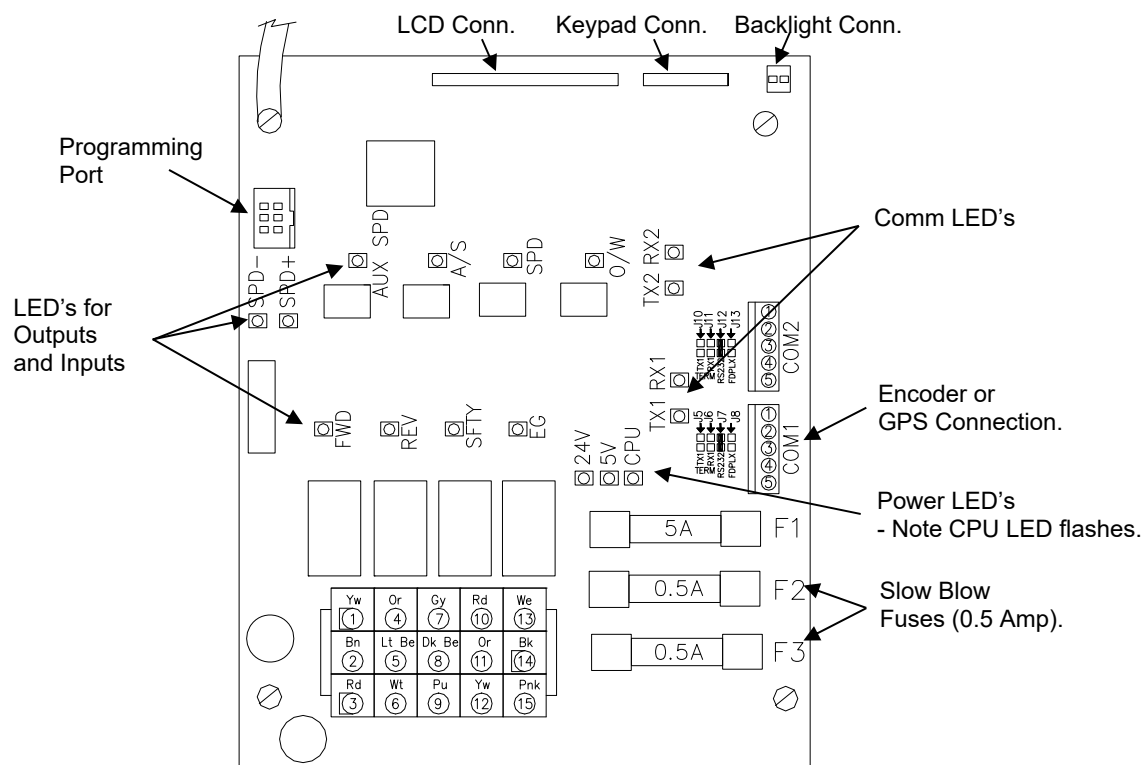


Figure 65. PPC III Circuit Board (2010-).

Precision Point II Circuit Board (2005 – 2010)

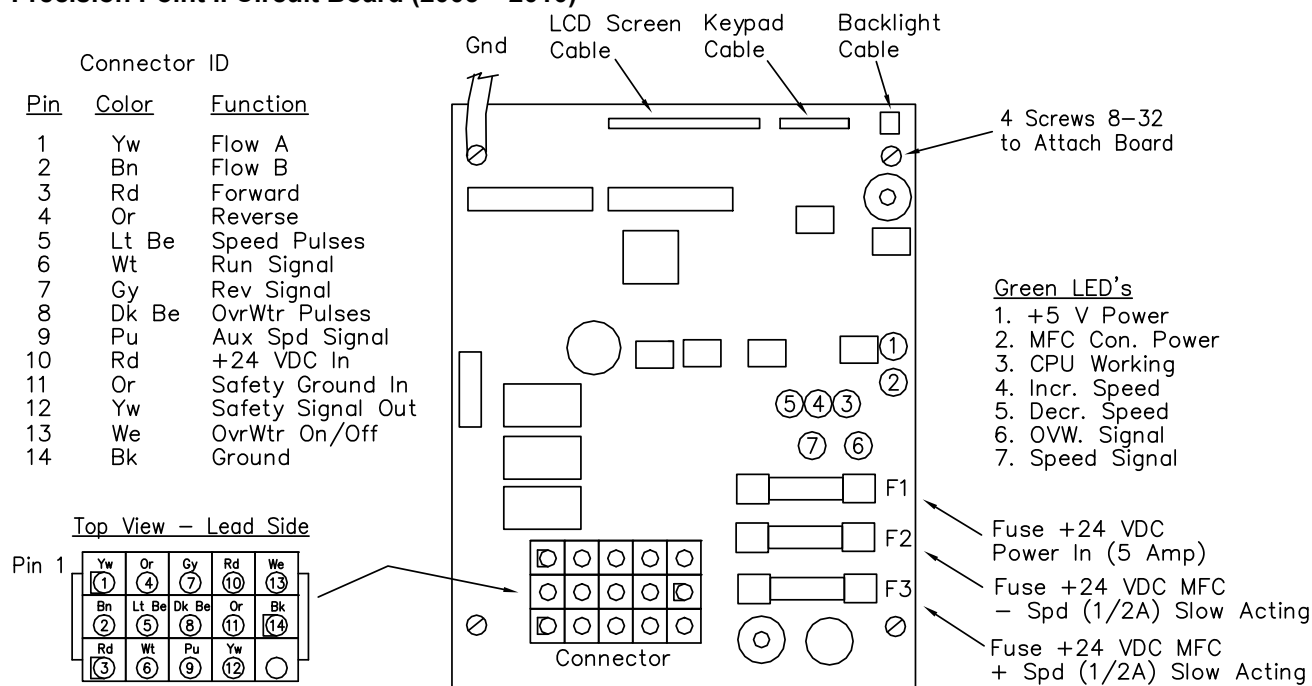


Figure 65.1. PPC II Circuit Board (2005 -2010).

End Tower Control Manifold Components (Figure 65)

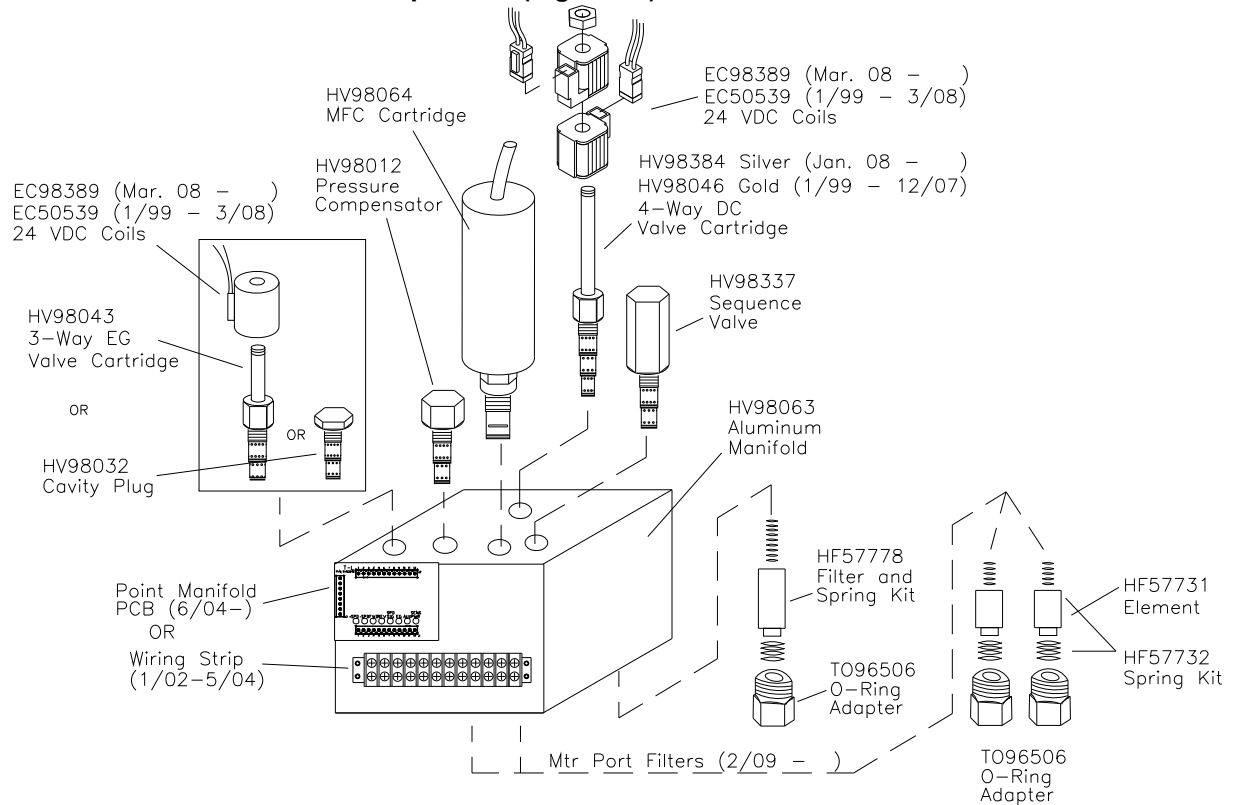


Figure 65.2 End Tower Control Manifold Components

Point Manifold PCB (Printed Circuit Board EH53240) June 2004 on

Observe the LED's on the Point Manifold PCB to tell what is happening at the end tower.

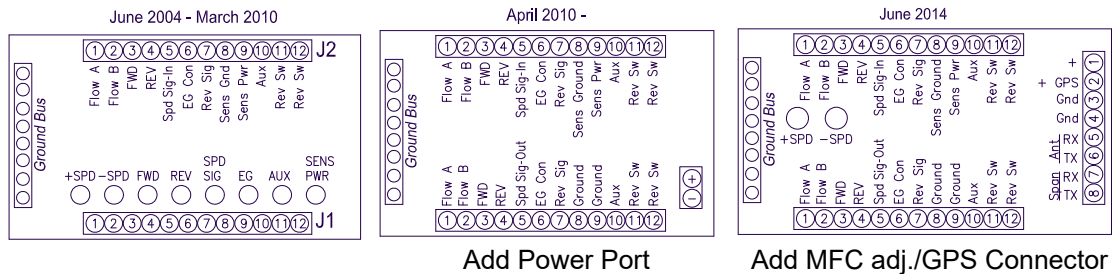


Figure 66.
Point Manifold PCB.

Ground Wire Checks (Especially if PPC Panel is Not on Pivot Point.)

Make sure panel grounds are in place from the #8 Terminal Block to the Pivot Frame. Use Star Washers.

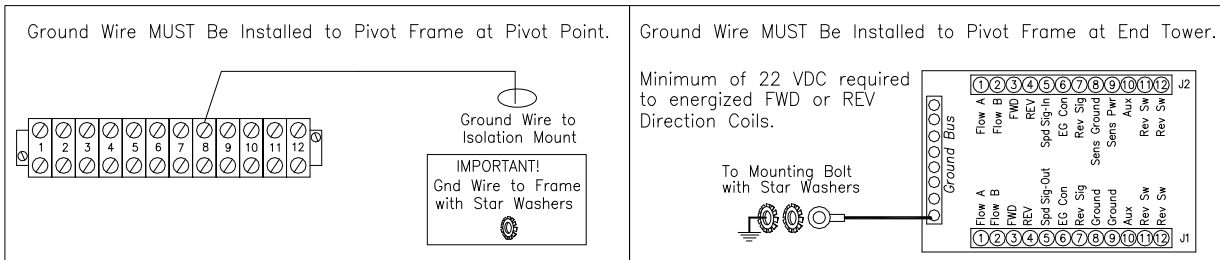


Figure 66.1. Check Ground Wires.

System Will Not Move in One Direction – Check Coils

Check for proper voltage (24 VDC) at the End Tower Terminal Block. See Figure 67.

NOTE: Voltage MUST BE at least 22 VDC for the Direction Solenoid to work properly.

CW = Forward, check between terminals 3 and 8 (ground)

CCW = Reverse, check between terminals 4 and 8 (ground).

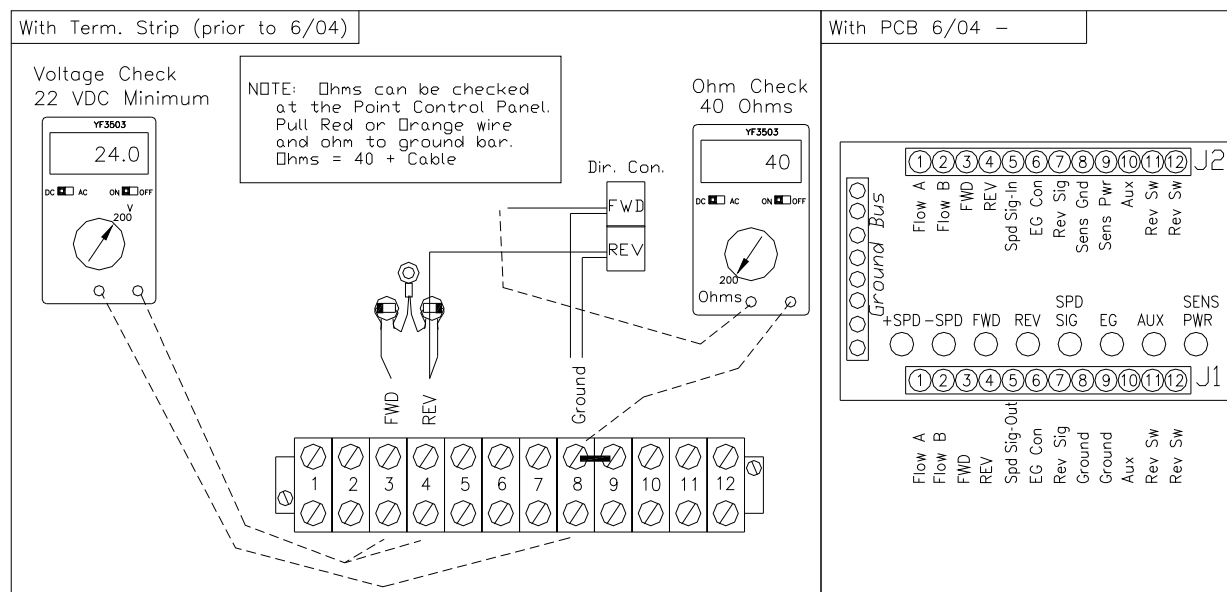


Figure 67. Check End Tower Solenoid Coils

System Will Not Move in One Direction – Check Solenoid Valve Cartridge

If the system will not stop when it should or if both coils can be removed and the system continues to walk; the 4-Way DC Valve Cartridge may be contaminated. Remove and check as follows:

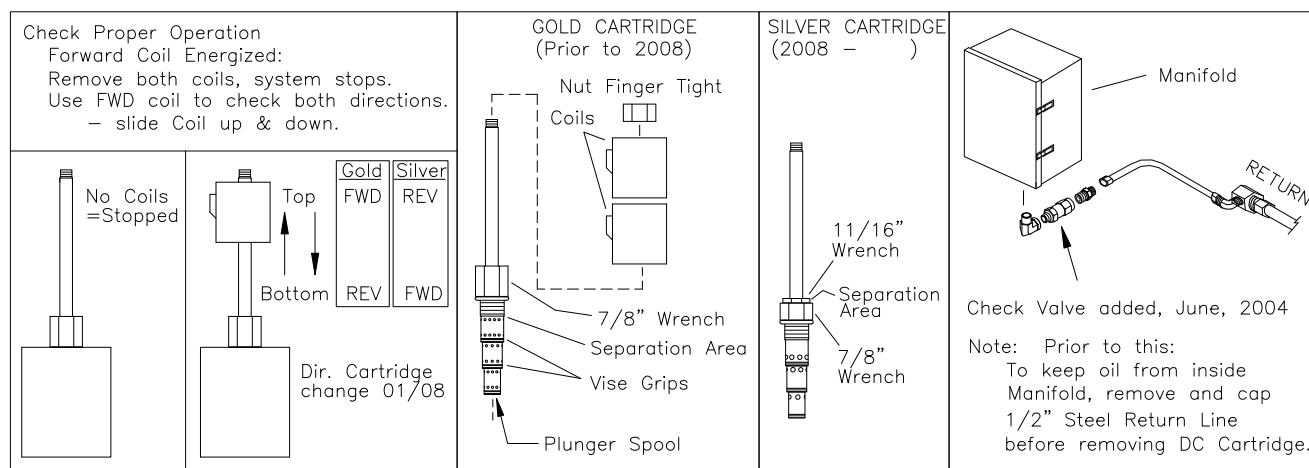


Figure 68. Check End 4-Way Solenoid Cartridge.

Remove the coils from the Valve Cartridge and then remove the Cartridge from the Body by unscrewing it. (To reduce oil loss, close the Ball Valve on the Pressure Line and if the system does not have a check valve, remove and cap the ½" Return Line.) Examine the Solenoid Valve ports for contamination. Blow the ports out, preferably with compressed air. Try to work the plunger, which is located inside the cartridge and can be accessed from the bottom of the cartridge.

The cartridge can be separated for contamination removal. The bottom portion will unscrew from the top portion. The threads have been sealed with lock-tight. See Figure 68.

System Will Not Move in Either Direction – Possible Reasons

Hydraulic reasons:

- No Hydraulic system pressure.
- Pressure Line Ball Valve at End Tower is closed.
- End Tower Filter is Plugged.
- DC Valve Cartridge is stuck, will not change position.

Electrical reasons:

- No 24VDC power to End Tower Coils.
- MFC (Motorized Flow Control) cartridge is in fully closed position.
- Ground Wires not in place at either the Pivot Point or the End Tower.

System Will Not Move in Either Direction – Check Power Supply

Electric Supplies - Figure 69. The power supply is located in the T-L Electric Panel and is regulated in the Precision Point Control III Panel. Power is supplied when the hydraulic pump is running.

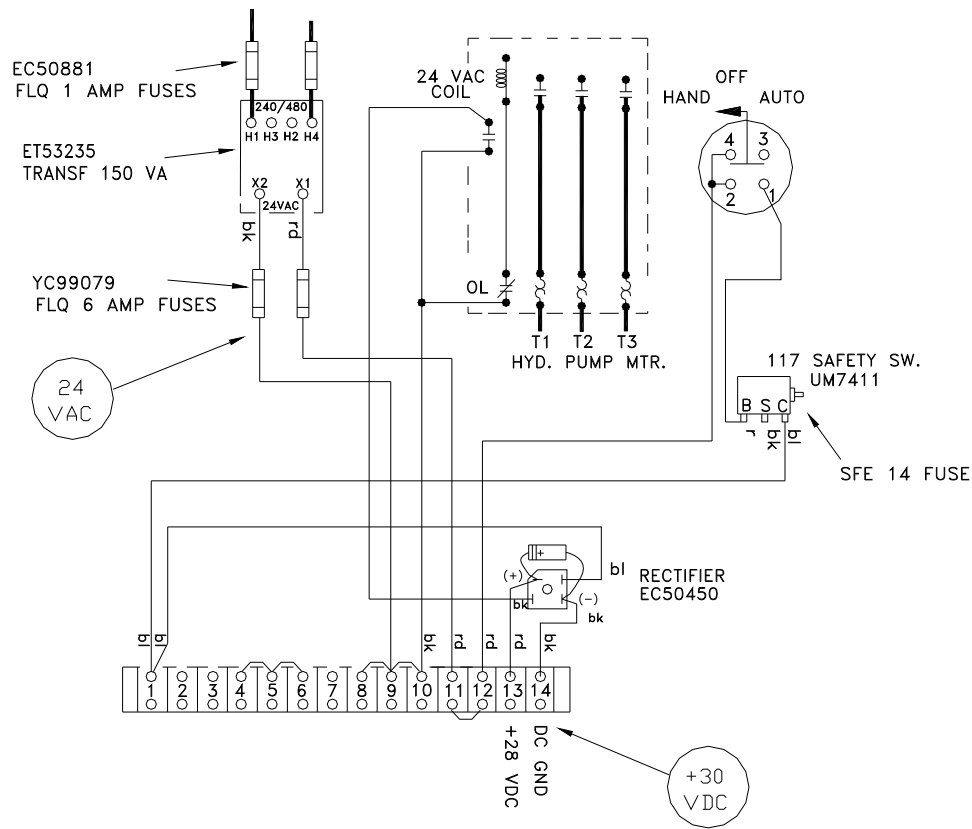


Figure 69. Power Supply Check - Electric Drive



Engine Supplies - Figure 70. The Engine Power Supply is located inside the Precision Point Control III Panel.

+12 VDC in from Murphy 117 "C", fused with AGC 15 amp fuse
 +28 VDC out, MDL 5 amp fuse
 T-L Part No. EA51055: DC-DC Power Supply

The voltage input to the DC-DC Power Supply board should be **at least +12VDC**, higher is better. The lower the input voltage, the harder the board has to work to develop the output. The engine alternator must be rated at **15 amps** continuous or higher.

Following are readings that can be taken to determine why the DC/DC Converter Board is defective, once a new board has been installed:

<u>Item</u>	<u>Amps</u>	<u>Voltage</u>
Direction Coil	0.58	24 VDC
End Gun Coil	0.58	24 VDC
Run Light	0.80	24 VDC
Motorized FC	0.25	12 or 24 VDC
DC/DC Bd Rating	2.75 output	28 VDC Maximum Rating on Board (80 watt version DC-DC)

Example: Point Control System, moving FWD, with End Gun On and Run Light on at 24 VDC:
 $0.58 \text{ amps} + 0.58 \text{ amps} + 0.80 \text{ amps} = 1.96 \text{ amps}$ on the output side
 Amps into the DC/DC Bd should roughly be 3 times the 24 VDC output current
 Amps into Bd = $1.96 \times 3 = 5.88 \text{ Amps}$ in at 12 VDC

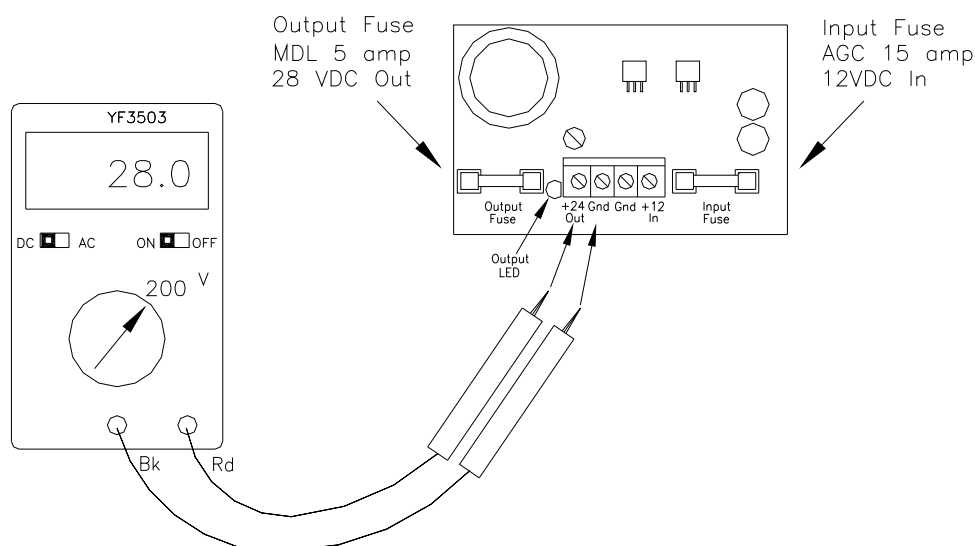


Figure 70. Power Supply Check - Engine Drive

System Will Not Move in Either Direction – Check Motorized Flow Control (MFC)

Make the following checks on the Precision Point Circuit Board:

- Verify CPU LED is flashing and that the Power LED's are lit.
- Verify integrity of F2 “- Spd Fuse” and F3 “+ Spd Fuse”.
 - Remove fuse from board and use ohm meter to check continuity.

With the system operational, verify speed change indication on Test Screen:

- Verify that the corresponding LED is lit on the Circuit Board.
- Review “Automatic Speed Control” from Operation section.
- Automatic Speed Adjustment occurs every 20 seconds for the 1st 2 minutes.
 - Note the “+” or “-” on the Test Screen every 20 seconds for the 1st 2 minutes.

SET IPM	038	+
ACT IPM	036	

- Power to Terminals 1 & 2 of the 12 terminal block in the PPC panel will only be present when speed adjustments are being made.

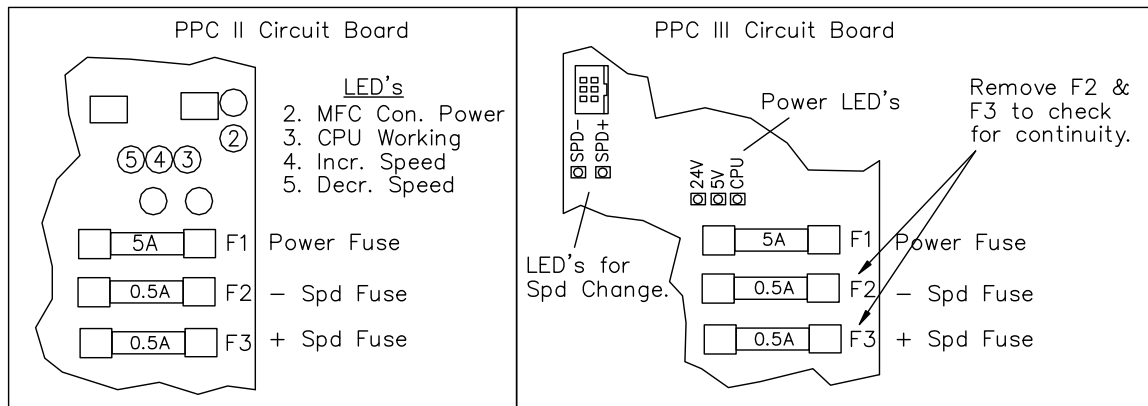


Figure 72. Check LED's and Fuses on PPC Board.

Check Voltages to MFC In PPC Panel and at End Tower.

NOTE: If either F2 or F3 fuse is blown, you may get voltage readings on both terminals 1 & 2. You also may see both “+ Spd” and “- Spd” LED's lit on the End Tower Manifold board..

Verify that 24 VDC is present at terminals 1 or 2 in the Precision Point Control Panel while the speed is being adjusted. See Figure 73. Then verify that 24 VDC is present at terminals 1 or 2 in the End Tower Panel while the speed is being adjusted.

It may be easier to Disable the “ET SEN SPD CON” in the System Setup and use the “+” and “-” keys on the keypad so you can make the speed changes manually instead of waiting every 20 seconds for the automatic speed change to occur.

If the voltage to the MFC checks good, then remove the Motorized Flow Control (MFC) from the aluminum body for visual inspection and operation. See Figure 74.

Manual speed adjustments at the End Tower Manifold can also be performed to check operation of the MFC Cartridge. See Figure 75.

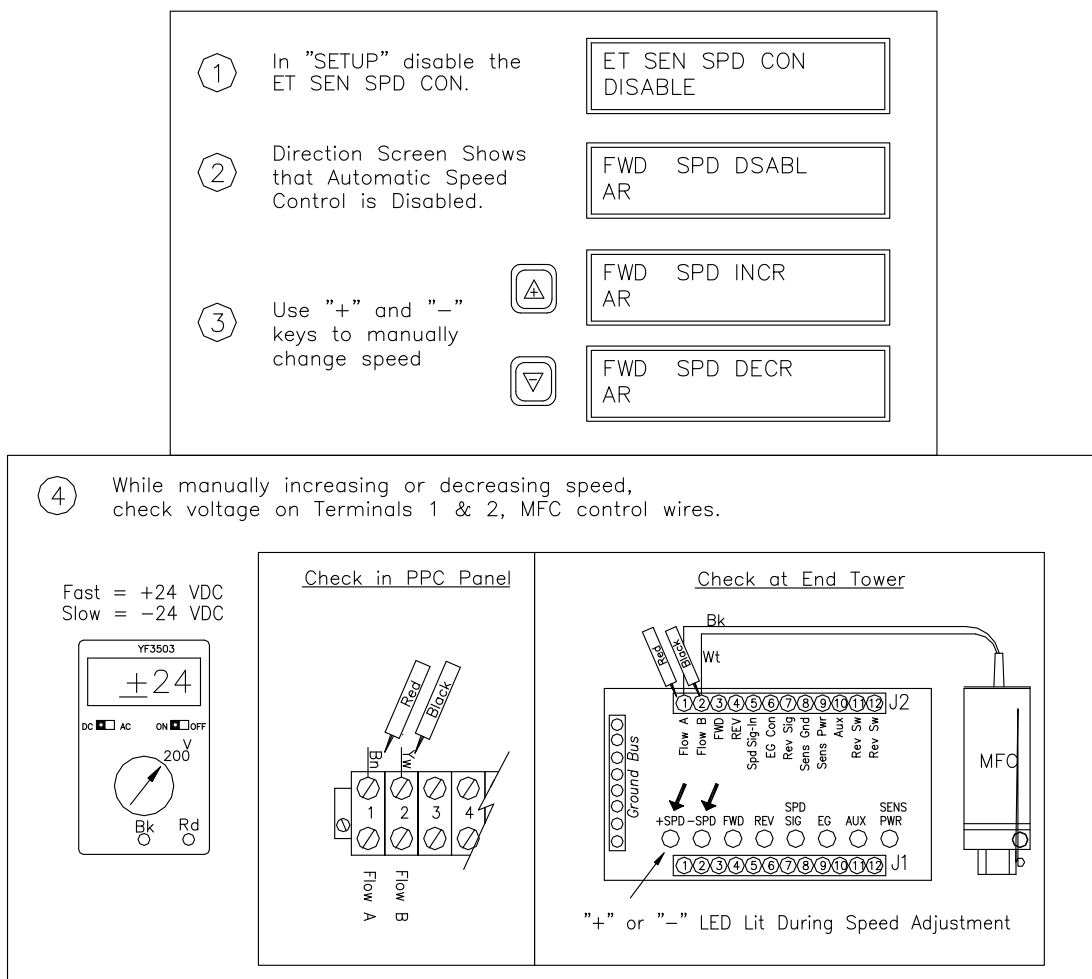


Figure 73. Check Voltage for MFC Operation in PPC II Panel and at End Tower.

Visually check the response of the Motorized Flow Control. Connect the 2 wires to a battery +12 VDC or to +24 VDC to see if the MFC adjusts. As the speed is being adjusted, you should visually see the internal sleeve rotating, until either full open or full closed position is achieved.

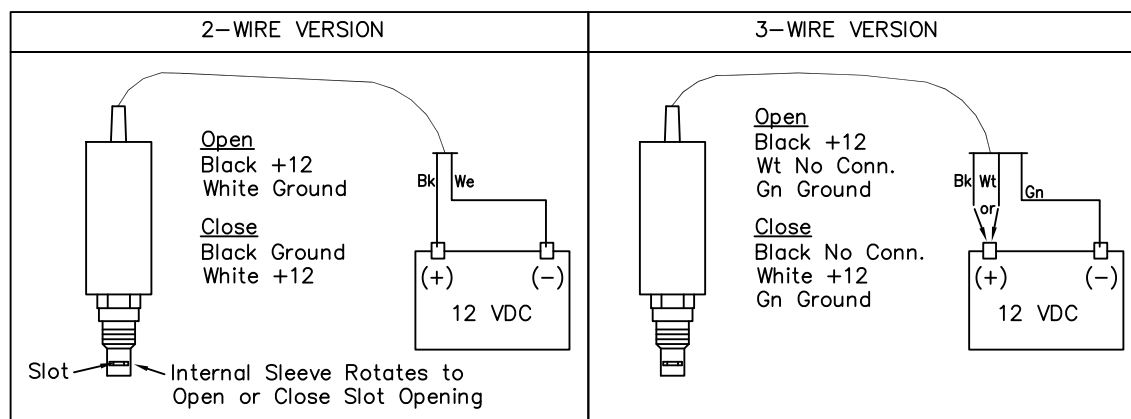


Figure 74. Check Motorized Flow Control by Removing and Connecting to a Battery.

Manually Adjust Speed at End Tower, System Moving Either FWD or REV

- Pushbuttons added to ET Manifold Board June 2014.
- Green Wire added to MFC in 2008. Green Wire disconnected from Gnd Bar June 2020.
- If Green Wire not connected, temporarily connect Green Wire to Gnd as troubleshooting aid to manually adjust the MFC End Tower Speed.

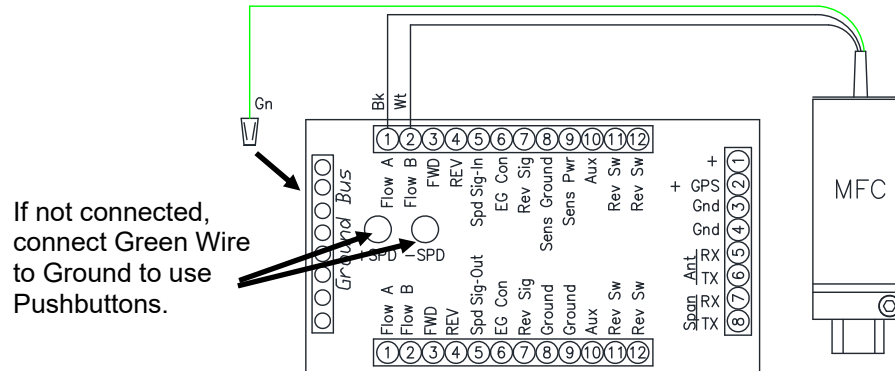


Figure 74.1 Use +SPD and –SPD Buttons to adjust.

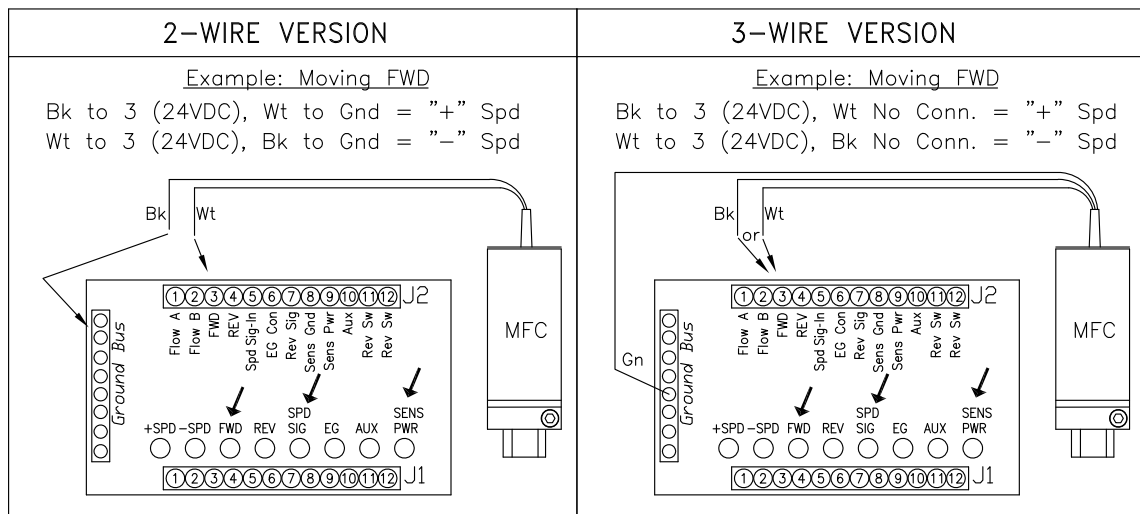


Figure 75. Manually Adjust End Tower Speed using Bk & Wt wires of MFC Valve.

Mechanically set MFC so End Tower Creeps Slowly

- If the end tower must run really slow, the MFC can be mechanically adjusted so it cannot close all the way.
- Electrically close MFC.
- Loosen Set Collar.
- Manually rotate MFC Body CW.
- Tighten Set Collar.

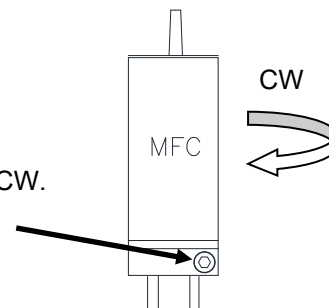


Figure 75.1. Mechanically set MFC.



Intermediate Tower in Bypass, ET Sensor Safety

NOTE: Use Caution, refer to Pivot Operators Manual.

Option 1: End Tower movement not needed to re-align:

1. Do not use PPC, all toggle switches in PPC Panel down.
2. Start Hydraulic Pump, safety switches to "Start" position.
3. Re-align tower, start PPC and operate normally.

Option 2: End Tower movement desired as tower is re-aligned:

1 PPC Panel Switches to Down position.	5 Re-align Tower in Bypass. – System should start moving
2 Start Hydraulic Pump – Safety Switches to "Start" position	6 Use "+" or "-" keys on PPC to Adjust Speed of End Tower.
3 On PPC, Disable ET Sensor Safety in Set-Up. – this also disables the Speed Control. ET SEN SPD CON DISABLE	 FWD SPD INCR AR  FWD SPD DECR AR
4 On PPC, System Switch to "RUN" position. Slow down End Tower so it is not full speed when Intermediate Tower is re-aligned. Direction Screen FWD SPD DSABL AR Use "-" key for 5 secs.  FWD SPD DECR AR	7 On PPC, Enable ET Sensor Safety in Set-Up – remember the System switch must be in "Stop" to make changes in Setup. ET SEN SPD CON ENABLE
	8 Use PPC per Normal Operation. – double check Set vs. Act IPM  SET IPM 038 ACT IPM 038

Figure 76. Procedure to Re-align Tower in Bypass.

ET Sensor Safety (Actual Speed Displays “0” When “TEST” Button is Pressed)

- Precision Point Control III Board not receiving pulses from the End Tower Speed Sensor
 - for 60 secs. (vers 2.02 on) with “System” switch in the “RUN” position,
- The end tower may be stopped, or it may be moving with a defective sensor or wiring problem.

OPERATION NOTE: If Overwater Speed Sensor works, those pulses can also be used as the End Tower Sensor Signal until the ET Sensor is changed. (Jumper 9 to 5 in PPC Panel, remove blue wire from 5).

STEP 1: Initial Checks at PPC Control Panel

Result: If no pulses, disable ET Sensor Safety.

Note: Need to slow End Tower Speed to 6 – 10 IPM.

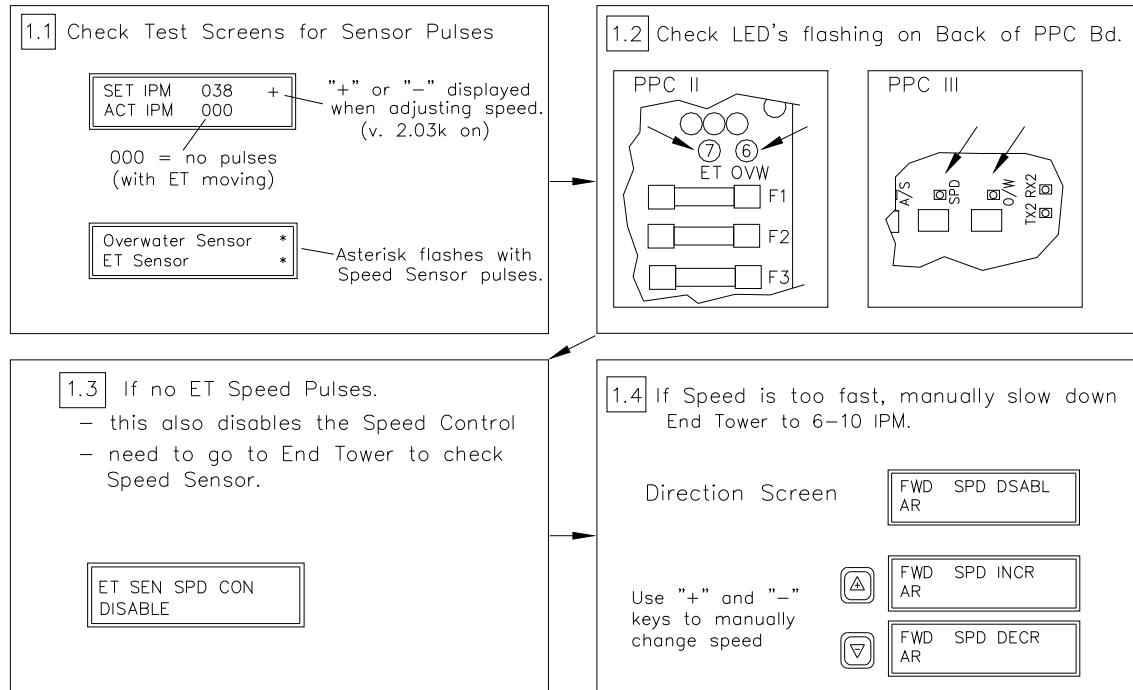


Figure 77. Initial Check's to Make at Precision Point Control Panel.

Speed Pulse Testing Notes for the following Steps:

- Make sure that End Tower is moving 6 - 10 IPM.
- Some “Auto-Ranging” meters cannot react quickly enough to correctly measure the pulsing voltage during the above tests. The use of Auto-Ranging meters is not recommended.
- If sensor pulses are present but the low voltage is above 1.0 VDC during any of the tests, the typical cause is poor system grounding. Be sure to check all grounds at components and if there are any available wires in the span cable add them to ground.

SENSOR PULSE TESTING TYPICAL VOLTMETER SETTINGS

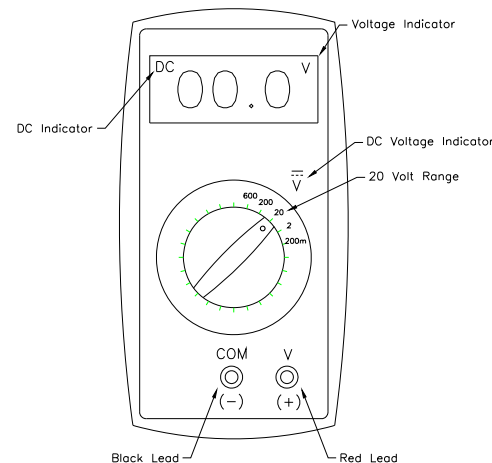


Figure 77.a. Voltmeter Settings

STEP 2: Check Pulse Voltage in PPC Control Panel – Blue Wire Connected to #5.

Result: If Pulse Voltage is good at Term #5, Speed Sensor and wiring back to PPC must be good and PPC Board must be defective.

- Replace PPC Board.

Note: Need End Tower Speed to be 6 – 10 IPM for test.

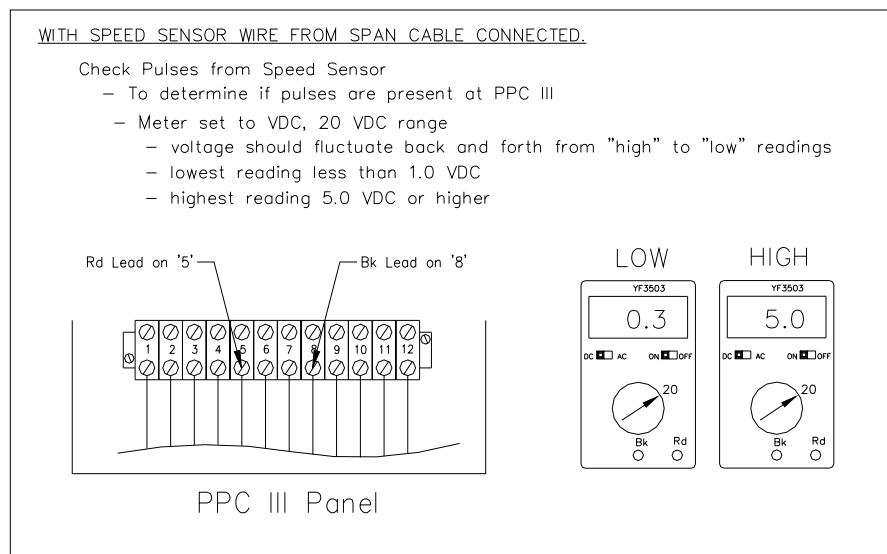


Figure 77.b Check Speed Pulses at PPC III

STEP 3: Check Pulse Voltage in PPC Control Panel – Blue Wire NOT Connected to #5.

Result: If Pulse Voltage is good at Blue wire when NOT connected to Term #5, Speed Sensor and wiring back to PPC must be good and PPC Board must be defective.

- Replace PPC Board.

Note: Need End Tower Speed to be 6 – 10 IPM for test.

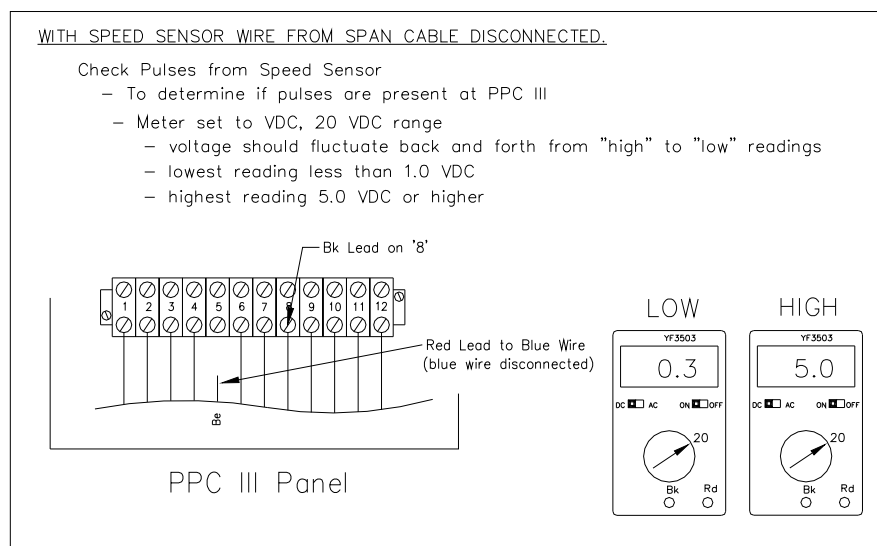


Figure 77.c Check Speed Pulses at PPC III – Blue Wire Disconnected

STEP 4: At End Tower, Check Power to Speed Sensor

Result: If 19-22 VDC supplied to Speed Sensor, continue to Step 5.

If no voltage, verify Fwd or Rev Voltage, replace defective component.

Notes: Use respective diagram for type of wiring at End Tower.

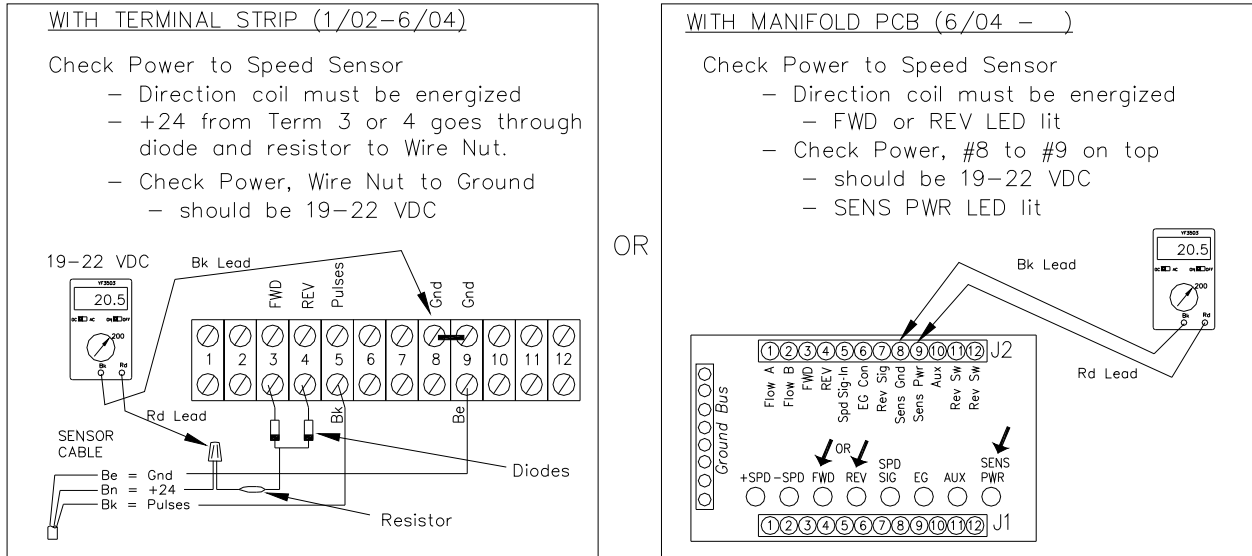


Figure 77.1 Check Speed Sensor Power at End Tower.

STEP 5: Check Speed Sensor Pulses at End Tower

Result: With Terminal Strip: If no pulses, go to Step 6. If pulses, check wiring back to PPC.

With Manifold PCB: If no pulses on top, go to step 6. If pulses on top and not on bottom, go to step 6. If pulses both top & bottom, check wiring back to PPC.

Note: Need End Tower Speed to be 6 – 10 IPM for test.

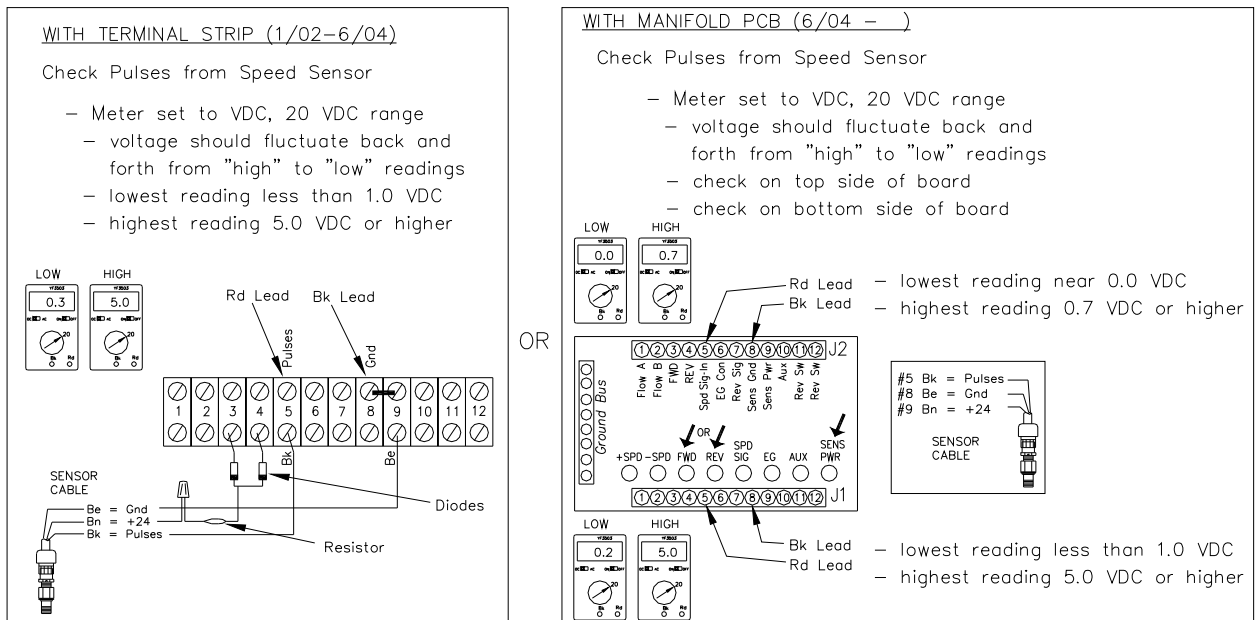


Figure 77.2 Check Speed Pulses from Sensor at End Tower.

STEP 6: Check Speed Sensor Pulses at End Tower, Black Pulse wire NOT Connected to Term 5.

Result: If no pulses, then Speed Sensor or Cable is defective, go to Step 8. If pulses when not connected but no pulses when connected, then pulses must be shorted somewhere – check all connection points back to PPC Panel.

If Manifold PCB, go to Step 7.

Note: Need End Tower Speed to be 6 – 10 IPM for test.

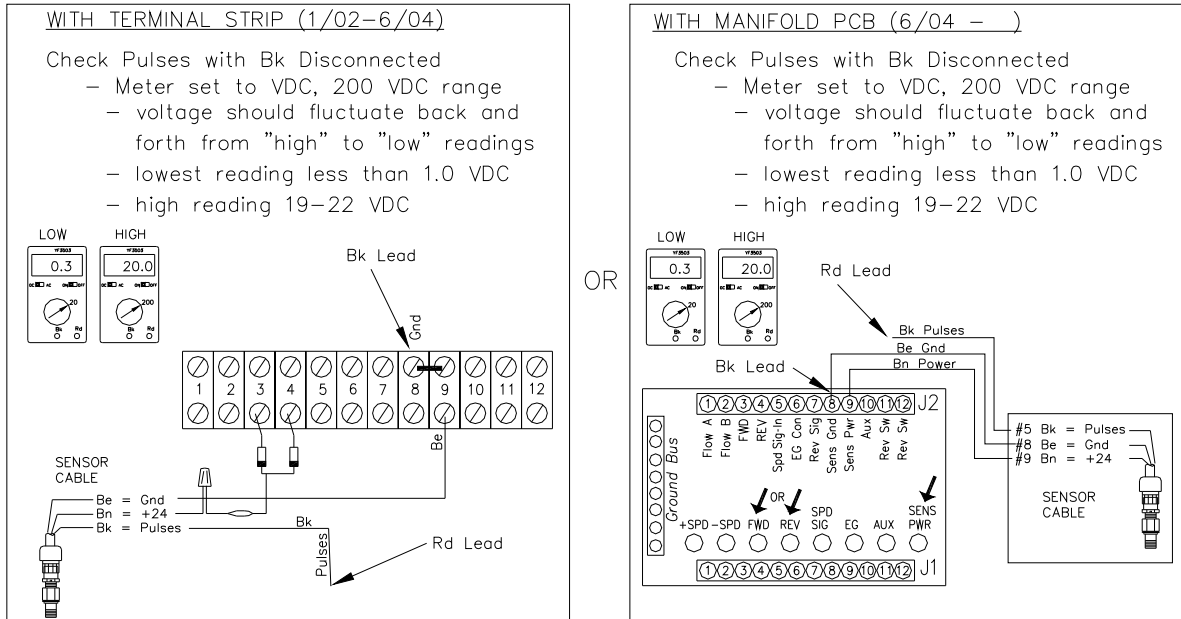


Figure 77.3 Check Speed Pulses from Sensor with Bk Signal Wire Disconnected at End Tower.

STEP 7: With Manifold PCB – Connect Bk Pulse Wire direct to Span Cable, bypass Circuit Board.

Result: If no pulses, then pulses must be shorted somewhere – check all connection points back to PPC Panel. If pulses, then replace Manifold PCB, T-L PN EH53240.

Note: Need End Tower Speed to be 6 – 10 IPM for test.

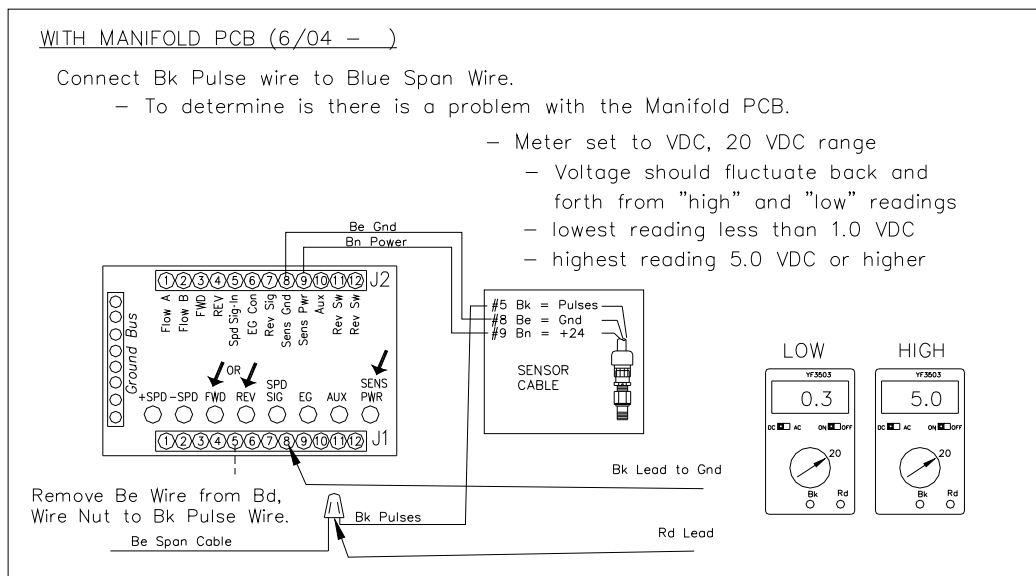


Figure 77.4. Bypass Manifold PCB with Signal Pulse Wire.

STEP 8: Check Speed Sensor on system using T-L Speed Sensor Tester – EA532180.

Result: If Speed Sensor checks good, replace Sensor Cable (ES53219)

If no sensor pulses, and alignment is good, replace Speed Sensor – see Step 10.

Note: See Sensor Tester Notes below.

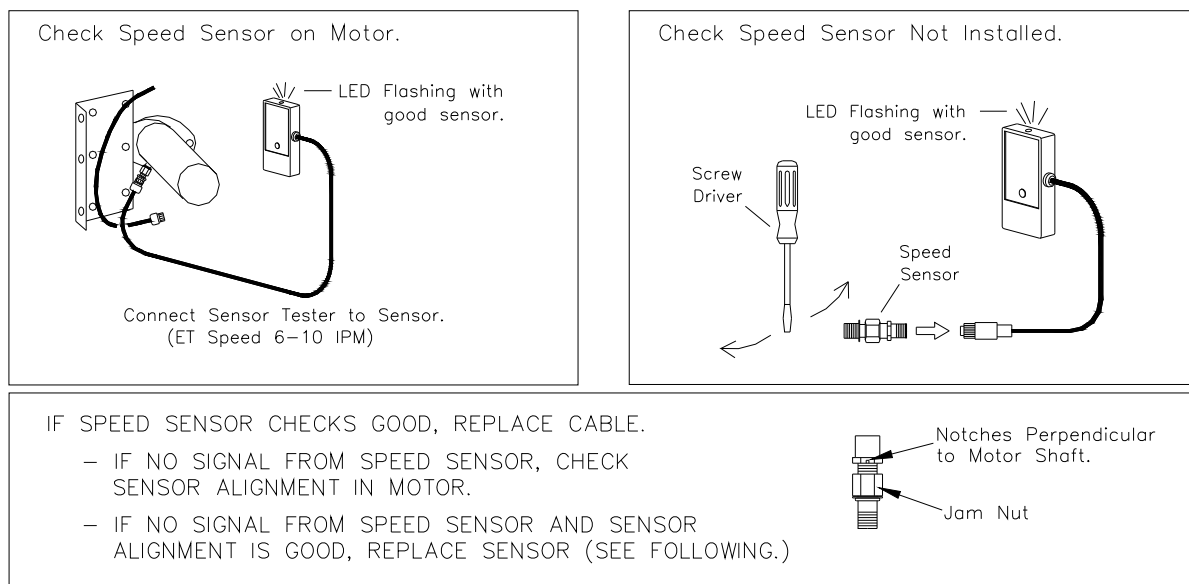


Figure 77.5. Use Sensor Tester to Check Speed Sensor.

T-L Sensor Tester (EA532180) Notes:

- Make sure that End Tower is moving 6 - 10 IPM.
- Press the Battery Test button on tester before connecting to sensor.
 - LED on top of Tester should light.
- Sensors that are not installed can be tested by connecting sensor to Tester and then placing “Ferrous” metal (screwdriver, pliers, etc) next to sensing face of sensor. Then move back and forth across the face of sensor to simulate motor shaft teeth going by sensor.

STEP 9: Check for additional Ground wire if Remote Panel, offset from Pivot Point.

SEE FIG. 35 FOR ADDL UNDERGROUND CABLE NOTES.

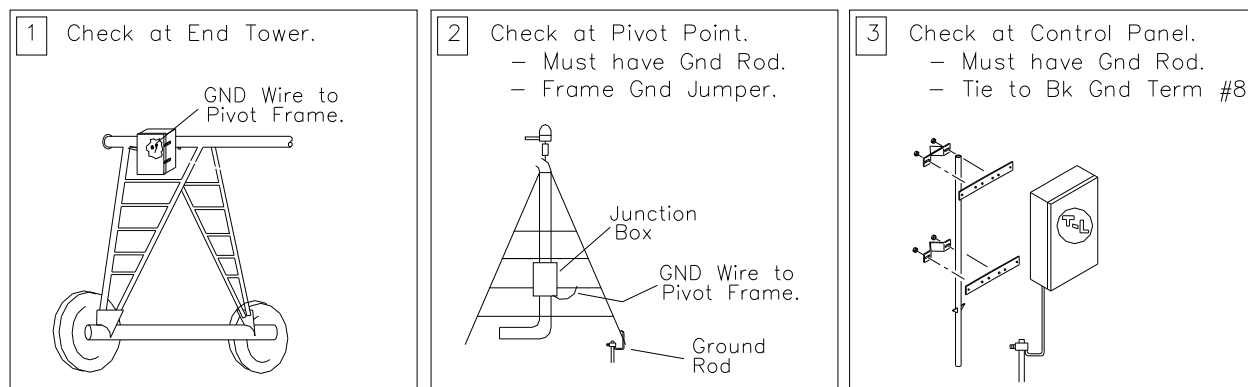


Figure 77.6. Remote Panel Sensor Checks.



STEP 10: Replace Speed Sensor.

If Speed Sensor does not generate pulses with +24 power to the sensor, it will need to be replaced using the following procedure:

1. Remove existing speed sensor wires from end tower junction box OR if separate yellow cable comes off sensor, unscrew cable from sensor
2. Unscrew existing speed sensor from hydraulic motor.
3. Install new speed sensor.
 - turn speed sensor in till it bottoms out on shaft (do NOT force) (be sure jam nut is backed off sufficiently)
 - back speed sensor out at least $\frac{1}{4}$ turn CCW plus the amount necessary to align the notches. Notches on speed sensor hex nut must end up perpendicular to motor shaft.
 - back speed sensor out only enough to align notch (at least $\frac{1}{4}$ turn)
 - hold in place and tighten jam nut.
 - Parker Sensor motors do not require alignment

NOTE: If the Speed Sensor at the End Tower is defective or pulses are not received by the Precision Point Control III Board, the Board will see this as the pivot not moving and thus keep trying to increase the speed of the pivot until the "ET Sensor Safety" occurs. (0 IPM actual vs. Set speed)

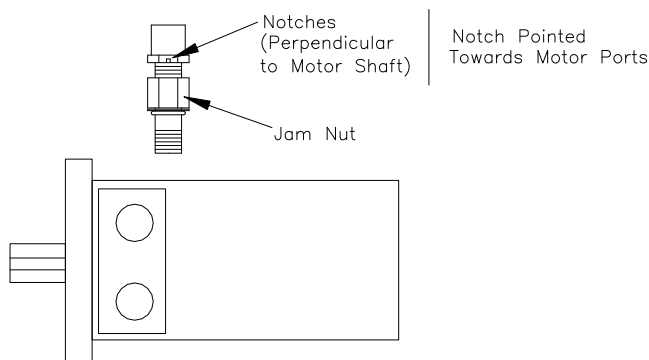


Figure 78. Speed Sensor Replacement.

Overwater Safety Occurs Continuously (Sensor Motor)

If the system is moving and Sensor Motor is being used for Overwater Timing, and “Overwater Safety” occurs, try the following:

- a. Verify Overwater Pulses on Test Screen or check “OVW SIG” LED on PCB.
 - see “Overwater Timer Operation” in the Operation Section
- b. Lengthen the “Overwater Timer” value in “Set-Up”
- c. Check the Overwater Speed Sensor for proper operation and pulses
 - see Figures 77 to 77.6.
 - if the Overwater speed sensor is defective, replace as shown previously.
 - Check ground jumper wires to frame. Use Star Washers for good grounding.

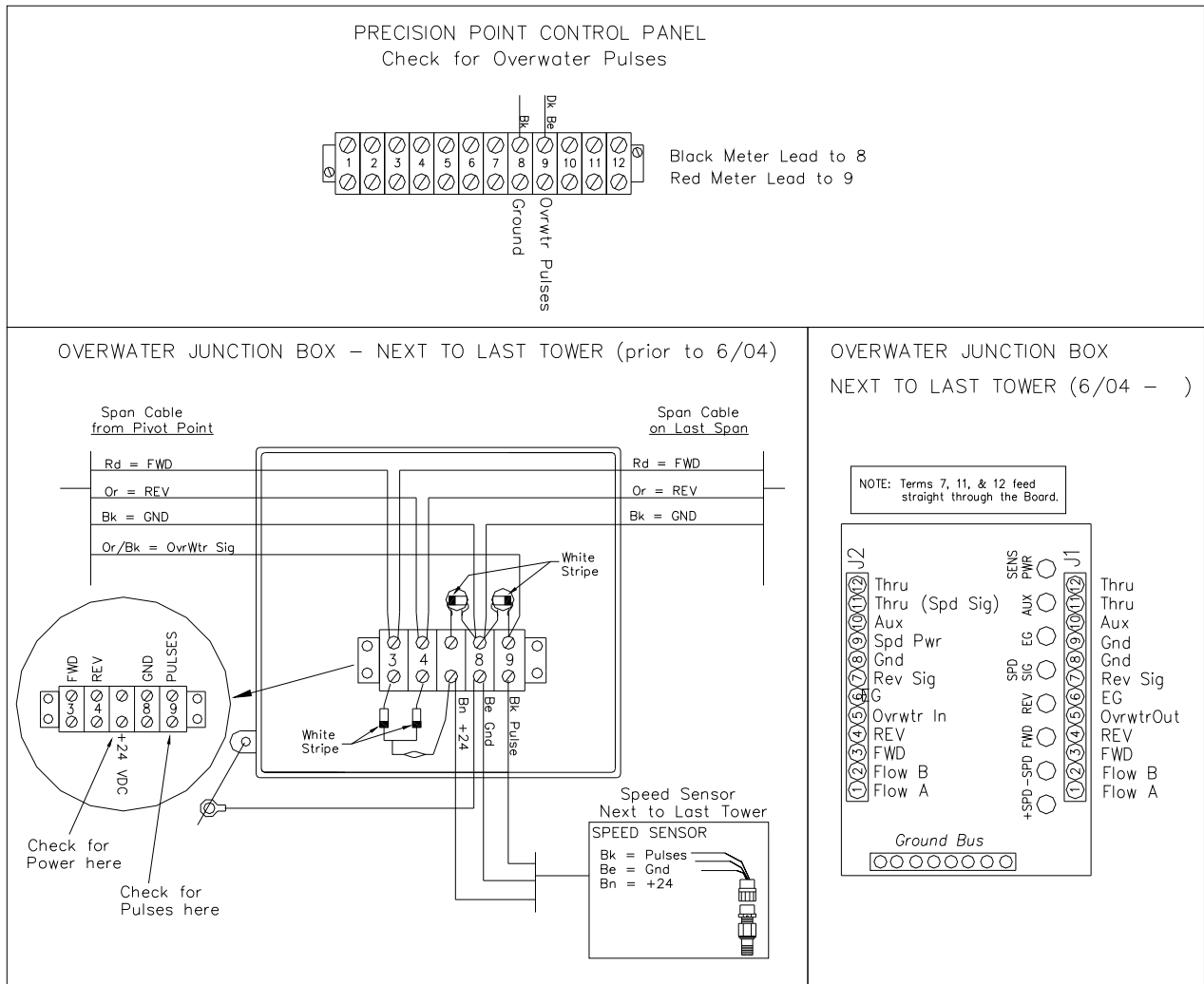


Figure 79. Check Overwater Speed Sensor Operation

Checking and Cleaning Collector Ring

Over time, buildup can occur on the Collector Rings and Brushes, causing a high resistance. This may cause erratic speed readings. See the following diagrams for checking and cleaning the Collector Ring.

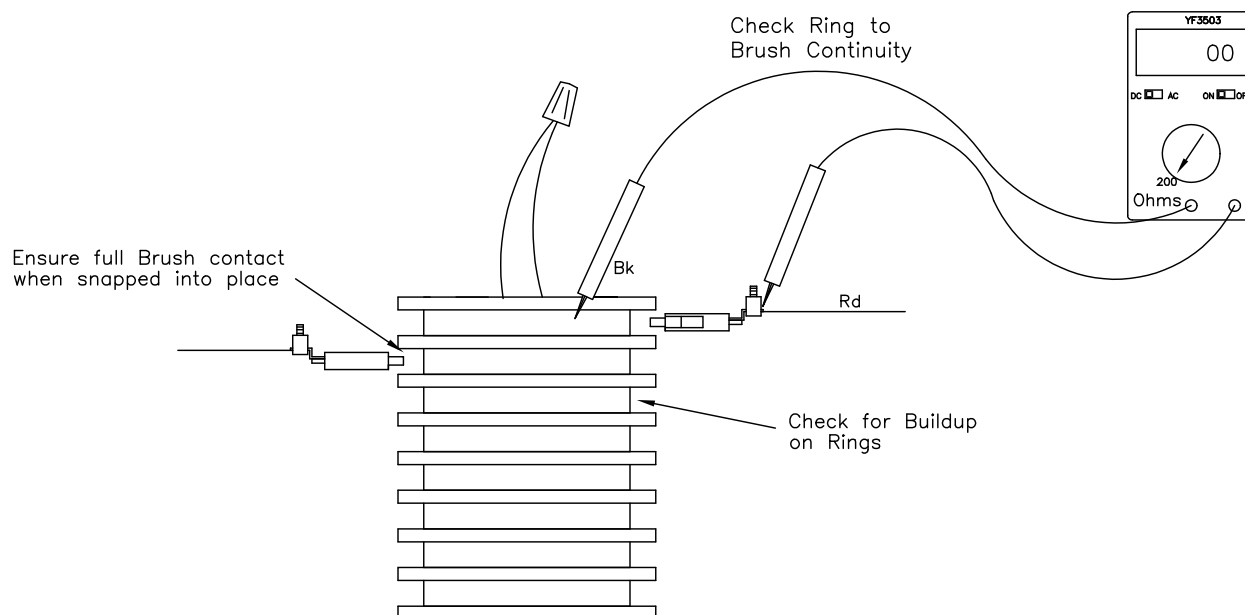


Figure 80. Check Collector Ring Continuity

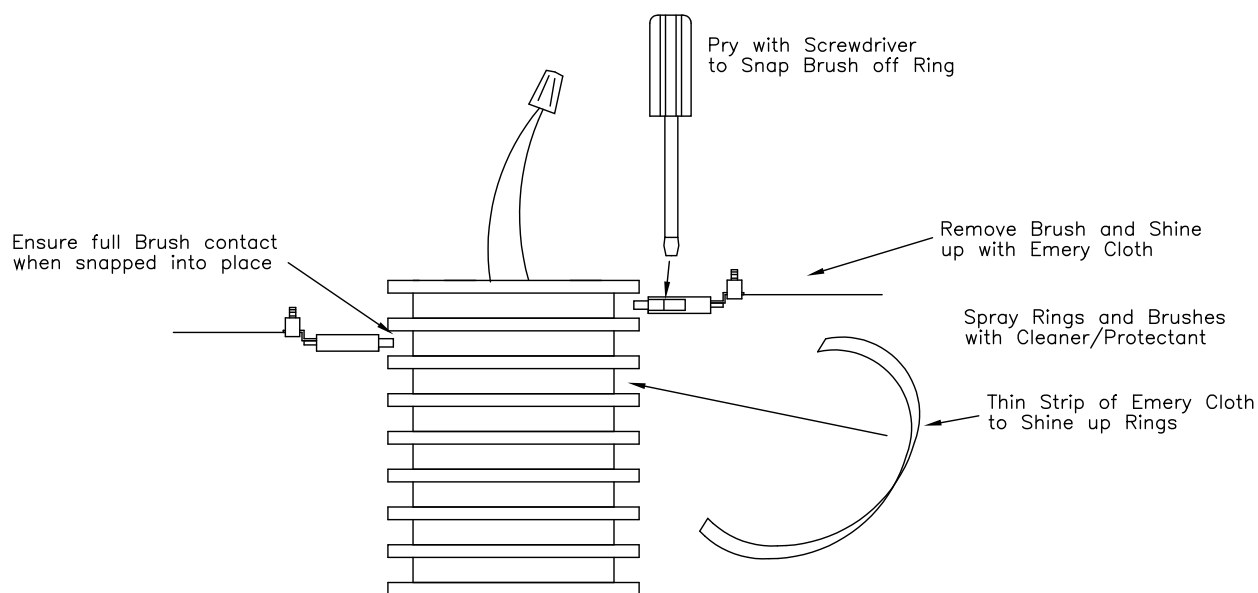


Figure 81. Cleaning Collector Ring and Brushes

End Gun Not Working

1. Check Wiring, Voltages, and LED's on Circuit Boards.
 - With Position Sensor, use End Gun Relay on PPC III Board:

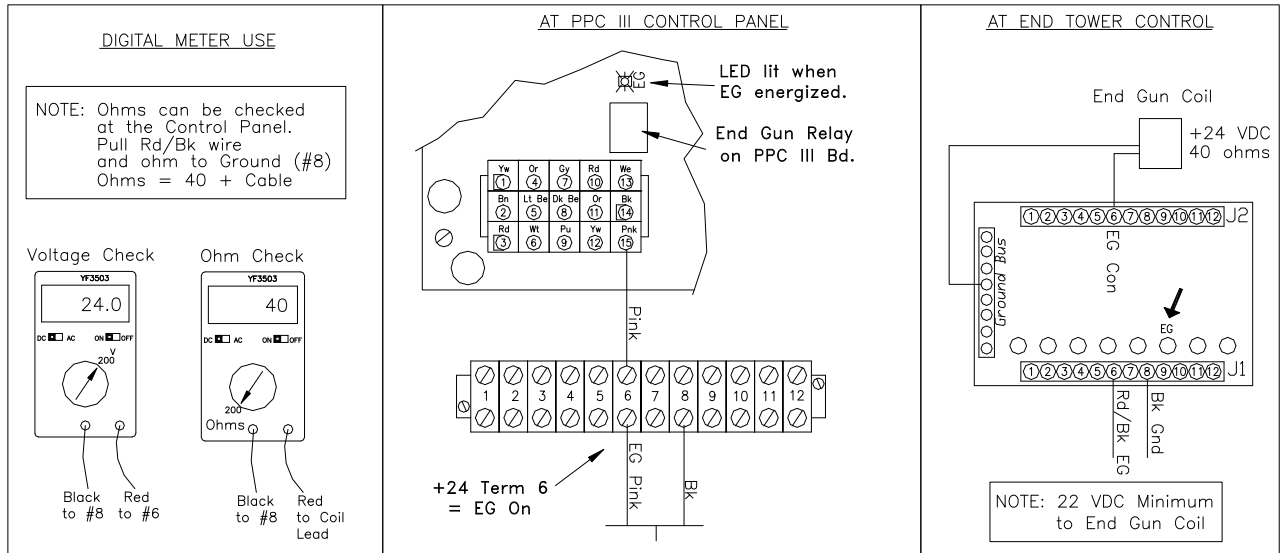


Figure 82. Check EG Operation, PPC III with Position Sensor.

- Without Position Sensor, use Roller Switch at Pivot Point>

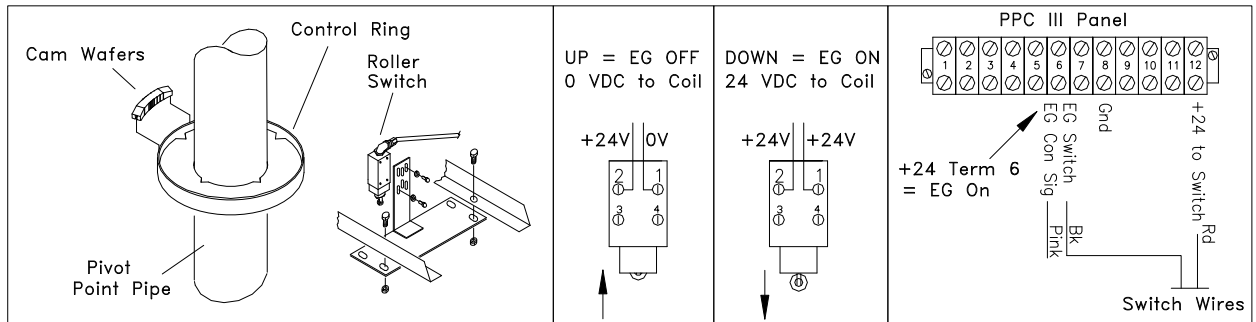


Figure 82.1. Check EG Operation, PPC II or III without Position Sensor.

2. Check Proper Operation at End Tower:
 - proper voltage to End Gun Coil (+24 VDC)
 - ohm the Coil (40 ohms)
 - Nut should be on FINGER TIGHT (3 ft. lbs. torque)
 - remove the Cartridge and check for contamination
 - Separate top from bottom if Plunger Spool will not shift.

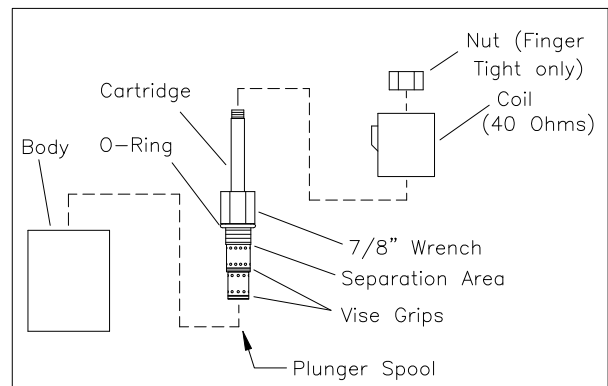


Figure 82.2. Check End Gun Solenoid.

3. Check Operation of Cylinder Shutoff Valve (without Booster).

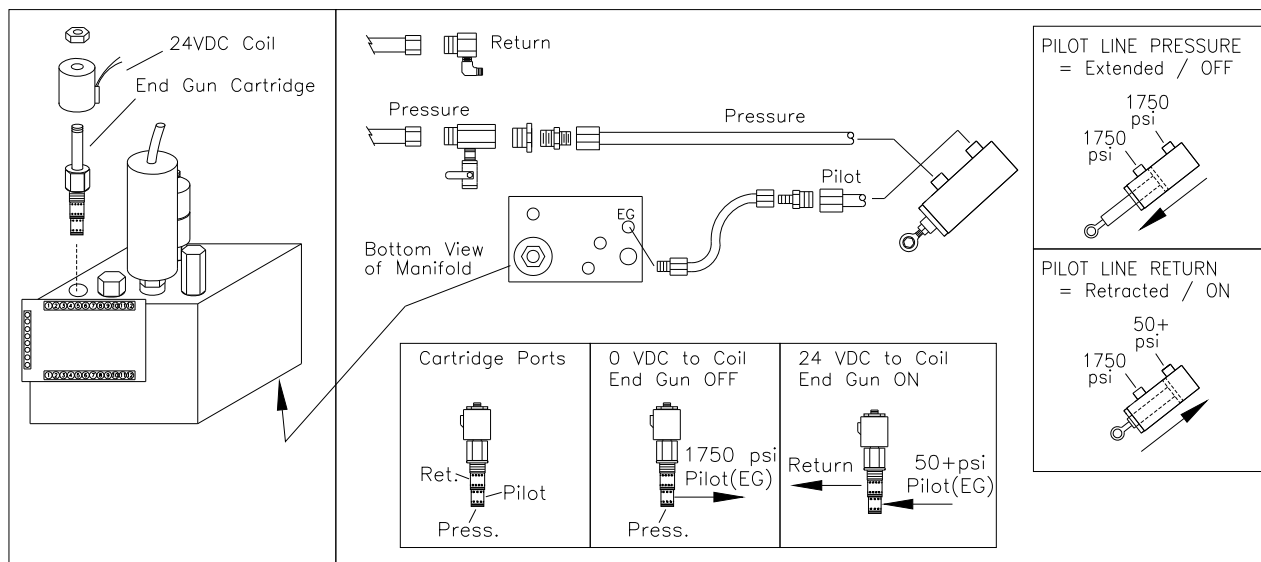


Figure 82.3. Check Operation of Cylinder Shutoff Valve.

4. Check Operation of Cylinder Shutoff Valve (with Booster).

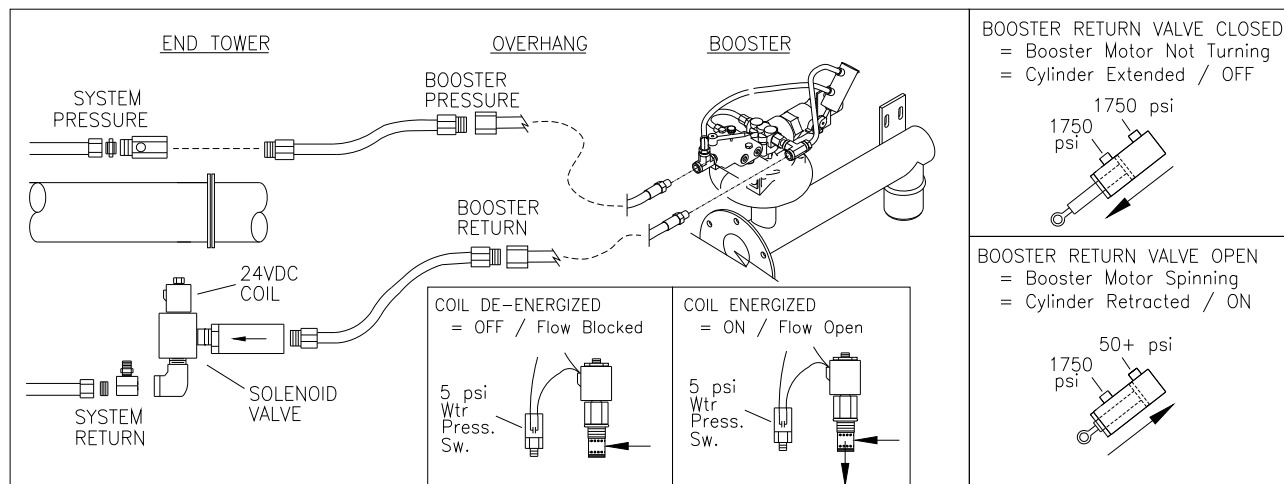


Figure 82.4. Check End Gun with Booster Solenoid.

Determine if the problem is hydraulic or electrical.

- Check that the direction coils are being activated correctly (see previous sections).
- If the coils are activated correctly, the problem is more than likely hydraulic and probably contamination in the directional control solenoid valve cartridge.
- If the coils are not switching, the problem can be assumed to be electrical. Start at the Control Panel and trace the direction control wires from the Control Panel to the End Tower.
- Trace the Reversing Signal from the Reversing Switch to the Control Panel. The Auto Reverse or Auto Stop function of the Precision Point Control III Panel can be checked by using a jumper wire in the control panel as shown in Figure 83.
- The A/S LED on the PPC III Bd will be lit when the Auto Reverse Signal is present.

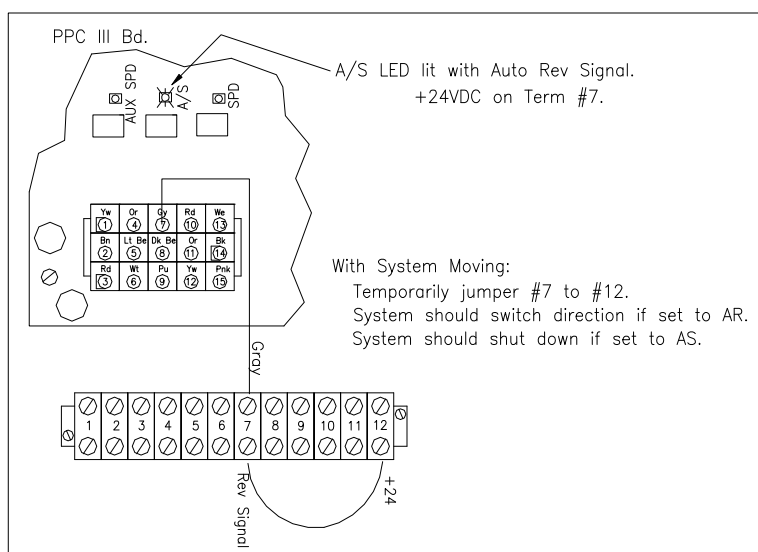


Figure 83. Check Reversing Operation in Precision Point Control III Panel.

- The function of the switch can be checked by verifying continuity as described below.
- Once activating the A/R function of the PPC III panel, you must wait at least 30 seconds before attempting to activate the function again.

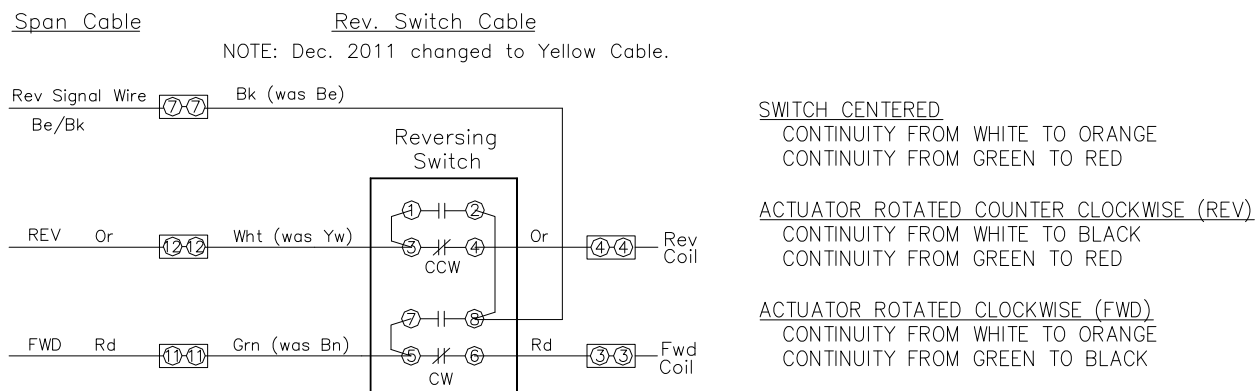


Figure 83a. Check Reversing Switch Functionality.



Communications Error (Encoder Comm Lost or GPS Comm Lost)

- Check all electrical connections:
 - power to Encoder or GPS.
 - comm wires connected properly at Com 1 Term Block on PPC III Bd.
 - comm wires connected properly at Encoder or GPS Antenna..
 - System must have good grounding for comm. to work. Use ET sensor low pulse voltage to verify good system grounding (less then 1.0 VDC low) at PPC III panel. Check Panel, J-box, ET manifold, grounding wire's with Star washers, etc..
 - Com1 Jumpers set correctly on PPC III Bd (Encoder = FDPLX, GPS = RS232, FDPLX After 06/14)
 - Check RX1 LED is flashing for Com 1 of PPC III Bd.
 - RX1 LED flashes on 1 second, off 1 second, etc. GPS only flashes when system is in RUN.

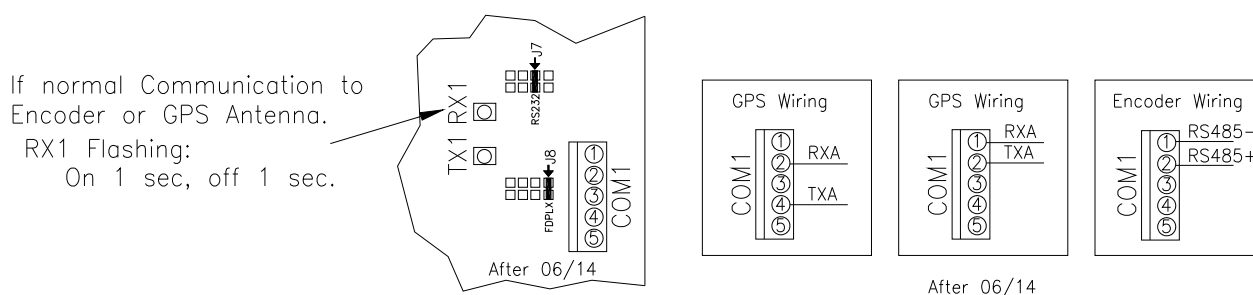


Figure 83.1 Check RX1 LED and Comm. wiring.

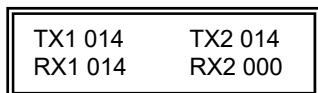
- Perform "Com Loop Test", First at the Board, then using the system wires.
Step 1: In System Setup, set "Com Loop Test" to YES, press Enter. Set the RS232 Jumpers on Com 1 and Com 2 of the PPC III Bd as shown in the figure below, set to YES and Enter.



- Step 2: Connect jumper wires on Com1 and Com2 term blocks, 2 & 4, of PPC III Bd (see figure below), set to YES and press Enter. This will test the Comm Ports on the PPC III Board.



- Step 3: Watch and compare TX1 to RX1 (Comm 1), and TX2 to RX2 (Comm 2)



TX1 or TX2 = packets set out.
RX1 or RX2 = packets received back at board.

Example above: RX1 = TX1 so communication is good for Com 1 Port.
RX2 = 000, which means the Com 2 Port is not working. Replace board.

Step 4: Repeat by jumpering the Com wires used on the system. If Encoder, move Swivel Cable wires to 2 & 4, connect loop in Collector Ring (Gn/Wt to Be/Wt). If Com Loop Test is good, the problem is with the GPS or Encoder. Otherwise, check wiring and connections.

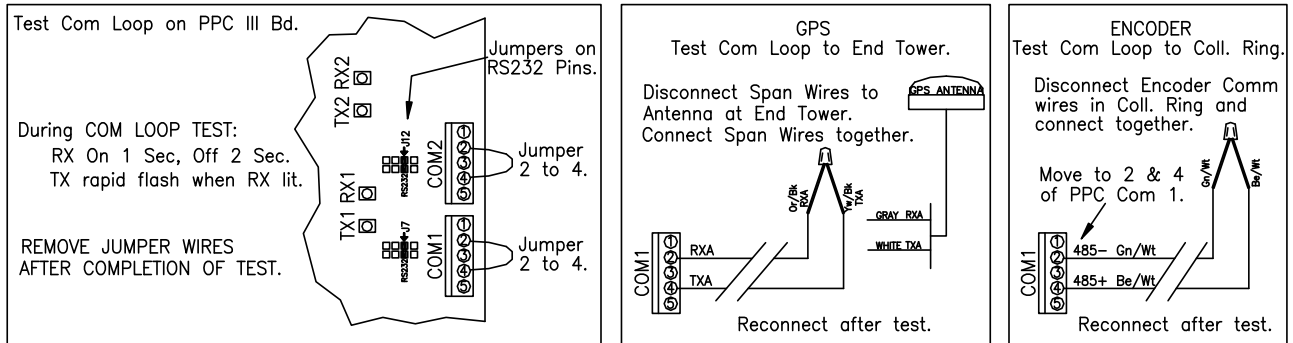


Figure 83.2 Configure Jumpers and Wiring for Loop Back Test.

GPS Comm Lost – Check Voltage on Comm Lines.

Voltage between terminal #2 & #4 or #1 & #2 should measure 3 to 5 VDC and fluctuate if communication is taking place. A steady voltage usually means there is no communication.

(NOTE: System Toggle Switch must be in “RUN” position.)

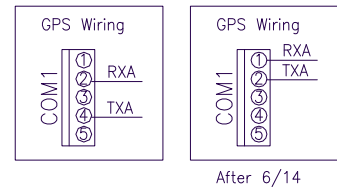


Figure 83.3. Check Voltage on Comm. Terminals.

End Tower Speed Limit – Orifice Disk Installation

To limit the top end speed, an Orifice Disk can be placed underneath the MFC Cartridge. The standard MFC flow rating is 3 GPM. Orifice sizes are available in 1.5, 2.0, and 2.5 GPM.

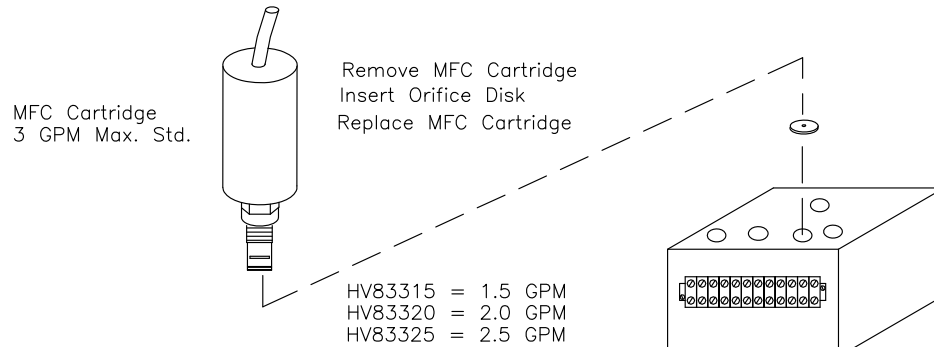


Figure 84. MFC Orifice Disk Installation.



Annual Maintenance Items

In addition to the regular maintenance items required for the pivot system (checking tire pressure, gear lube, filter changes, greasing pivot point, etc.), there are a few maintenance items required for the Point Control:

All Precision Point Control Systems

- check Control Panel at pivot point for loose wires
- check electrical cables and eliminate wear points
- open the End Tower Junction Box and inspect
- inspect Speed Sensor Cable and insure that it is not damaged

Reversing Point Control Systems

- check slide tube at End Tower for free movement in both directions

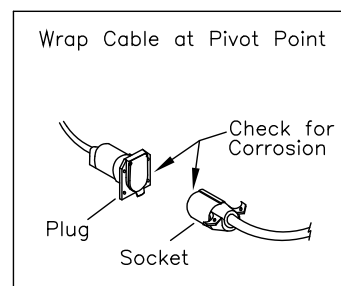
Collector Ring Systems

- Inspect Rings and Brushes for corrosion or buildup (see previous Figures)

Wrap Cable at Pivot Point

- check Plug and Socket contacts at pivot point for corrosion.
- clean and use electrician's di-electric grease to prevent corrosion.

Figure 85. Check for Corrosion.





Service and/or Maintenance Parts Listing

EA51055	DC-DC Converter, Engine 12 to 24 VDC
EA52513	Rev. Switch Assy - Yellow Cable (Dec. 2011 -)
EA53468	Diode Assy with Resistor (power for speed sensor, '98 – 2004)
EA53470	ZNR (MOV) with Diode Assy (in ET Control Panel, '98 - 2004)
EC50450	Rectifier Assy for Electric Power Supply
EC98389	Sterling Coil 24 VDC with Deutsche Connector and wire leads (black coil CCS024H)
EC98370	Coil Wires 36" Long with Deutsche Connector
EC98371	Coil Only with Deutsche Connector.
EC50746	PPC III Circuit Board (backwards compatible to PPC in gray panel)
EC50747	PPC III Face Plate, Keypad, Circuit Board Assembly, & Screen, less toggle switches
EC50714	Precision Point Control III Wiring Harness Cable
EC50718	LCD Display Replacement for PPC in Gray Box, with backlighting
EC50719	LCD Display Replacement for PPC II or III in Yellow Box with backlighting
EC50764	Fuse 15 amp AGC
EC5096	Fuse 5 amp AGC
EC50953	Fuse 5 amp MDL slow acting
EC50951	Fuse ½ amp MDL slow acting
EC53204	Brush Assy for Collector Ring
EC5322	Encoder
EC53231	GPS Antenna
EH53240	Point Manifold PCB (6/04 -)
EH53613	Weather Guard Corrosion Spray Protectant
EH53614	Collector Ring Spray Cleaner
ES53218	Speed Sensor for Eaton Motor – M12
ES532180	Sensor Tester
ES53219	Separate Sensor Cable for M12
EW53680	#15 Pink Wire for Wiring Harness, to add End Gun Control to existing Harness.
FW56512	Star Washer 1/4"
FW56513	Star Washer 5/16"
HF57731	Replacement Element for Motor Port Filters in Manifold Body (2/09 -)
HF57778	Replacement Element for Pressure Filter in Manifold Body
HV98012	Pressure Compensator Cartridge
HV98032	Cavity Plug 8-3 (for 3-way cartridge)
HV98043	End Gun Solenoid Valve Cartridge
HV98063	Aluminum Manifold, Body Only, for End Tower Controls
HV98064	MFC Cartridge Only – 3 GPM
HV98065	MFC Cartridge Only – 5 GPM
HV98337	Pressure Sequence Valve Cartridge
HV98382	Sterling Cartridge 2-Way for Booster Control Valve
HV98384	DC Solenoid 4-Way Valve Cartridge - Silver Plated (1/08 -)
HV98386	Body with ¾" Ports for 2-Way Booster Control Valve
MB53243	Retrofit Kit – Manifold PCB & Snap Track for Manifold system with 12 terminal block.



Adjustment if Measured End Tower Speed does not Match Actual Displayed.

Problem: Time it takes to make 1 Revolution does not match Hr/Rev on screen.

Solution: Rolling Radius needs to be adjusted in set-up to match actual field conditions.

1 Disable ET Sensor Safety in Set-Up – this disables the MFC Speed Control ET SEN SFTY SPD CON DISABLE (v. 2.03k)	OR	1 Remove Bn Wire from Term 1 in PPC – so MFC does not adjust speed
2 Start system moving, measure End Tower Speed		
A. At PPC, set System Switch to Run Position SYSTEM RUN STOP	B. Measure End Tower Speed – average IPM over 5 minutes	
3 Adjust Rolling Radius in Set-up of PPC so Actual Speed equals Measured Speed		
A. At PPC, check "ACT IPM" – with system moving, press "Test/Enter" key. SYSTEM RUN STOP TEST ENTER SET IPM 038 ACT IPM 038	B. Adjust "Last Tower Tire" in Setup (System switch=Stop) SYSTEM RUN STOP SET UP LAST TOWER TIRE 024.0 IN	C. Check "ACT IPM" again SYSTEM RUN STOP TEST ENTER SET IPM 038 ACT IPM 043
4 Repeat process of adjusting Last Tower Tire until "ACT IPM" equals what was measured at ET. Example: Measured Speed at End Tower = 43 IPM ACT Speed read on PPC = 38 IPM Rolling Radius needs to be increased.		
5 Enable ET Sensor Safety in Set-Up – to enable the MFC Speed Control ET SEN SFTY SPD CON ENABLE (v. 2.03k)	OR	5 Replace Bn Wire on Term 1 in PPC

Figure 86. Adjust Rolling Radius in Setup so Measured Speed = Actual Displayed Speed.

SETUP RECORDS



System: _____
Date: _____

POSITION SENSOR

NONE

BEI Encoder

GPS

Com Port to Use *Port 1* _____

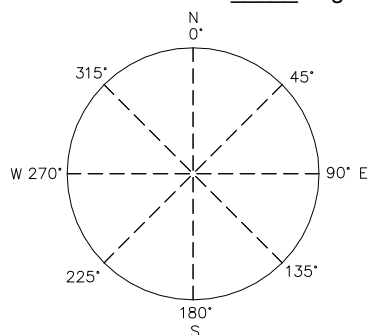
Com Port to Use *Port 1* _____

Current Angle _____ deg

PP1 Location: _____ ° _____ N or S

Direction Offset _____ deg

_____ ° _____ W or E



PP2 Location: _____ ° _____ N or S

_____ ° _____ W or E

PP3 Location: _____ ° _____ N or S

_____ ° _____ W or E

SYSTEM SETUP

Item

Example

Setting

Choices/Units

Circle to Use

Circle 1

Circle 1, Circle 2, Circle 3

Units

English

English or Metric

Last Wheel Track

1246

Ft

Irrigated Radius

1350

Ft

Water Flow

800

GPM

Last Tower Gear

68:1

Last Tower Tire

23.5

11.0 = 20.0, 14.9 = 23.5, 16.9=24.0, 11.2x38=27.0

Overwater Timer

Encoder

Encoder, Sensor Motor, GPS

Overwater Time

60

Min.

Max Speed

.20"

in.

96

IPM

Spd Adj Rate of MFC

.018

sec/IPM

Wrong Dir Safety

Enable

Enable / Disable

ET Sensor Safety

60

Sec.

ET SEN SPD CON

Enable

Enable / Disable
(If "Disable", MFC Adjusts for 45 secs.)

Com Loop Test

No

No/Yes

Run Time

(cannot change)

AgSense Link

No

No/Yes

AUTO STOP/REV

AUTO STOP/REV

No

AR DELAY TIMER

Min

FWD AS/AR

By Switch or By Angle and Angle

REV AS/AR

By Switch or By Angle and Angle

SETUP RECORDS



EG AREAS

EG AREAS _____ Enable/Disable

EG Table _____ 1, 2 or 3 (Up to 3 tables can be entered)

Table 1

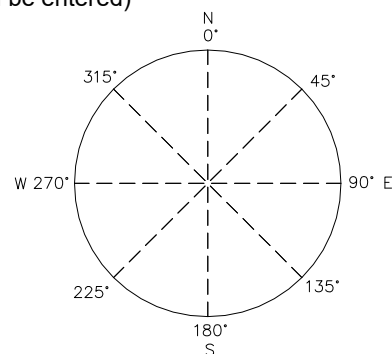
EG Table 1 Area 1 _____ deg To _____ deg
 EG Table 1 Area 2 _____ deg To _____ deg
 EG Table 1 Area 3 _____ deg To _____ deg
 EG Table 1 Area 4 _____ deg To _____ deg
 EG Table 1 Area 5 _____ deg To _____ deg
 EG Table 1 Area 6 _____ deg To _____ deg

Table 2

EG Table 2 Area 1 _____ deg To _____ deg
 EG Table 2 Area 2 _____ deg To _____ deg
 EG Table 2 Area 3 _____ deg To _____ deg
 EG Table 2 Area 4 _____ deg To _____ deg
 EG Table 2 Area 5 _____ deg To _____ deg
 EG Table 2 Area 6 _____ deg To _____ deg

Table 3

EG Table 3 Area 1 _____ deg To _____ deg
 EG Table 3 Area 2 _____ deg To _____ deg
 EG Table 3 Area 3 _____ deg To _____ deg
 EG Table 3 Area 4 _____ deg To _____ deg
 EG Table 3 Area 5 _____ deg To _____ deg
 EG Table 3 Area 6 _____ deg To _____ deg



AUX APPL

AUX APPL _____ Enable/Disable

Enabled with No Position Sensor

Fwd Aux Appl _____ in. _____ IPM

Rev Aux Appl _____ in. _____ IPM

Enabled with Position Sensor (either GPS or BEI Encoder)

Fwd Aux1 Appl _____ in. _____ deg To _____ deg

Fwd Aux2 Appl _____ in. _____ deg To _____ deg

Fwd Aux3 Appl _____ in. _____ deg To _____ deg

Fwd Aux4 Appl _____ in. _____ deg To _____ deg

Rev Aux1 Appl _____ in. _____ deg To _____ deg

Rev Aux2 Appl _____ in. _____ deg To _____ deg

Rev Aux3 Appl _____ in. _____ deg To _____ deg

Rev Aux4 Appl _____ in. _____ deg To _____ deg

Disbled with Position Sensor and AgSense Precision Link

VRI Table _____ Enable/Disable



Quick Start Operation of PPC III Panel.

Main Direction Screen

FWD APPL01.00" C1

AS at 180°157°

"Direction" key to change Dir.
 "+" or "-" to change Application.



"+" or "-" to change to next Setup Group.

SETUP GROUP

System Setup

SETUP GROUP

End Gun Areas

SETUP GROUP

Auto Stop/Reverse

SETUP GROUP

Aux Appl

SETUP GROUP

Position Sensor

SETUP GROUP

Safety History

"Enter" to access each Group.



Access Test & Diagnostic information..

SET IPM038 +

ACT IPM036

"+" or "-" to change to next Test screen.

OVERWATER20 min

ET SENSOR*

HR/REV128.2*

EG TBL1-3 ON

CURRENT RUN TIME

000:00:00

Etc.